SUPPLEMENT: THE SKETCHED STEPS OF BAŁABAN'S PROPOSITION 1 AT BLOCK FACTOR 5

Abstract. This supplement to the paper on the Osterwalder–Schrader axioms for the three-dimensional abelian Higgs model at fixed photon mass writes out the proof of Bałaban's perturbative estimate for localised renormalised classes, [3, Prop. 1, (1.33)], at block factor 5. The printed proof on [3, pp. 420–445] consists of twenty sketched steps and seven further sketched places; each is stated here in the form in which the proposition uses it and proved, with constants independent of the lattice spacing, the step, the domains, the background and the torus. The main paper cites these steps; nothing here is used elsewhere in it.

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1. The sketched steps of Proposition 1, I: slices, external-field kernels, the summation by parts and the induction

The main paper reads the perturbative estimate [3, Prop. 1, (1.33)] at block factor $L = 5$ and uses it in the form

$$|E(G_{\mathrm{ren}}, \Lambda, \Phi)| \;\le\; O(1)\, e_k^{d_v} \lambda_k^{d_s}\, e^{-\delta_0 d(\Lambda)} \prod \|h\Phi\|_{1,\alpha_0},$$

for a localised renormalised class G_{ren} with localisation $\Lambda = \{\square(v)\}$, $d(\Lambda)$ the length of a shortest tree through the cubes of Λ. Its proof on [3, pp. 420–445] consists of twenty sketched steps and seven further sketched places. This supplement writes them out. We number the sketched steps S1–S21 in the order in which they are invoked in the printed proof (S12 is the printed cube split of [3, p. 433], which needs no supplement), and the seven further places F1–F7. This part contains the first five:

step	[3] page	printed object
S1	414	the kernel [3, (1.16)], $[G_k(\Omega,\tilde{B})V_k]^n G_k(\Omega,\tilde{A}+\tilde{B})[V_k G_k(\Omega,\tilde{B})]^{n'}$, as an external field
S2	414, 424	$\delta G_k(\Omega,\Omega_2,\tilde{B}) = G_k(\Omega,\tilde{B}) - G_k(\Omega_2,\tilde{B})$ as an external field: [3, (2.5)], first alternative
S3	424, 426	the slice decomposition [3, (2.6)] and the slice bounds [3, (2.10)–(2.12)]
S4	425	the summation by parts [3, (2.8)–(2.9)] at a vertex of the graph [3, (2.4)]
S5	428	the induction [3, (2.16)]

S1 and S2 rest on S3, so the order below is S3 (section 1.2), two common tools (section 1.3), S1 (section 1.4), S2 (section 1.5), S4 (section 1.6) and S5 (section 1.7). Everything is at $L = 5$, $d = 3$, on the scale-k lattice with $k \ge 1$, in Bałaban's own model (no shell mass).

1.1. Setting and inputs.

Units and scales. $\eta := L^{-k}$ with $L = 5$ and $k \ge 1$, so $\eta \le \frac{1}{5}$; $t := t_k = L^k\varepsilon$; $e_k := e\, t^{1/2}$ is the running charge [1, (2.23)]. The slice scales are $s_j := L^j\eta = L^{j-k}$, $0 \le j \le k-1$, so $\eta = s_0 \le s_j \le \frac{1}{L}$. Scalar fields take values in $\mathbb{R}^N$; $U(A_b) = \exp(q e_k \eta A_b)$ with q antisymmetric, $\|q\| \le 1$ [1, p. 605]; all U commute and are orthogonal. The covariant derivative is $(D_{A,\mu}\phi)(x) = \eta^{-1}(U(A_{x,\mu})\phi(x+\eta e_\mu) - \phi(x))$ [1, (1.7)]; inner products carry the weight η^d and kernels are taken with respect to η^d. $|\cdot|_\infty$ is the torus sup metric [1, (1.3)], used for every distance, depth and exponential; $|\cdot|$ is the Euclidean torus norm, used only in the Hölder quotients of [3, (1.32)]; $|\cdot|_\infty \le |\cdot| \le \sqrt{3}|\cdot|_\infty$.

Blocks and domains. Unit blocks $B^k(y)$ have side 1; large blocks have side M [1, p. 607]. $\Omega = B^k(\Omega^{(k)})$ with $\Omega^{(k)}$ a union of large blocks; Ω_1, Ω_2 are of the same kind [3, p. 414]. The depth of x in X is $\rho_X(x) := \mathrm{dist}_\infty(x, X^c)$.

Averages and propagators. $(Q_k(A)\phi)(y) = \sum_{x\in B^k(y)} \eta^d U(A(\Gamma^{(k)}_{y,x}))\phi(x)$ along the composite contours of [1, (2.1)–(2.2)]; $P_k = Q_k^* Q_k$; $H_X(A) := -\Delta^{\eta,N}_{A,X} + m^2t^2 + a_k P_k(A)$ and $G_k(X,A) := H_X(A)^{-1}$ [1, (2.22)], where X is a union of unit blocks with Neumann conditions or the torus T_η. We assume $0 \le m^2t^2 \le c_m$. The vector field is the case $N = d$, $A = 0$, $X = T_\eta$, with μ_0 in place of m [1, p. 608]. $a_j = a(1-L^{-2})/(1-L^{-2j}) \in [\frac{24}{25}a, a]$ [1, (2.15)]; put $a_* := \frac{24}{25}a$.

The hypotheses of [3, p. 412]. $|\tilde{A}|$, $|d^\eta\tilde{A}|$, $|d^\eta\tilde{B}|$ and their Hölder norms of exponent α_0 are $\le c_1 p(L^k\varepsilon)$, and $\mathrm{dist}(\mathrm{supp}\,\tilde{A}, \partial\Omega) > 2r(L^k\varepsilon)$. Put

$$\sigma := e_k(\|\tilde{A}\|_\infty + \|d\tilde{A}\|_\infty) \le 2c_1 e_k p(t), \qquad \theta(\tilde{B}) := e_k\|d\tilde{B}\|_\infty \le c_1 e_k p(t), \qquad \theta(\tilde{A}+\tilde{B}) \le 2c_1 e_k p(t).$$

Smallness. We assume throughout

$$2c_1 e_k^{1-\beta} p(t) \le \min\{c, 1\}, \tag{E}$$

with $\beta \in (0,1)$ and c the constants of [1, (2.23)]. It holds for $t \le t_0$, since $e_k^{1-\beta} p(t) = e^{1-\beta} t^{(1-\beta)/2} b_0 (1 + \ln t^{-1})^{\bar p} \to 0$; it is the "$e(L^k\varepsilon)$ sufficiently small" of [1, Prop. 2.1].

Depth hypotheses. The lemmas below are stated with explicit depth hypotheses on the supports of the vertex localisations and of $\tilde A$ (S1: $\operatorname{supp}\tilde A$ at depth $\ge R_0 + 2$ and vertex supports at depth $\ge R_0 + 3$ in Ω; S2: vertex supports at depth $\ge R_0 + 12$ in Ω_2; S3: scalar line ends at depth $\ge R_0$). At the uses these depths are supplied by the nesting of Bałaban's regions: the vertex localisations lie within 1 of Ω_1 or of $\operatorname{supp}\tilde A$ ([3, p. 412]: the A'-legs carry g_k with $\operatorname{supp} g_k \subset \Omega_1$; [3, p. 414]: the block vertices [3, (1.13)–(1.15)] are paired at a common y), $\operatorname{dist}(\operatorname{supp}\tilde A, \partial\Omega) > 2r(t) \ge 2R \ge R_0 + 2$ under $R \ge 3R_0 + 15$, and at the domain changes of [2, pp. 579–580] the vertex supports have depth $\ge 5r(t) - 9$ in the smaller domain. This supply is part of the main paper's reading of the step; here the depths are hypotheses.

Inputs. Three propagator estimates are used, all of them statements about $G_k(X, A)$ and the fluctuation covariance at a single scale, proved in the main paper's appendix on the torus estimates (group (A) there) from [4]:

(i) **(P)** [1, Prop. 2.1] = the theorem [4, (1.9)–(1.12)]: for $\Omega = B^k(\Omega^{(k)})$ a union of large blocks and A regular in the sense of [1, (2.23)], $|(D^\eta_{A,\mu} G_k(\Omega, A) f)(x)|$ and $|(G_k(\Omega, A) f)(x)|$ are $\le c_0 e^{-\delta_0 \operatorname{dist}(x, \operatorname{supp} f)} \|f\|_\infty$ for $\operatorname{dist}(x, \Omega^c) \ge R_0$ [1, (2.25)], and the Hölder quotient $|x - x'|^{-\alpha} |U(A(\Gamma_{x,x'}))(DGf)(x') - (DGf)(x)|$ obeys the same bound with $\operatorname{dist}(\{x, x'\}, \operatorname{supp} f)$ in the exponent, for $\operatorname{dist}(\{x, x'\}, \Omega^c) \ge R_0$ [1, (2.24)]. The constants δ_0, c_0, R_0 are independent of A, k, Ω and of the torus, and the depth restriction is omitted for $\Omega = T_\eta$ [1, pp. 610–611].

(ii) **(C)** [1, Prop. 2.3, (2.33)–(2.34)] at $L = 5$: $|C^{(k)}_\Lambda(X, A; y, y')| \le c_2 e^{-\theta_0 |y - y'|_\infty}$ with no restriction on the points, and $(C^{(k)}_\Lambda)^{-1} \ge \gamma_X > 0$; the constants depend on L (through γ_L) and not on k, X, ε or the torus.

(iii) **(H0)** [4, (1.8)] on every union X of unit blocks and on T_η: $H_X(A) - m^2 t^2 \ge \gamma_0 := \frac{1}{4}\min\{2, a_*\}$, for A regular, which (E) supplies.

The norm [3, (1.32)] for kernels. For a kernel $Y(x, x')$ "extended in a natural way" to two variables [3, p. 420] we use the strongest natural reading: $\|Y\|_{1,\alpha}$ is the maximum over $a, b \in \{0, 1\}$ of the suprema of $\|Y_{ab}\|$, $Y_{ab} := D^a Y D^{b*}$, and of the α-Hölder quotients in x (for $a = 1$), in x' (for $b = 1$) and of the mixed second difference (for $ab = 1$), with partners $0 < |x - \tilde x| \le 1$ and $\tilde B$-transport along shortest contours. Quotients over larger distances are at most twice a supremum, and the summed form of [3, (1.32)] is at most 16 times this maximum.

1.2. **S3: the slice decomposition** [3, (2.6)] **and the slice bounds** [3, (2.10)–(2.12)]**.** [3, p. 424] decomposes every scalar line of G' by

(1)

$$G_k(\Omega, \tilde B) = C^{(0),\eta}(\Omega, \tilde B) + \sum_{j=1}^{k-1} a_j^2 (L^j\eta)^{-4} G_j^\eta(\Omega, \tilde B) Q_j^*(\tilde B) C^{(j), L^j\eta}(\Omega, \tilde B) Q_j(\tilde B) G_j^\eta(\Omega, \tilde B) = \sum_{j=0}^{k-1} G^\eta_{(j)}(\Omega, \tilde B),$$

"and the similar equality for the vector field propagator", and [3, p. 426] asserts the slice bounds [3, (2.10)–(2.12)] "by rescaling from the η-lattice to the L^{-j}-lattice and application of Propositions I.2.1 and I.2.3". The consumed form is [3, (2.13)–(2.14)]: every line l of G' carries one slice index j_l and contributes a factor $(L^{j_l}\eta)^{-(d-2)}$, times $(L^{j_l}\eta)^{-1}$ per derivative and $L^{j_l}\eta$ per contour leg, and an exponential with rate $\delta_1 (L^{j_l}\eta)^{-1}$.

Definitions in scale-k units. For $1 \le j \le k-1$ and $s := s_j$: $(Q_j^\eta(A)\phi)(z) := s^{-d} \sum_{x \in B_s(z)} \eta^d U(A(\Gamma^{(j)}_{z,x}))\phi(x)$, the average over the s-block $B_s(z)$; it acts on $\ell^2(\eta^d)$, its range carries the weight s^d, and $Q_j^\eta Q_j^{\eta *} = 1$. $P_j^\eta := Q_j^{\eta *} Q_j^\eta$. $H_0 := -\Delta^{\eta, N}_{A,\Omega} + m^2 t^2$, $G_j^\eta := (H_0 + a_j s^{-2} P_j^\eta)^{-1}$ (this is [1, (2.20)] with $\varepsilon \to \eta$, $L^k\varepsilon \to s$), $\Delta^{(j),s} := a_j s^{-2} - a_j^2 s^{-4} Q_j G_j^\eta Q_j^*$ [1, (2.21)], $C^{(j),s} := (a(Ls)^{-2} P^{(j)} + \Delta^{(j),s})^{-1}$ [1, (2.30)], where $P^{(j)}$ is the L-block projection on the s-lattice. $C^{(0),\eta} := G_1^\eta$ (this is [1, (2.30)]

at $k=0$ with [1, (2.17)]: $C^{(0)}=(-\Delta+m^2+a(L\varepsilon)^{-2}P)^{-1}=G_1$, the identity "$G_1=C^{(0)}$" of [1, p. 612]). Finally $G^\eta_{(0)}:=C^{(0),\eta}$ and $G^\eta_{(j)}:=a_j^2 s^{-4}G^\eta_j Q^*_j C^{(j),s}Q_jG^\eta_j$.

Lemma 1.1 (composite contours)**.** *$Q^\eta_{j+1}(A)=Q^{(j)}(A)Q^\eta_j(A)$, with $Q^{(j)}$ the L-block average on the s_j-lattice. Hence $P^\eta_{j+1}=Q^{\eta*}_jP^{(j)}Q^\eta_j$.*

Proof. By [1, (2.2)], $\Gamma^{(j+1)}_{z,x}=\Gamma_{z,x_j}\cup\Gamma^{(j)}_{x_j,x}$ with $x\in B^j(x_j)$, $x_j\in B(z)$; U is a homomorphism, so $U(A(\Gamma^{(j+1)}_{z,x}))=U(A(\Gamma_{z,x_j}))U(A(\Gamma^{(j)}_{x_j,x}))$; the sum over $x\in B^{j+1}(z)$ splits as a sum over $x_j\in B(z)$ and $x\in B^j(x_j)$, and $(Ls)^{-d}=L^{-d}s^{-d}$. The second claim is $Q^{(j)*}Q^{(j)}=P^{(j)}$. □

Lemma 1.2 (one renormalisation step; [1, (2.42)] at scale s)**.** *For $1\le j\le k-1$ every inverse below exists and $G^\eta_{j+1}=G^\eta_j+a_j^2s^{-4}G^\eta_jQ^*_jC^{(j),s}Q_jG^\eta_j$.*

Proof. On $\ell^2(\Omega^{(j)},s^d)\oplus\ell^2(\Omega,\eta^d)$ let $\mathsf{M}:=\begin{pmatrix}a_js^{-2}+a(Ls)^{-2}P^{(j)} & X\\ X^* & H_0+a_js^{-2}P_j\end{pmatrix}$, $X:=-a_js^{-2}Q_j$. Its quadratic form is $a_js^{-2}\|\psi-Q_j\phi\|^2+a(Ls)^{-2}\langle\psi,P^{(j)}\psi\rangle+\langle\phi,H_0\phi\rangle$, since $\langle Q_j\phi,Q_j\phi\rangle=\langle\phi,P_j\phi\rangle$. The (ϕ,ϕ) block $H_0+a_js^{-2}P_j=(G^\eta_j)^{-1}$ is $\ge\gamma_0s^{-2}$ by (H0) after the rescaling of lemma 1.4; its Schur complement is $a(Ls)^{-2}P^{(j)}+\Delta^{(j),s}=(C^{(j),s})^{-1}\ge\gamma_Xs^{-2}>0$ by (C), rescaled. So $\mathsf{M}>0$. The other Schur complement: on $\ell^2(\Omega^{(j)})$, $(a_js^{-2}+a(Ls)^{-2}P^{(j)})^{-1}=s^2[a_j^{-1}(1-P^{(j)})+(a_j+aL^{-2})^{-1}P^{(j)}]$, hence

$$a_js^{-2}P_j-a_j^2s^{-4}Q^*_j(a_js^{-2}+a(Ls)^{-2}P^{(j)})^{-1}Q_j=s^{-2}Q^*_jP^{(j)}Q_j\Big(a_j-\frac{a_j^2}{a_j+aL^{-2}}\Big)=a_{j+1}(Ls)^{-2}P_{j+1},$$

by the recursion $a_j-a_j^2/(a_j+aL^{-2})=aa_jL^{-2}/(a_j+aL^{-2})=a_{j+1}L^{-2}$ [1, (2.13)], lemma 1.1 and $s_{j+1}=Ls$; so this complement is $(G^\eta_{j+1})^{-1}$. The (ϕ,ϕ) entry of M^{-1} is both the inverse of this complement and $D^{-1}+D^{-1}X^*(A-XD^{-1}X^*)^{-1}XD^{-1}$ with D the (ϕ,ϕ) block and $A-XD^{-1}X^*=(C^{(j),s})^{-1}$, which is the claim. □

Corollary 1.3 ((1), scalar and vector)**.** *Iterating lemma 1.2 from $j=1$ to $k-1$ gives $G^\eta_k=\sum_{j=0}^{k-1}G^\eta_{(j)}$, and $G^\eta_k=G_k(\Omega,\tilde B)$ since $s_k=1$ [1, (2.22)]. The vector propagator is $G_k(T_\eta,0)\otimes\mathbf{1}_d$ with mass $\mu_0^2t^2$ ([1, p. 608], the Feynman-gauge action [1, (1.11)] being componentwise), so it has the same decomposition, with slices $G^{\mathrm{vec},\eta}_{(j)}$. These are the covariances of the independent slice fields $A'^{(j),\eta}=a_j(L^j\eta)^{-2}G^\eta_jQ^*_jA'_j$ of [3, (1.2)], which is [3, p. 414]: "the field A' has the covariance G_k".*

Lemma 1.4 (rescaling)**.** *Fix $0\le j\le k-1$, $s:=s_j$, and let $(S_sf)(x):=f(x/s)$ map functions on the L^{-j}-lattice (scale-j units) to functions on the η-lattice. Give the rescaled background $\tilde B^{[j]}$ the same bond phases, $e_jL^{-j}\tilde B^{[j]}_{b/s}:=e_k\eta\tilde B_b$, so $U^{[j]}_{b/s}=U_b$ and $\tilde B^{[j]}(y)=s^{1/2}\tilde B(sy)$ (as $e_k/e_j=s^{-1/2}$); rescale $\Omega^{[j]}:=\Omega/s$ and $m^2t_j^2=s^2m^2t^2$. Then $D^\eta S_s=s^{-1}S_sD^{(j)}$, $-\Delta^\eta S_s=s^{-2}S_s(-\Delta^{(j)})$, $Q^\eta_jS_s=TQ_j$, $Q^{\eta*}_jT=S_sQ^*_j$ (T the corresponding map on coarse functions, Q_j Bałaban's average over unit blocks at scale j), and*

$$G^\eta_j(x,x')=s^{2-d}G_j(x/s,x'/s),\qquad G^\eta_{(j)}(x,x')=s^{2-d}K_j(x/s,x'/s),\quad K_j:=a_j^2G_jQ^*_jC^{(j)}Q_jG_j\ (K_0:=C^{(0)}),$$

with $G_j=G_j(\Omega^{[j]},\tilde B^{[j]})$ [1, (2.22)] and $C^{(j)}=C^{(j)}(\Omega^{[j]},\tilde B^{[j]})$ [1, (2.31)]; kernels with respect to η^d on the left and L^{-jd} on the right. $\Omega^{[j]}$ is again a union of large blocks of side M at scale j (the pavings nest), and depths scale as $\rho_{\Omega^{[j]}}(x/s)=s^{-1}\rho_\Omega(x)$.

Proof. Each identity is a one-line computation from $\eta=sL^{-j}$, equal bond phases and $\langle S_sf,S_sh\rangle_{\eta^d}=s^d\langle f,h\rangle_{L^{-jd}}$. Composition: $a_j^2s^{-4}(s^2SG_jS^{-1})(SQ^*_jT^{-1})(s^2TC^{(j)}T^{-1})(TQ_jS^{-1})(s^2SG_jS^{-1})=s^2SK_jS^{-1}$, and the kernel of s^2SKS^{-1} with respect to η^d is $s^{2-d}K(x/s,x'/s)$. □

Lemma 1.5 (regularity transfers to every slice)**.** *Under (E), $\tilde B^{[j]}$ and $(\tilde A+\tilde B)^{[j]}$ satisfy [1, (2.23)] at index j, $|d^{L^{-j}}A^{[j]}|\le ce_j^{\beta-1}$, and the coercivity hypothesis of (H0) at index j.*

Proof. $d^{L^{-j}}\tilde{B}^{[j]}(y) = s^{3/2}(d^{\eta}\tilde{B})(sy)$, so $|d\tilde{B}^{[j]}| \le s^{3/2}c_1 p(t)$. With $e_j = e_k s^{1/2}$, condition [1, (2.23)] reads $s^{3/2}c_1 p(t) \le c e_k^{\beta-1} s^{(\beta-1)/2}$, i.e. $c_1 e_k^{1-\beta} p(t) s^{2-\beta/2} \le c$, which (E) gives since $s \le 1$. For $\tilde{A}+\tilde{B}$ use $2c_1$. □

Lemma 1.6 (unit-scale slice kernels)**.** *Let $1 \le j \le k-1$, $n, n' \in \{0,1\}$, $0 \le \alpha < 1$, $\delta_* := \min\{\delta_0, \theta_0\}$, $\delta_1 := \delta_*/2$. On the scale-j lattice, for y, y' with $\rho_{\Omega^{[j]}} \ge R_0$ at y, y' and at the Hölder partners:*

$$|D_y^n D_{y'}^{n'} K_j(y,y')| \le c_3 e^{-\delta_1|y-y'|_\infty}, \qquad |y_1-y_2|^{-\alpha}\big|U(\tilde{B}^{[j]}(\Gamma_{y_1,y_2}))D_{y_2}D_{y'}^{n'}K_j(y_2,y') - D_{y_1}D_{y'}^{n'}K_j(y_1,y')\big| \le c_3 e^{-\delta_1}$$

the same in y' and for the mixed second difference, with $c_3 := 2a^2c_0^2c_2e^{2\delta_0}S(\delta_1)^2$, $S(\delta) := \sum_{n\in\mathbb{Z}^d} e^{-\delta(|n|_\infty-1)_+}$. For $j=0$ the same holds for $K_0 = C^{(0)}$ with no depth restriction and constants from the unit-lattice coercivity and decay of (C).

Proof. Let δ_w be the unweighted coarse delta at $w \in \mathbb{Z}^d$; $Q_j^*\delta_w = U(\tilde{B}(\Gamma_{w,\cdot}))^T \mathbf{1}_{B(w)}$ has sup norm 1 and support $B(w)$; G_j is real symmetric. So $K_j(y,y') = a_j^2 \sum_{w,w'} (G_jQ_j^*\delta_w)(y)\, C^{(j)}(w,w')\, [(G_jQ_j^*\delta_{w'})(y')]^T$. (P) at index j with $f = Q_j^*\delta_w$ and lemma 1.5 gives $|D^n(G_jQ_j^*\delta_w)(y)| \le c_0 e^{-\delta_0 \operatorname{dist}_\infty(y,B(w))} \le c_0 e^{\delta_0} e^{-\delta_0|y-w|_\infty}$, and [1, (2.24)] the Hölder quotient with $\operatorname{dist}(\{y_1,y_2\},B(w))$ in the exponent; the same at y'. (C) gives $|C^{(j)}(w,w')| \le c_2 e^{-\theta_0|w-w'|_\infty}$. Since $|y-w|+|w-w'|+|w'-y'| \ge |y-y'|$, half of the exponents pays $e^{-\delta_1|y-y'|}$ and the other half the two lattice sums, each $\le S(\delta_1)$; mixed derivatives and quotients act on different outer factors. For $j=0$, $C^{(0)} = (-\Delta_A^{(0)} + m^2\varepsilon^2 + aL^{-2}P(A))^{-1}$ on the unit lattice is the fluctuation covariance at $k=0$, so (C) gives $|C^{(0)}(y,y')| \le c_2 e^{-\theta_0|y-y'|_\infty}$ with no depth restriction; a derivative at unit spacing is a difference of two kernel values (U orthogonal), and a Hölder quotient at unit spacing has $|y_1-y_2| \ge 1$. □

Lemma 1.7 ([3, (2.10)–(2.11)] in scale-k units)**.** *For $0 \le j \le k-1$, $n, n' \in \{0,1\}$, $x, x' \in \Omega$ with $\rho_\Omega \ge R_0$ at x, x' and at the Hölder partners (no restriction for $j=0$, none on $\Omega = T_\eta$):*

$$|D_x^n D_{x'}^{n'} G_{(j)}^{\eta}(x,x')| \le c_3 s_j^{-d+2-n-n'} e^{-\delta_1 s_j^{-1}|x-x'|_\infty}, \quad |x_1-x_2|^{-\alpha}\big|U(\tilde{B}(\Gamma_{x_1,x_2}))(DG_{(j)}^{\eta})(x_2,x') - (DG_{(j)}^{\eta})(x_1,x')\big| \le$$

and the same with the quotient in x' and for the mixed second difference (power $s_j^{-d-2\alpha}$). The constant c_3 is independent of ε, k, j, Ω, $\tilde{B}$ and the torus side.

Proof. Lemma 1.4 then lemma 1.6: $D^{\eta} = s^{-1}D^{(j)}$, $|x_1-x_2|^{-\alpha} = s^{-\alpha}|y_1-y_2|^{-\alpha}$, contours and phases correspond. The depth needed at scale j is R_0 there, i.e. $R_0 s_j \le R_0/5$ in scale-k units, so $\rho_\Omega \ge R_0$ suffices for every j. The constants of (P) and (C) do not depend on the index nor on the torus side. □

Lemma 1.8 (vector slices and [3, (2.12)])**.** (a) *The vector slices $G_{(j)}^{\mathrm{vec},\eta}$ obey lemma 1.7 at every point of T_η, with no depth condition, including one derivative at each end.* (b) *For a contour leg put $G_{(j)}^{\eta}(\Gamma,b) := \sum_{b'\in\Gamma} \eta\, G_{(j)}^{\mathrm{vec},\eta}(b',b)$ with $\Gamma = \Gamma_{x_{j+1},x}^{(j+1)}$ [1, (2.2)], [3, (1.3)]. Then for $n \in \{0,1\}$*

$$|D_b^n G_{(j)}^{\eta}(\Gamma_{x_{j+1},x}^{(j+1)}, b)| \le c_3\, dL\, e^{\delta_1 L}\, s_j^{-d+3-n} e^{-\delta_1 s_j^{-1}|x-b|_\infty}.$$

(c) *With contour legs at both ends the power is s_j^{-d+4} and the constant $c_3(dLe^{\delta_1 L})^2$. These imply* [3, (2.12)], *since* $\operatorname{dist}(B^j(x),b) \le |x-b|_\infty$*; at $L=5$ the constant is $15e^{5\delta_1}c_3$ (resp. its square).*

Proof. (a) is lemma 1.7 at $A=0$, $N=d$, $X=T_\eta$, μ_0 for m. (b) $\Gamma_{x_{j+1},x}^{(j+1)} = \Gamma_{x_{j+1},x_j} \cup \cdots \cup \Gamma_{x_1,x}$; the piece Γ_{x_{i+1},x_i} lies on the $L^i\eta$-lattice and has d straight segments of length $\le (L-1)L^i\eta$ [1, (2.1)], so the total length is $\le d(L-1)\sum_{i\le j} L^i\eta \le dLs_j$; every point of the contour lies in $B^{j+1}(x_{j+1})$, of sup-diameter $\le Ls_j$, so $|z-b|_\infty \ge |x-b|_\infty - Ls_j$ on Γ. Bound each term by (a) and sum: the factor η per bond times the length gives dLs_j, the "additional factor $L^j\eta$". (c) is the same at both ends. Since the slice fields are independent [3, p. 414], a contour leg $A'^{(j),\eta}(\Gamma^{(j+1)})$ contracts only with slice j: the contour index and the line index coincide, which is how [3, (2.14)] books the factor at the line's scale. □

Remark 1.9. (i) [3, (2.12)] prints $\mathrm{dist}(B^j(x), b)$ in the exponent; the bound above has the stronger $|x-b|_\infty$ at the cost of $e^{\delta_1 L}$, and since contour points lie up to Ls_j from x the printed form also needs a constant of this type inside its $O(1)$. (ii) The $j=0$ slice $C^{(0),\eta} = G_1^\eta$ is a unit-lattice inverse with an L-block pin; it needs neither (P) nor a depth condition. (iii) Scalar lines need depth $\geq R_0$ at both evaluation points and at the Hölder partners; vector lines and contour legs live on the torus and need none.

1.3. **Two tools: weighted ℓ^2 bounds and summed pointwise bounds.** Conventions. X is a union of unit blocks with Neumann conditions, or $X = T_\eta$ (for the pointwise bounds, a union of large blocks, as (P) requires). Bond functions are weighted at b_-. ρ is any function which is 1-Lipschitz for $|\cdot|_\infty$, so $|\rho(b_+) - \rho(b_-)| \leq \eta$ on every bond and $|\rho(x) - \rho(x')| \leq 1$ inside a unit block. For an operator Y put $Y^\beta := e^{\beta\rho} Y e^{-\beta\rho}$.

Lemma 1.10 (real-part Combes–Thomas bound)**.** *Let $Y = D^*D + v + aP$ on $\ell^2(X, \eta^d; \mathbb{R}^N)$, with arbitrary bond set and orthogonal transporters, $v \geq 0$ a multiplication operator, $a \geq 0$, $P \geq 0$ with $P(x,x') = 0$ unless x, x' lie in one unit block and $\sup_x \sum_{x'} \eta^d \|P(x,x')\| \leq 1$, and $Y \geq \gamma > 0$. Let $0 \leq \beta \leq 1$ with $c(\beta) := 3\beta^2 \cosh\beta + a(e^\beta - 1) \leq \gamma/2$. Then for every F the solution w of $Y^\beta w = F$ exists, $\|w\| \leq 2\gamma^{-1}\|F\|$ and $\|Dw\|^2 + \langle w, vw\rangle \leq 4\gamma^{-1}\|F\|^2$.*

Proof. Negative β is the case $-\rho$; all operators are real, so F and w may be taken real. For a bond b put $\tau := \beta(\rho(b_+) - \rho(b_-))$, $|\tau| \leq \beta\eta$, and $z := w(b_-) \cdot U_b w(b_+)$. The contribution of b to $\langle w, e^{\beta\rho} D^* D e^{-\beta\rho} w\rangle = \langle De^{\beta\rho} w, De^{-\beta\rho} w\rangle$ is $\eta^{d-2}[|w(b_+)|^2 + |w(b_-)|^2 - 2\cosh(\tau) z]$; at $\tau = 0$ this is $\eta^d |Dw(b)|^2$, and the difference is at most $(\cosh\tau - 1)(|w(b_+)|^2 + |w(b_-)|^2)\eta^{d-2} \leq \frac{1}{2}\beta^2 \cosh(\beta)\eta^d(|w(b_+)|^2 + |w(b_-)|^2)$. Each site lies in at most $2d = 6$ bonds, so the bond part is $\geq \|Dw\|^2 - 3\beta^2\cosh(\beta)\|w\|^2$. v commutes with $e^{\beta\rho}$. $P^\beta - P$ has kernel $(e^{\beta(\rho(x)-\rho(x'))} - 1)P(x,x')$ with $|\rho(x) - \rho(x')| \leq 1$, so $\|P^\beta - P\| \leq e^\beta - 1$ by Schur's test. Hence $\langle w, Y^\beta w\rangle \geq \langle w, Yw\rangle - c(\beta)\|w\|^2 \geq \frac{\gamma}{2}\|w\|^2$, and $\langle w, Y^\beta w\rangle = \langle w, F\rangle \leq \|w\|\|F\|$ gives the first bound; then $\|Dw\|^2 + \langle w, vw\rangle \leq \langle w, Yw\rangle \leq \|w\|\|F\| + \frac{\gamma}{2}\|w\|^2 \leq 4\gamma^{-1}\|F\|^2$. □

Lemma 1.11 (the weighted bounds (W))**.** *Assume (H0) for A on X, let $m^2 t^2 \geq 0$ and $G := G_k(X, A)$. Fix $0 < \beta_* \leq 1$ with $c(\beta_*) \leq \gamma_0/4$ (at $a = a_k$) and put $c' := \sqrt{d}\,\beta_* e^{\beta_*}$. Then for every 1-Lipschitz ρ and $|\beta| \leq \beta_*$: $\|G^\beta\| \leq 2/\gamma_0$, and $\|(DG)^\beta\|$, $\|(GD^*)^\beta\|$, $\|(DGD^*)^\beta\|$ are $\leq K_W := \max\{2\gamma_0^{-1/2} + 2c'\gamma_0^{-1}, (1+c'^2)/\min\{\frac{1}{2}, \gamma_0/4\}\}$. The bounds are uniform in η, k, ε, $m^2 \geq 0$, the shape of X (wrapping allowed) and the boundary condition. With $\mathfrak{D} := (1, D_1, D_2, D_3)$, $\|(\mathfrak{D} G \mathfrak{D}^*)^\beta\| \leq c_W := 2\gamma_0^{-1} + 3K_W$.*

Proof. Lemma 1.10 with $Y = H_X(A)$, $v = m^2t^2$, $P = P_k(A)$, $a = a_k$, $\gamma = \gamma_0$ gives $\|G^\beta\| \leq 2/\gamma_0$ and $\|DG^\beta\| \leq 2\gamma_0^{-1/2}$. Next $(DG)^\beta = D^\beta G^\beta$ with $D^\beta = D + \beta K_\beta$, $(\beta K_\beta \phi)(b) = \eta^{-1}(e^{-\beta(\rho(b_+)-\rho(b_-))} - 1) U_b \phi(b_+)$; the coefficient is at most βe^β in modulus and $\phi \mapsto (\phi(b_+))_b$ has norm $\sqrt{d}$, so $\|\beta K_\beta\| \leq c'$. G is real symmetric, so $(GD^*)^\beta = ((DG)^{-\beta})^*$. For $(DGD^*)^\beta$ let $w := G^\beta (D^*)^\beta g$; the adjoint of $(D^*)^\beta$ is $D^{-\beta}$. By the argument of lemma 1.10 with $H \geq D^*D$ and $H \geq \gamma_0$, $\mathrm{Re}\langle w, H^\beta w\rangle \geq \frac{1}{2}\|Dw\|^2 + \frac{\gamma_0}{4}\|w\|^2$, while $\mathrm{Re}\langle w, H^\beta w\rangle = \mathrm{Re}\langle D^{-\beta} w, g\rangle \leq (\|Dw\| + c'\|w\|)\|g\|$; with $s^2 := \|Dw\|^2 + \|w\|^2$ this gives $\min\{\frac{1}{2}, \gamma_0/4\} s \leq (1+c'^2)^{1/2}\|g\|$ and $\|D^\beta w\| \leq (1+c'^2)^{1/2} s$. Only $|\rho(b_+) - \rho(b_-)| \leq \eta$ and the block range of P_k were used, and (H0) holds on every union of unit blocks and on T_η. □

At $\beta = 0$ and $\rho = \mathrm{dist}(\cdot, \mathrm{supp}\, f')$ this is [4, Cor. 2.3, (2.30)] for DGD^*, with no restriction on the supports.

Lemma 1.12 (slice sums at $L = 5$)**.** *Let $p > 0$, $\delta > 0$, $z \geq 0$, $r := \max(z, \eta)$. Then $\sum_{j=0}^{k-1} s_j^{-p} e^{-\delta z/s_j} \leq c_\Sigma(p,\delta)\, r^{-p} e^{-\delta z/2}$ with $c_\Sigma(p,\delta) := (1-5^{-p})^{-1} + \sup_{1\leq u\leq 5} \sum_{m\geq 0} (u5^m)^p e^{-\delta u 5^m/2}$.*

Proof. Slices with $s_j \geq r$ contribute at most $r^{-p}\sum_m 5^{-mp} e^{-\delta z}$. Slices with $s_j < r$ occur only when $r = z$; then $u_j := z/s_j > 1$ has the form $u5^m$ with $u \in (1, 5]$, $m \geq 0$, each term is $z^{-p} u_j^p e^{-\delta u_j}$, and $e^{-\delta u_j} \leq e^{-\delta u_j/2} e^{-\delta z/2}$ because $u_j \geq z$. □

Kernel classes. For $0<\kappa<3$ put $K^\delta_\kappa(z):=\max(|z|_\infty,\eta)^{\kappa-3}e^{-\delta|z|_\infty}$; $K^\delta_3(z):=(1+\log_+(1/\max(|z|_\infty,\eta)))e^{-\delta|z|_\infty}$; $K^\delta_\kappa(z):=e^{-\delta|z|_\infty}$ for $\kappa>3$.

Lemma 1.13 (the summed pointwise bounds (K)). *Let X be a union of large blocks or T_η, $A\in\{\tilde B,\tilde A+\tilde B\}$ (regular by (E)), $G:=G_k(X,A)$, $0<\alpha<1$, $\delta_K:=\delta_1/2$. For x,w with $\rho_X(x),\rho_X(w)\ge R_0$ (on T_η: all x,w):*
(K0) $\|G(x,w)\|\le c_K K_2^{\delta_K}(x-w)$;
(K1) $\|D_xG(x,w)\|\le c_K K_1^{\delta_K}(x-w)$, *and* $\|(GD^*)(w,x)\|$ *obeys the same bound;*
(K2) *for* $|x-\tilde x|\le 1$ *with* $\rho_X(\tilde x)\ge R_0$: $\|U(A(\Gamma_{\tilde x,x}))D_xG(x,w)-D_{\tilde x}G(\tilde x,w)\|\le c_K(\alpha)|x-\tilde x|^\alpha\big(K^{\delta_K}_{1-\alpha}(x-w)+K^{\delta_K}_{1-\alpha}(\tilde x-w)\big)$;
with $c_K:=c_3\max\{c_\Sigma(1,\delta_1),c_\Sigma(2,\delta_1)\}$ *and* $c_K(\alpha):=2c_3c_\Sigma(2+\alpha,\delta_1)$.

Proof. $G=\sum_j G^\eta_{(j)}$ by corollary 1.3. (K0), (K1): sum lemma 1.7 ($d=3$: powers s^{-1}, s^{-2}) with lemma 1.12 at $p=1,2$. (K2) slice by slice: for $s_j\ge|x-\tilde x|_\infty$ use the Hölder clause of lemma 1.7 with $\operatorname{dist}(\{x,\tilde x\},w)\le\min(|x-w|,|\tilde x-w|)$; for $s_j<|x-\tilde x|_\infty$ bound the two terms separately (U orthogonal) with $|x-\tilde x|^{-\alpha}\le s_j^{-\alpha}$; in both cases the slice quotient is $\le c_3 s_j^{-2-\alpha}(e^{-\delta_1|x-w|/s_j}+e^{-\delta_1|\tilde x-w|/s_j})$, and lemma 1.12 at $p=2+\alpha$ sums it. The column bound: G is real symmetric, so $(GD^*)(w,x)=(D_xG(x,w))^T$. □

The mixed kernel $D_xGD^*_w$ has s^{-3} per slice, which is not summable uniformly in η at $|x-w|\lesssim\eta$. It never enters pointwise below: it enters only through lemma 1.11.

1.4. **S1: the kernel** [3, (1.16)] **as an external field.** [3, p. 414] expands $G_k(\Omega,\tilde A+\tilde B)$ by [1, (3.44)–(3.45)] and treats the last term,

$$K_{n,n'}:=[G_k(\Omega,\tilde B)V_k(\tilde A,\tilde B)]^n\,G_k(\Omega,\tilde A+\tilde B)\,[V_k(\tilde A,\tilde B)G_k(\Omega,\tilde B)]^{n'},\tag{2}$$

as an external field: "for n,n' sufficiently large, a kernel of the operator (1.16) is a sufficiently regular function of both variables … uniformly bounded by $O(1)(e(L^k\varepsilon)^{1-\alpha})^{n+n'}$". The consumed form is the second alternative of [3, (2.5)]: $\|hK_{n,n'}h'\|_{1,\alpha}\le O((e(L^k\varepsilon)p(L^k\varepsilon))^{n+n'})e^{-\delta_0\operatorname{dist}(\operatorname{supp}h,\operatorname{supp}h')}$ for the localisation functions h,h' of the two end vertices. Here h,h' are the smooth localisers of [3, (1.33), p. 420] ($h=1$ on $\square(v)$, $h=0$ outside a neighbourhood, $|dh|,|ddh|\le c_h$), not the sharp cubes of the vertex localisations. At every use $m:=n+n'\ge\bar n+1\ge14$, since [1, (3.45)] stops at $(n,n')=(\bar n+1,0)$ and $\bar n>12$ [3, p. 421]; the proof needs only $m\ge6$ (or $m\ge4$ with $\alpha_0<\frac12$).

Write $G:=G_k(\Omega,\tilde B)$, $G':=G_k(\Omega,\tilde A+\tilde B)$, $V:=G^{-1}-G'^{-1}=V_k(\tilde A,\tilde B)$ (the resolvent identity), so $K_{n,n'}=L_0VL_1V\cdots VL_m$ with $L_n=G'$ and $L_i=G$ for $i\ne n$.

Lemma 1.14 (the vertex operator). *Put $D:=D_{\tilde B}$, $\phi_\mu(x):=\eta^{-1}(1-U(\tilde A_{x,\mu})^{-1})$, $\psi_\mu:=\phi^*_\mu\phi_\mu$, $\Pi:=P_k(\tilde A+\tilde B)-P_k(\tilde B)$, $\mathfrak D:=(1,D_1,D_2,D_3)$, $\operatorname{supp}\tilde A$ the set of endpoints of bonds with $\tilde A_b\ne0$, and $\mathcal U:=\{y:\operatorname{dist}_\infty(y,\operatorname{supp}\tilde A)\le1\}$.*
(i) $V=-[\sum_\mu(D^*_\mu\phi_\mu+\phi^*_\mu D_\mu+\psi_\mu)+a_k\Pi]=-\mathfrak D^*\mathfrak E\mathfrak D$, *with* $\mathfrak E_{00}=\sum_\mu\psi_\mu+a_k\Pi$, $\mathfrak E_{0\mu}=\phi^*_\mu$, $\mathfrak E_{\mu0}=\phi_\mu$, 0 *otherwise.*
(ii) *(Right-pushed form.)* $V=-(R_0+\sum_\mu R_{1,\mu}D_\mu)=:-\mathfrak r\mathfrak D$, *where* $(R_{1,\mu}f)(x):=\phi_\mu(x)^*f(x)-\phi_\mu(x-\eta e_\mu)U(\tilde B_{x-\eta e_\mu,\mu})^{-1}f(x-\eta e_\mu)$ *and* $R_0:=\sum_\mu(\psi_\mu-\mathsf m_\mu)+a_k\Pi$, $\mathsf m_\mu$ *the multiplication by* $(d^\eta_\mu\phi_\mu)(x-\eta e_\mu)$.
(iii) *(Left-pushed form.)* $V=V^*=-\mathfrak D^*\mathfrak r^*$.
(iv) *Let $\sigma\le1$. Then $|\phi_\mu|,|d_\mu\phi_\mu|\le\sigma$, $|\psi_\mu|\le\sigma^2$, $|\Pi(y,y')|\le2d\sigma$ and $\sum_{y'}\eta^d|\Pi(y,y')|\le2d\sigma$; hence $\|\mathfrak r\|,\|\mathfrak E\|\le c_V\sigma$ on $\ell^p(\eta^d)$, $p\in\{1,2,\infty\}$, $c_V:=4d+2da+1$. The kernels have sup-range ≤1, and $V=\mathbf1_{\mathcal U}V\mathbf1_{\mathcal U}$, $\mathfrak r=\mathbf1_{\mathcal U}\mathfrak r\mathbf1_{\mathcal U}$, $\mathfrak E=\mathbf1_{\mathcal U}\mathfrak E\mathbf1_{\mathcal U}$.*
On $X=\Omega$ with Neumann conditions, (i)–(iv) hold if $\operatorname{dist}_\infty(\operatorname{supp}\tilde A,\Omega^c)\ge2$.

Proof. $U(\tilde A+\tilde B)=U(\tilde A)U(\tilde B)$, so $D_{\tilde A+\tilde B,\mu}f=U(\tilde A)(D_\mu f+\phi_\mu f)$, and since U is pointwise orthogonal, $-\Delta_{\tilde A+\tilde B}=\sum_\mu(D_\mu+\phi_\mu)^*(D_\mu+\phi_\mu)$ on the same bond set; the masses cancel and the pins differ by Π: this is (i). (ii): $D^*_\mu(\phi_\mu f)(x)=-\phi_\mu(x-\eta e_\mu)U^{-1}_{x-\eta e_\mu,\mu}(D_\mu f)(x-\eta e_\mu)-$

$(d_\mu\phi_\mu)(x-\eta e_\mu)f(x)$ (the computation of [4, (2.32)], redone in lemma 1.22). (iii): both inverses are real symmetric. (iv): $|e^{q\theta}-e^{q\theta'}|\le|\theta-\theta'|$ gives $|\phi|\le e_k|\tilde A|$, $|d_\mu\phi_\mu|\le e_k|d\tilde A|$; for y,y' in one block $B(z)$, $\Pi(y,y')=U(\tilde B\cdots)^T[U(\tilde A(\Gamma_{z,y}))^TU(\tilde A(\Gamma_{z,y'}))-1]U(\tilde B\cdots)$ with contours of length $\le d$, so $|\Pi|\le 2de_k\|\tilde A\|$; the Laplacian part lives on bonds with $\tilde A_b\neq 0$ plus one η-shift, Π on unit blocks whose contours meet $\operatorname{supp}\tilde A$, of sup-diameter $1-\eta$. On Ω every such bond and block lies in Ω once the distance is ≥ 2. □

Lemma 1.15 (convolution and smearing; $d=3$, sup metric)**.** (a) *For $\kappa_1,\kappa_2\in(0,3)$ and $0<\delta'<\delta\le 1$: $\sum_w \eta^d K^\delta_{\kappa_1}(z-w)K^\delta_{\kappa_2}(w)\le c(\kappa_1,\kappa_2)(1+(\delta-\delta')^{-3})K^{\delta'}_{\kappa_1+\kappa_2}(z)$.* (b) *$K^\delta_\kappa\le c(1+(\delta-\delta'')^{-1})K^{\delta''}_{\kappa'}$ for $\kappa\ge\kappa'$, $\delta''<\delta$.* (c) *If R is a multiplication, a one-bond shift, or a kernel supported in single unit blocks, with $|R(z,z')|\le r_R$ and column sums $\le r_R$, then $(|R|K^\delta_\kappa(\cdot-y))(z)\le c(\kappa)e^{2\delta}r_RK^\delta_\kappa(z-y)$.*

Proof. (a) $e^{-\delta|z-w|-\delta|w|}\le e^{-\delta'|z|}e^{-(\delta-\delta')(|z-w|+|w|)}$. Put $r:=\max(|z|_\infty,\eta)$ and split into $|w|\le r/2$, $|z-w|\le r/2$ and the rest. In the two inner regions one factor is $\le cr^{\kappa_i-3}$ and the other is summed with $\sum_{|w|\le R}\eta^d\max(|w|,\eta)^{\kappa-3}\le c(\kappa)R^\kappa$ $(R\ge\eta)$; the outer region is $\le c\int_{r/2}^\infty u^{\kappa_1+\kappa_2-4}e^{-(\delta-\delta')u}du$, which gives the stated classes (with the conventions at $\kappa_1+\kappa_2\ge 3$). (b) For $|z|\le 1$ the larger exponent gives the smaller kernel; for $|z|>1$ the power is absorbed by $e^{-(\delta-\delta'')|z|}$. (c) For $|z-y|_\infty\ge 2$ every z' with $R(z,z')\neq 0$ has $\max(|z'-y|,\eta)\ge\frac12\max(|z-y|,\eta)$; for $|z-y|_\infty<2$ the block sum of $K_\kappa(\cdot-y)$ is $\le c(\kappa)$ and $K_\kappa(z-y)\ge ce^{-2\delta}$. □

Lemma 1.16 (end factors)**.** *Let $L_0,L_1,L_2\in\{G,G'\}$, $a\in\{0,1\}$, $E^{(2)}_a:=D^aL_0VL_1$, $E^{(3)}_a:=D^aL_0VL_1VL_2$, every V in right-pushed form; $\delta':=\delta_K/2$, $0\le\beta\le\delta_K/4$. Assume (K) at every pair of points used (automatic on T_η).*

(i) *For $y\in\mathcal U$: $\|E^{(2)}_a(x,y)\|\le c_E\sigma K^{\delta'}_2(x-y)$ and $\|E^{(3)}_a(x,y)\|\le c'_E\sigma^2K^{\delta'}_3(x-y)$.*

(ii) *For $a=1$ and $|x-\tilde x|\le 1$ the $\tilde B$-transported difference in x of $E^{(2)}_1$ is $\le c_E(\alpha)\sigma|x-\tilde x|^\alpha(K^{\delta'}_{2-\alpha}(x-y)+K^{\delta'}_{2-\alpha}(\tilde x-y))$; the same for $E^{(3)}_1$ with σ^2 and $K_{3-\alpha}$.*

(iii) *Put $J_\kappa:=(\sum_y\eta^dK^{\delta'}_\kappa(y)^2e^{2\beta|y|_\infty})^{1/2}$. Then $\|e^{\beta|\cdot-x|}\mathbf 1_{\mathcal U}E^{(2)}_a(x,\cdot)\|_2\le c_EJ_2\sigma$; the transported difference of (ii) has weighted ℓ^2 norm $\le 2e^\beta c_E(\alpha)J_{2-\alpha}\sigma|x-\tilde x|^\alpha$; for $E^{(3)}$ the same with J_3, $J_{3-\alpha}$, σ^2. $J_\kappa<\infty$ uniformly in η and in the torus side iff $\kappa>\frac32$: so $J_{2-\alpha}<\infty$ iff $\alpha<\frac12$, while J_3, $J_{3-\alpha}$ are finite for every $\alpha<1$.*

Proof. (i) $E^{(2)}_a=-(D^aL_0)R_0L_1-\sum_\mu(D^aL_0)R_{1,\mu}(D_\mu L_1)$: every line carries at most one derivative (L_0 the external one, L_1 the vertex one). For a line of type G' write $D_{\tilde B,\mu}=U(\tilde A)^{-1}D_{\tilde A+\tilde B,\mu}-\phi_\mu$, so (K) for G' gives $\|D_{\tilde B}G'(x,w)\|\le c_K(K_1+\sigma K_2)\le cK^{\delta'}_1$ by lemma 1.15(b). Lemma 1.15(c) moves R ($r_R\le c_V\sigma$) onto the kernels and (a), (b) give $K_{2-a}*K_1\to K_{3-a}\le cK_2$, $K_{2-a}*K_2\to K_2$; for $E^{(3)}_a=(D^aL_0)\mathfrak r(\mathfrak DL_1)\mathfrak r(\mathfrak DL_2)$ one gets $K_{2-a}*K_1*K_1\to K_3$. (ii) The difference acts on the first factor only. For $L_0=G$ use (K2), then $K_{1-\alpha}*K_1\to K_{2-\alpha}$ (resp. $K_{1-\alpha}*K_1*K_1\to K_{3-\alpha}$). For $L_0=G'$ write $U(\tilde B(\Gamma))=U((\tilde A+\tilde B)(\Gamma))U(\tilde A(\Gamma))^{-1}$: the quotient is at most the (K2) quotient of $D_{\tilde A+\tilde B}G'$, plus $3\sigma c_KK_1$ from $|U(\tilde A(\Gamma))^{-1}U(\tilde A_x)^{-1}-U(\tilde A_{\tilde x})^{-1}|\le e_k(\|\tilde A\||\Gamma|+\eta|\tilde A_x-\tilde A_{\tilde x}|)\le 3\sigma|x-\tilde x|$, plus the ϕ-terms, telescoped along Γ as in lemma 1.18, which give $c\sigma|x-\tilde x|(K_1(x-\cdot)+K_1(\tilde x-\cdot))$; since $|x-\tilde x|\le|x-\tilde x|^\alpha$ and $K_1\le cK_{1-\alpha}$, (K2) holds for $D_{\tilde B}G'$ with constant $c_K(\alpha)+10\sigma c_K$. (iii) Square and sum, with $|y-x|\le|y-\tilde x|+1$ for the $\tilde x$-term. Near 0, $\sum_y\eta^d\max(|y|,\eta)^{2\kappa-6}$ is bounded uniformly in η iff $2\kappa-6>-3$; the tail converges because $\delta'-\beta\ge\delta_K/4$. □

Proposition 1.17 (the ℓ^2 sandwich)**.** *Assume (E); then (H0), (K) for G and G', and $\sigma\le 1$ hold. Let $0<\beta\le\beta_U:=\min\{\beta_*,\delta_K/4\}$, and either $m\ge 4$ and $0<\alpha<\frac12$, or $m\ge 6$ and $0<\alpha<1$. Put $\mathfrak N_{\alpha,\beta}(Y):=\max_{a,b}\sup_{x,x'}e^{\beta|x-x'|_\infty}\max\{\|Y_{ab}(x,x')\|,aH_x,bH_{x'},abH_{xx'}\}$, H the α-Hölder quotients of Y_{ab} with $\tilde B$-transport.* (a) *On $X=T_\eta$: $\mathfrak N_{\alpha,\beta}(K_{n,n'})\le C_U(\alpha)(C_V\sigma)^m$, with $C_V:=e^{2\beta}c_V\max\{c'_W,1\}$, $c'_W:=(1+\sigma)^2c_W$.* (b) *On $X=\Omega$ with Neumann conditions and $\operatorname{dist}_\infty(\operatorname{supp}\tilde A,\Omega^c)\ge R_0+2$ the same bound holds with the supremum restricted to x,x' such that*

x, x', their bond partners, their Hölder partners and the points of the contours $\Gamma_{\tilde{x},x}$, $\Gamma_{\tilde{x}',x'}$ have depth $\geq R_0$. $C_U(\alpha)$ depends only on $d, a, M, L, \gamma_0, K_W, c_K, c_K(\alpha), \alpha, \beta$; the bound is uniform in η (hence k, ε), in the torus side, in $0 \leq m^2t^2 \leq c_m$ and in the backgrounds within (E).

Proof. $K_{n,n'}$ has m vertices and $m+1$ lines, and each vertex is first order on each side (lemma 1.14(i)), so D^aKD^{b*} carries $m+2$ derivatives on $m+1$ lines and some line carries two; that line is put in the middle factor, which enters only through lemma 1.11. No pointwise bound on DGD^* is used.

Two-line ends ($m \geq 4$). $Y := D^aKD^{b*} = E_L\mathcal{N}E_R$ with $E_L := E_a^{(2)} = D^aL_0VL_1$, $\mathcal{N} := VL_2V\cdots VL_{m-2}V$ (vertices $2,\dots,m-1$) and $E_R := L_{m-1}VL_mD^{b*} = (D^bL_mVL_{m-1})^*$ (by $V = V^*$ and the symmetry of the L_i). Put the first vertex of $\mathcal{N}$ in right-pushed, the last in left-pushed and the others in symmetric form: $\mathcal{N} = (-1)^{m-2}\mathfrak{r}(\mathfrak{D}L_2\mathfrak{D}^*)\mathfrak{E}\cdots\mathfrak{E}(\mathfrak{D}L_{m-2}\mathfrak{D}^*)\mathfrak{r}^*$. By lemma 1.14(iv) $\mathcal{N} = \mathbf{1}_{\mathcal{U}}\mathcal{N}\mathbf{1}_{\mathcal{U}}$, so $E_L(x,\cdot)$, $E_R(\cdot,x')$ are needed only on $\mathcal{U}$. For $L = G'$ each $\mathfrak{D}_{\tilde{B}} = U(\tilde{A})^{-1}\mathfrak{D}_{\tilde{A}+\tilde{B}} - \Phi$ costs $(1+\sigma)$ per side; the kernels of $\mathfrak{r}, \mathfrak{E}$ have sup-range $\leq 1+\eta \leq 2$, so their conjugates by $e^{\pm\beta\rho}$ are bounded by $e^{2\beta}c_V\sigma$; lemma 1.11 bounds the blocks $\mathfrak{D}L\mathfrak{D}^*$. Hence $\|e^{\mp\beta\rho}\mathcal{N}e^{\pm\beta\rho}\| \leq (e^{2\beta}c_V\sigma)^{m-2}c_W'^{m-3}$. Fix x, x', $\rho := |\cdot - x|_\infty$, and write $Y(x,x') = \langle e^{\beta\rho}\mathbf{1}_{\mathcal{U}}E_L(x,\cdot)^*, (e^{-\beta\rho}\mathcal{N}e^{\beta\rho})e^{-\beta\rho}\mathbf{1}_{\mathcal{U}}E_R(\cdot,x')\rangle$; Cauchy–Schwarz, $e^{-\beta|y-x|} \leq e^{-\beta|x-x'|}e^{\beta|y-x'|}$ and lemma 1.16(iii) (for E_R by transposition) give $e^{\beta|x-x'|}\|Y(x,x')\| \leq (c_EJ_2\sigma)^2(e^{2\beta}c_V\sigma)^{m-2}c_W'^{m-3}$. Y is linear in $E_L(x,\cdot)$ and in $E_R(\cdot,x')$ separately, so H_x is the same pairing with $E_L(x,\cdot)$ replaced by $U(\tilde{B}(\Gamma_{\tilde{x},x}))E_L(x,\cdot) - E_L(\tilde{x},\cdot)$, similarly $H_{x'}$, $H_{xx'}$; lemma 1.16(iii) bounds these with $2e^{\beta}c_E(\alpha)J_{2-\alpha}$, finite iff $\alpha < \frac{1}{2}$.

Three-line ends ($m \geq 6$, any $\alpha < 1$). Repeat with $E_L := E_a^{(3)}$, $\mathcal{N} := VL_3V\cdots VL_{m-3}V$ (vertices $3,\dots,m-2$, at least two since $m \geq 6$) and $E_R := L_{m-2}VL_{m-1}VL_mD^{b*}$; now J_3, $J_{3-\alpha}$ are finite. Since $c_V \geq 1$ both cases give $\leq C_U(\alpha)(C_V\sigma)^m$: (a).

(b) (H0) and lemma 1.11 hold on Ω with no depth condition, and lemma 1.14 holds since dist ≥ 2. Every pointwise use of (K) above is at a point of $\mathcal{U}$ or a bond partner of one (depth $\geq R_0 + 2 - 1 - \eta \geq R_0$), or at $x, x', \tilde{x}, \tilde{x}'$, their bond partners and the contour points of the telescoping in lemma 1.16(ii), which are deep by the restriction. So (K) applies and the argument is verbatim. □

Lemma 1.18 (localisation of two-variable bounds)**.** *Let Y be a kernel, $\beta \geq 0$, $M_Y \geq 0$, and $\mathcal{P}, \mathcal{P}'$ sets of lattice points such that $e^{\beta|y-y'|_\infty}\max\{\|Y_{ab}(y,y')\|, aH_y, bH_{y'}, abH_{yy'}\} \leq M_Y$ for all a, b, all directions, all $y, \tilde{y} \in \mathcal{P}$, $y', \tilde{y}' \in \mathcal{P}'$ with partners $0 < |y-\tilde{y}|, |y'-\tilde{y}'| \leq 1$. Let $0 \leq h, h' \leq 1$ with $|d_\mu h|, |d_\nu d_\mu h| \leq c_h$ (and the same for h'), and assume that every point at sup-distance $\leq 1+\eta$ from $\operatorname{supp} h$ lies in $\mathcal{P}$, and likewise for h', $\mathcal{P}'$. Then $\|hYh'\|_{1,\alpha} \leq e^{3\beta}(1+3\sqrt{3}c_h)^2M_Ye^{-\beta\operatorname{dist}_\infty(\operatorname{supp}h,\operatorname{supp}h')}$.*

Proof. Leibniz: $D_\mu(hf) = F_1D_\mu f + F_2f$ with $F_1 := h(\cdot + \eta e_\mu)$, $F_2 := d_\mu h$; for the second variable, $(ZD_\mu^*)(x,x') = ((D_\mu Z^\dagger)(x',x))^*$. Along a shortest monotone contour $\Gamma_{\tilde{x},x}$, of ℓ^1-length $\leq \sqrt{3}|x-\tilde{x}|$, every φ with $|d\varphi| \leq c$ obeys $|\varphi(x) - \varphi(\tilde{x})| \leq \sqrt{3}c|x-\tilde{x}|$, and $U(\Gamma)W(x) - W(\tilde{x})$ is a sum over the bonds of Γ of transported $\eta D_\nu W$ terms, so $\|U(\Gamma)W(x) - W(\tilde{x})\| \leq |x-\tilde{x}|_1\max_{z\in\Gamma,\nu}\|D_\nu W(z)\|$. In one variable, $U(hY)_1(x) - (hY)_1(\tilde{x})$ splits into $F_1(x)[UY_1(x) - Y_1(\tilde{x})]$, $(F_1(x) - F_1(\tilde{x}))Y_1(\tilde{x})$, $F_2(x)[UY_0(x) - Y_0(\tilde{x})]$ (telescoped, $D_\nu Y_0$ being a component of Y_1) and $(F_2(x) - F_2(\tilde{x}))Y_0(\tilde{x})$; with $|x-\tilde{x}| \leq |x-\tilde{x}|^\alpha$ the quotient of $(hY)_1$ is $\leq (1+3\sqrt{3}c_h)$ times the largest component or quotient of Y at points of $\{x,\tilde{x}\}\cup\Gamma$, and a term vanishes unless x or $\tilde{x}$ is within η of $\operatorname{supp}h$, so every evaluated point is within $1+\eta$ of $\operatorname{supp}h$ and lies in $\mathcal{P}$. The expansions in x and x' commute (different variables, transports from opposite sides); the factors are $(1+c_h)^2$ per component, $(1+3\sqrt{3}c_h)(1+c_h)$ per one-variable quotient and $(1+3\sqrt{3}c_h)^2$ for the mixed one. Finally $|y-y'|_\infty \geq \operatorname{dist}_\infty(\operatorname{supp}h, \operatorname{supp}h') - 2(1+\eta)$. □

Corollary 1.19 ([3, (2.5)] for [3, (1.16)] at $L = 5$)**.** *Assume* (E) *and* $\operatorname{dist}_\infty(\operatorname{supp}\tilde{A}, \Omega^c) \geq R_0+2$*; let $m = n+n' \geq 6$, $\alpha_0 \in (0,1)$, $0 < \beta \leq \beta_U$, and let h, h' be as in lemma 1.18 with supports at depth $\geq R_0+3$ in Ω (on T_η: no condition). Then*

$$\|hK_{n,n'}h'\|_{1,\alpha_0} \leq e^{3\beta}(1+6c_h)^2C_U(\alpha_0)\,(2c_1C_Ve_kp(t))^{n+n'}\,e^{-\beta\operatorname{dist}_\infty(\operatorname{supp}h,\operatorname{supp}h')}.$$

Proof. Apply lemma 1.18 with $\mathcal{P} = \mathcal{P}' := \{z : \rho_\Omega(z) \geq R_0 + 1 + \eta\}$: for points of $\mathcal{P}$ the bond partners and the Hölder partners (and their bond partners, and the contour points between them) have depth $\geq R_0$, so proposition 1.17(b) gives the hypothesis with $M_Y = C_U(C_V\sigma)^m$. A point within $1+\eta$ of a set of depth $\geq R_0+3$ has depth $\geq R_0+2-\eta \geq R_0+1+\eta$ since $\eta \leq \frac{1}{5}$. Finally $\sigma \leq 2c_1 e_k p(t)$ and $3\sqrt{3} \leq 6$. □

Remark 1.20. (i) This is the second alternative of [3, (2.5)] with $\delta_0 := \beta \leq \beta_U = \min\{\beta_*, \delta_K/4\}$; shrinking the rate is free. Since [3, p. 426] extracts half of each line's rate for $\exp[-\frac{1}{2}\delta_1 \operatorname{dist}(\square(v), \square(v'))]$, the rate δ_0 of Proposition 1 is $\min\{\delta_1/2, \beta\} > 0$. (ii) The corollary holds for every $\alpha_0 < 1$, in particular $\alpha_0 = \frac{1}{2}$; for $m \in \{4,5\}$ it holds for $\alpha_0 < \frac{1}{2}$. It implies the [3, p. 414] form read as $(e(L^k\varepsilon))^{1-\alpha'}$ raised to $n+n'$: $(2c_1C_V e_k p(t))^m \leq (e_k^{1-\alpha'})^m$ as soon as $2c_1C_V p(t) \leq e_k^{-\alpha'}$, i.e. for $t \leq t_0(\alpha')$, since p is polylogarithmic and $e_k = et^{1/2}$. (The other parsing, $e\,(L^k\varepsilon)^{1-\alpha}$, is not consumed: [3, (1.33)] counts powers of $e(L^k\varepsilon)$.) (iii) "n, n' sufficiently large" is necessary: at $m=0$, with $g := \eta^{-d}\mathbf{1}_x$ and $\Lambda := 4d\eta^{-2} + m^2t^2 + a \geq \|H\|$, $(D_\mu G' D_\mu^*)(x,x) = \langle D_\mu^* g, G' D_\mu^* g\rangle \geq \Lambda^{-1}\|D_\mu^* g\|^2 = 2\eta^{-2-d}/\Lambda \sim \eta^{-3}/6$, not bounded uniformly in η. (iv) The depths used are $\operatorname{supp}\tilde{A}$ at $\geq R_0+2$ and vertex supports at $\geq R_0+3$, both in the sup metric and uniform in $k \geq 1$.

1.5. **S2: the two-domain propagator $\delta G_k(\Omega, \Omega_2, \tilde{B})$ as an external field.** [3, p. 414] replaces Ω by $\Omega_2 \subset \Omega$ "satisfying the same conditions as Ω", and treats $\delta G_k(\Omega, \Omega_2, \tilde{B}) = G_k(\Omega, \tilde{B}) - G_k(\Omega_2, \tilde{B})$ as an external field; [3, (2.5)], first alternative, asserts $\|h\,\delta G_k(\Omega, \Omega_2, \tilde{B})\,h'\|_{1,\alpha} \leq O(e^{-\delta_0 \operatorname{dist}(\Omega_2, \partial\Omega)}) e^{-\delta_0 \operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')}$, and [3, p. 426] uses it in [3, (2.13)] as an $O(1)$ bound with the two-point decay; the small depth factor is what the domain changes of [3, pp. 421–422] use. δG is defined on $\Omega_2 \times \Omega_2$, $G_k(\Omega_2, \tilde{B})$ being extended by 0.

The reading of the depth factor. The printed factor $e^{-\delta_0 \operatorname{dist}(\Omega_2, \partial\Omega)}$, read as a distance between the sets Ω_2 and $\partial\Omega$, cannot be what is meant: δG is of order one at points near $\partial\Omega_2$ whatever the distance to $\partial\Omega$. (For $\Omega = T_\eta$ there is no $\partial\Omega$ at all; in one dimension at $\eta = \frac{1}{5}$, $A=0$, $m=0$, $a_k = 1$, with Ω_2 a union of 20 blocks, $\delta G(x,x)$ takes the values -0.43, -0.28, -0.05, -0.0008 at depth 0, 0.2, 1, 3 in Ω_2, against $G(x,x) = 0.52$; the same values arise with Ω a Neumann segment at distance 90 from Ω_2.) The factor holds in the form of [1, Prop. 2.1, (2.26)], where the extra decay is in the depth of the arguments in the *smaller* set: here, the depth of the vertex supports in Ω_2. This is the form proved below, and it is the form the uses need.

Setting. $X_{\text{in}} \subset X_{\text{out}}$ are unions of large blocks of T_η, X_{out} may be T_η, $X_{\text{in}} \neq T_\eta$; A is real and regular by (E), $0 \leq m^2t^2 \leq c_m$; $H_{\text{in/out}} := H_{X_{\text{in/out}}}(A)$, $G_{\text{in/out}} := H_{\text{in/out}}^{-1}$, $\delta := G_{\text{out}} - G_{\text{in}}$, $\rho := \operatorname{dist}_\infty(\cdot, X_{\text{in}}^c)$; then $\rho_{X_{\text{out}}} \geq \rho$.

Lemma 1.21 (collars)**.** *Fix $f \in C^\infty(\mathbb{R})$ with $0 \leq f \leq 1$, $f = 0$ on $(-\infty, 0]$, $f = 1$ on $[1, \infty)$, and $\varphi \in C_c^\infty([-\frac{1}{4}, \frac{1}{4}]^3)$, $\varphi \geq 0$, $\int \varphi = 1$. For a unit block b with lowest corner y_b put $\hat{b} := y_b + [0,1)^3$ and $\hat{X}_{\text{in}} := \bigcup_{b \subset X_{\text{in}}} \hat{b}$; let ρ^c be the continuum sup-distance to the complement of $\hat{X}_{\text{in}}$, $\tilde{\rho} := \rho^c * \varphi$, and $\chi_s := f(\tilde{\rho} - s)$ at lattice points; $c_1' := \|f'\|_\infty$, $c_2' := \|f''\|_\infty + \|f'\|_\infty \max_\nu \|\partial_\nu \varphi\|_{L^1}$. Then:* (i) *$\tilde{\rho}$ is 1-Lipschitz for $|\cdot|_\infty$ and $\rho - \frac{1}{4} - \eta \leq \tilde{\rho} \leq \rho + \frac{1}{4}$ at lattice points;* (ii) *$\chi_s = 0$ on $\{\rho \leq s - \frac{1}{4}\}$ and $\chi_s = 1$ on $\{\rho \geq s + \frac{5}{4} + \eta\}$;* (iii) *$|d_\mu^\eta \chi_s| \leq c_1'$ and $|d_\mu^\eta d_\nu^\eta \chi_s| \leq c_2'$, uniformly in η, s, X_{in};* (iv) *if $\chi_s(u) \neq \chi_s(u')$ and $|u - u'|_\infty \leq l$, then $\tilde{\rho}(u), \tilde{\rho}(u') \in (s - l, s + 1 + l)$.*

Proof. (i) A lattice point outside X_{in} lies in a cube $\hat{b}'$ with $b' \not\subset X_{\text{in}}$, so $\rho^c \leq \rho$; the infimum defining $\rho^c(z)$ is attained on the closure of such a cube, whose lattice points are η-dense for $|\cdot|_\infty$, so $\rho \leq \rho^c + \eta$; the mollifier preserves the Lipschitz constant and moves values by at most $\frac{1}{4}$. (ii) follows from (i). (iii) Lattice differences are averages of derivatives along segments; $|\partial_\mu \tilde{\rho}| \leq 1$ and $\partial_\mu \partial_\nu \tilde{\rho} = (\partial_\mu \rho^c) * \partial_\nu \varphi$. (iv) If $\tilde{\rho}(u) \leq s - l$ then $\tilde{\rho}(u') \leq s$ and both values of χ_s are 0; symmetrically both are 1 if $\tilde{\rho}(u) \geq s + 1 + l$. □

Lemma 1.22 (commutators)**.** *Let $s \geq 1$, $\chi := \chi_s$, $\phi_\mu := d_\mu^\eta \chi$.* (i) *$H_{\text{out}}(\chi g) = H_{\text{in}}(\chi g)$ for every g on X_{in} (extended by 0).* (ii) *$C_\chi := H_{\text{out}}\chi - \chi H_{\text{in}} = [H_{\text{in}}, \chi] = -\mathfrak{r}_\chi \mathfrak{D} = \mathfrak{D}^* \mathfrak{r}_\chi^*$, where $\mathfrak{r}_0 := \sum_\mu (d_\mu \phi_\mu)(\cdot - \eta e_\mu) - a_k[P_k(A), \chi]$ and $(\mathfrak{r}_\mu g)(x) := \phi_\mu(x) g(x) + \phi_\mu(x - \eta e_\mu) U_{x-\eta e_\mu, \mu}(A)^{-1} g(x - \eta e_\mu)$.* (iii) *$\sum_{i=0}^3 \sup_z \sum_w \eta^d \|\mathfrak{r}_i(w,z)\| \leq c_{\mathfrak{r}} := 6c_1' + 3c_2' + ac_1'$.* (iv) *If $\mathfrak{r}_i(w,z) \neq 0$ then $|w - z|_\infty \leq 1$ and $w, z \in S_s := \{s - \frac{5}{4} - \eta < \rho < s + \frac{9}{4} + 2\eta\}$.*

Proof. (i) If $\chi(y)\neq 0$ then $\rho(y)>s-\frac14\geq\frac34>\eta$, so every bond at y and the unit block of y lie in $X_{\rm in}$; a bond of $X_{\rm out}$ which is not a bond of $X_{\rm in}$ has both ends at depth $\leq\eta$, outside $\operatorname{supp}\chi$; P_k acts inside unit blocks. (ii) $D_\mu(\chi f)=\chi(\cdot+\eta e_\mu)D_\mu f+\phi_\mu f$ and $D^*_\mu(\chi(\cdot+\eta e_\mu)g)=\chi D^*_\mu g-\phi_\mu g$, so $[D^*_\mu D_\mu,\chi]=D^*_\mu(\phi_\mu\,\cdot)-\phi_\mu D_\mu$, and $D^*_\mu(\phi_\mu f)(x)=\eta^{-1}(\phi_\mu(x-\eta e_\mu)U^{-1}f(x-\eta e_\mu)-\phi_\mu(x)f(x))=-\phi_\mu(x-\eta e_\mu)U^{-1}_{x-\eta e_\mu,\mu}(D_\mu f)(x-\eta e_\mu)-(d_\mu\phi_\mu)(x-\eta e_\mu)f(x)$ ([4, (2.32)]); collecting terms gives $-\mathfrak{r}_\chi\mathfrak{D}$, and $C^*_\chi=-C_\chi$ since $H_{\rm in}$ is self-adjoint and χ real. (iii) The two parts of each $\mathfrak{r}_\mu$ give $2c_1'$, so $6c_1'$ over three directions; the multiplication part of $\mathfrak{r}_0$ gives $3c_2'$; $[P_k,\chi](w,z)=P_k(w,z)(\chi(z)-\chi(w))$ with $|\chi(z)-\chi(w)|\leq c_1'|\tilde\rho(z)-\tilde\rho(w)|\leq c_1'$ and $\sum_w\eta^d\|P_k(w,z)\|\leq 1$. (iv) Lemma 1.21(iv) with $l=2\eta$ (bond and multiplication parts) or $l=1-\eta$ (block part) gives $\tilde\rho\in(s-1+\eta,s+2-\eta)$; convert to ρ by lemma 1.21(i). □

Choices. $\chi:=\chi_{R_0+3}$, $\psi:=\chi_{R_0+13/2}$, $\mathcal{P}_D:=\{z:\rho(z)\geq R_0+10+2\eta\}\supset\{\rho\geq R_0+10.4\}$. With $\eta\leq\frac15$, lemmas 1.21 and 1.22 give: $S_\chi\subset\{R_0+1.55<\rho<R_0+5.65\}$, $S_\psi\subset\{R_0+5.05<\rho<R_0+9.15\}$; $\psi=0$ on $\{\rho\leq R_0+6.25\}\supset S_\chi$; $\chi=1$ on $\{\rho\geq R_0+4.45\}$ and $\psi=1$ on $\{\rho\geq R_0+7.95\}$, both containing $\mathcal{P}_D$ and every bond partner of a point of $\mathcal{P}_D$; for $z\in\mathcal{P}_D$, $\operatorname{dist}_\infty(z,S_\chi)>\max\{\rho(z)-R_0-7,4.75\}$ and $\operatorname{dist}_\infty(z,S_\psi)>\max\{\rho(z)-R_0-10,1.25\}$; every point of S_χ, S_ψ and its bond partners has depth $>R_0+1.5$ in $X_{\rm in}$, hence in $X_{\rm out}$.

Proposition 1.23 (the two-domain theorem)**.** *Let $0<\alpha<1$, $0<\beta\leq\beta_D:=\min\{\beta_*,\delta_K/4\}$, $a,b\in\{0,1\}$, $Y:=D^a\delta D^{b*}$. For $x,x'\in\mathcal{P}_D$ with Hölder partners in $\mathcal{P}_D$,*

$$e^{\beta|x-x'|_\infty}\max\{\|Y(x,x')\|,aH_x,bH_{x'},abH_{xx'}\}\leq C_D(\alpha)\exp\Big(-\frac{\delta_K}{2}\big[(\rho(x)-R_0-7)+(\rho(x')-R_0-10)\big]\Big),$$

with $C_D(\alpha):=c_W c_{\rm end}(\alpha)^2$, $c_{\rm end}(\alpha):=c_{\mathfrak r}e^{2\delta_K}\max\{c_K,2c_K(\alpha)\}c_3'^{1/2}$, $c_3':=\sup_\eta\sum_z\eta^d e^{-(\delta_K/2)|z|_\infty}\leq(\frac15+4/\delta_K+1)^3$. The constants depend only on $d,a,M,L,\gamma_0,K_W,c_K,c_K(\alpha),f,\varphi$, not on η,k,ε, the torus side, $X_{\rm in}$, $X_{\rm out}$, A, $m^2t^2\in[0,c_m]$ or R_0.

Proof. $G_{\rm out}C_\chi G_{\rm in}=\chi G_{\rm in}-G_{\rm out}\chi$, so at y,y' with $\chi(y)=\chi(y')=1$, $\delta(y,y')=-(G_{\rm out}C_\chi G_{\rm in})(y,y')$, the inner variable ranging over S_χ only (lemma 1.22(iv)), where $\psi=0$. Likewise $G_{\rm in}C_\psi G_{\rm in}=\psi G_{\rm in}-G_{\rm in}\psi$ gives $G_{\rm in}(z,y')=-(G_{\rm in}C_\psi G_{\rm in})(z,y')$ when $\psi(z)=0$, $\psi(y')=1$. Inserting and using lemma 1.22(ii), at all points of $\mathcal{P}_D$ and their bond partners (the only points at which D^a, D^{b*} evaluate δ),

$$\delta=-G_{\rm out}\mathfrak{r}_\chi(\mathfrak{D}G_{\rm in}\mathfrak{D}^*)\mathfrak{r}_\psi^*G_{\rm in},\qquad Y(x,x')=-\langle E_L(x,\cdot),\mathfrak{M}E_R(\cdot,x')\rangle,$$

$E_L:=D^aG_{\rm out}\mathfrak{r}_\chi$, $\mathfrak{M}:=\mathfrak{D}G_{\rm in}\mathfrak{D}^*$, $E_R:=\mathfrak{r}_\psi^*G_{\rm in}D^{b*}$. *Left end.* $E_L(x,z)=\sum_w\eta^dD^a_xG_{\rm out}(x,w)\mathfrak{r}_\chi(w,z)$ with $w,z\in S_\chi$, $|w-z|_\infty\leq1$; (K) applies on $X_{\rm out}$ at (x,w) (both deep, $|x-w|_\infty\geq4.75\geq1$), so $K^{\delta_K}_{2-a}(x-w)\leq e^{-\delta_K|x-w|_\infty}$ and $\|E_L(x,z)\|\leq c_{\mathfrak r}c_Ke^{\delta_K}e^{-\delta_K|x-z|_\infty}\mathbf{1}_{S_\chi}(z)$; with $2(\delta_K-\beta)\geq\frac32\delta_K$, $\|e^{\beta|\cdot-x|}E_L(x,\cdot)\|_2\leq c_{\mathfrak r}c_Ke^{\delta_K}e^{-(\delta_K/2)\operatorname{dist}(x,S_\chi)}c_3'^{1/2}\leq c_{\rm end}e^{-(\delta_K/2)(\rho(x)-R_0-7)}$; for $a=1$ and a partner $\tilde x\in\mathcal{P}_D$, (K2) with $|\tilde x-w|\geq1$ gives the transported difference with the factor $2c_K(\alpha)e^{\delta_K}|x-\tilde x|^\alpha$. *Right end.* $(G_{\rm in}D^{b*})(v,x')=(D^b_{x'}G_{\rm in}(x',v))^T$ with $v\in S_\psi$, and (K) on $X_{\rm in}$ at (x',v) gives $\|e^{\beta|\cdot-x'|}E_R(\cdot,x')\|_2\leq c_{\rm end}e^{-(\delta_K/2)(\rho(x')-R_0-10)}$, and the same times $|x'-\tilde x'|^\alpha$ for the transported difference. *Middle.* Lemma 1.11 on $X_{\rm in}$ with weight $|\cdot-x|_\infty$ gives $\|e^{-\beta|\cdot-x|}\mathfrak{M}e^{\beta|\cdot-x|}\|\leq c_W$; then $e^{-\beta|u-x|}\leq e^{-\beta|x-x'|}e^{\beta|u-x'|}$ and Cauchy–Schwarz give the bound. The quotients follow by replacing one or both ends by their transported differences, Y being linear in each end. No kernel is evaluated at points less than 1 apart, and the doubly differentiated line enters only through its operator norm. □

Corollary 1.24 (the consumed form at $L=5$)**.** *Let $0\leq h,h'\leq1$ with $|dh|,|ddh|\leq c_h$ (the same for h'), the smooth localisers of* [3, (1.33)]*; put $d_h:=\operatorname{dist}_\infty(\operatorname{supp}h,X_{\rm in}^c)$, $d_{h'}$ likewise, and assume $d_h,d_{h'}\geq R_0+12$. Then, at any mutual distance of the supports (overlap included),*

$$\|h\,\delta\,h'\|_{1,\alpha}\leq C_{\rm loc}\exp\Big(-\frac{\delta_K}{2}(d_h+d_{h'}-2R_0-19.4)\Big)e^{-\beta\operatorname{dist}_\infty(\operatorname{supp}h,\operatorname{supp}h')},\qquad C_{\rm loc}:=e^{3\beta}(1+6c_h)^2C_D(\alpha).$$

Since $d_h+d_{h'}-2R_0-19.4\geq4.6>0$, this is the first alternative of [3, (2.5)] *in the $O(1)$ form, and it carries the depth factor $\exp(-\frac{\delta_K}{2}(d_h+d_{h'}))$, the form of* [1, (2.26)] *with the complement of the smaller set.*

Proof. Lemma 1.18 with $\mathcal{P} := \{\rho \geq d_h - 1 - \eta\}$, $\mathcal{P}' := \{\rho \geq d_{h'} - 1 - \eta\}$, both contained in $\mathcal{P}_D$ since $R_0 + 12 - 1.2 = R_0 + 10.8 \geq R_0 + 10.4$; on $\mathcal{P} \times \mathcal{P}'$ proposition 1.23 gives the hypothesis with $M_Y := C_D \exp(-\frac{\delta_K}{2}[(d_h - 1.2 - R_0 - 7) + (d_{h'} - 1.2 - R_0 - 10)])$. □

Remark 1.25. The three instances in [1, 2] are $X_{\text{out}} = \Omega$, $X_{\text{in}} = \Omega_2$; $X_{\text{out}} = \Omega$, $X_{\text{in}} =$ the cube; and $X_{\text{out}} = T_\eta$, $X_{\text{in}} =$ the cube, $A = 0$. In the first the vertex supports have depth $\geq 5r(t) - 9$ in Ω_2 (the nesting of [2, pp. 579–580]: the summations in the vertices are restricted to $\Lambda_7^{(k)}$ while the propagator domain is $B^k(\Lambda_2^{(k)})$), which exceeds $R_0 + 12$ under $R \geq 3R_0 + 15$; at $L = 5$ the depth $R_0 + 12$ is uniform in $k \geq 1$, there being no separate $k = 1$ case ($\eta_1 = \frac{1}{5}$). Near $\partial\Omega_2$ nothing is proved, and nothing follows from (P) alone (Neumann boundary regularity at the edges of a union of large blocks).

1.6. **S4: the summation by parts** [3, (2.8)–(2.9)] **at a vertex of** [3, (2.4)]**.** The graph [3, (2.4)] is two vertices [3, (1.8)] with $n + n' = 1$, joined by one scalar line carrying the derivative at both ends, with two external scalar and two external vector legs; its degree is 0. [3, p. 425] replaces such graphs by graphs of positive degree "by integration by parts" at one of the two vertices, [3, (2.8)], applied "for all graphs $G_1, G_2^{(2)}, G_3^{(3)}, \dots$" of a fixed ordering, so that G' becomes a sum of graphs $\{G'^*\}$ whose components $G_1, \dots, G_m = G'^*$ along the ordering have positive degrees. Two things are to be proved: the identity, as an identity of multilinear forms in the legs, and the degree statement, which is the hypothesis of S5.

Lemma 1.26 (the identity)**.** *Let $X \subset T_\eta$, $\phi_1, \phi_2 \colon X \to \mathbb{R}^N$, and h_μ real functions with $\operatorname{supp} h_\mu \subset \{x : \operatorname{dist}_\infty(x, X^c) \geq 2\eta\}$; $\tilde{B}$ arbitrary and real. Put $D_\mu^{\leftarrow}\phi(x) := U(\tilde{B}_{x-\eta e_\mu,\mu})^{-1}(D_\mu\phi)(x - \eta e_\mu) = \eta^{-1}(\phi(x) - U^{-1}\phi(x - \eta e_\mu))$ and $d_\mu^* h(x) := \eta^{-1}(h(x - \eta e_\mu) - h(x))$. Then*

$$(-e)\sum_\mu \sum_x \eta^d[(D_\mu\phi_1)(x)\cdot q\phi_2(x)]h_\mu(x) = (-e)\sum_x \eta^d[\phi_1(x)\cdot q\phi_2(x)]\Big(\sum_\mu d_\mu^* h_\mu\Big)(x) + e\sum_x \sum_\mu \eta^d[\phi_1(x)\cdot q(D_\mu^{\leftarrow}\phi_2)$$

With $h_\mu = gA'_\mu$, $d_\mu^*(gA'_\mu)(x) = g(x)(d_\mu^* A'_\mu)(x) + (d_\mu^* g)(x)A'_\mu(x - \eta e_\mu)$ gives the first two terms of [3, (2.8)] exactly; the third printed term carries the derivative at the bond $(x - \eta e_\mu, x)$ transported to x, and its sign is as above.

Proof. For $F(x) := q\phi_2(x)h_\mu(x)$, supported where both bonds at x in direction μ lie in X, $\sum_x \eta^d(D_\mu\phi_1)(x) \cdot F(x) = \sum_x \eta^d \phi_1(x) \cdot (D_\mu^* F)(x)$ with $(D_\mu^* F)(x) = \eta^{-1}(U^{-1}_{x-\eta e_\mu,\mu}F(x - \eta e_\mu) - F(x))$, no boundary term arising because $\operatorname{supp} F$ is at depth $\geq 2\eta$. U commutes with q, so $D_\mu^*(q\phi_2 h_\mu) = qD_\mu^*(\phi_2 h_\mu)$, and $D_\mu^*(\phi_2 h_\mu)(x) = -(D_\mu^{\leftarrow}\phi_2)(x)h_\mu(x - \eta e_\mu) + \phi_2(x)(d_\mu^* h_\mu)(x)$. Multiply by $-e$ and sum over μ. □

Remark 1.27. With $D^\eta_{\tilde{B},\mu}$ the forward covariant difference [1, (1.7)] evaluated at x, the printed third term of [3, (2.8)] has the opposite sign; it is exactly right with $D^{\eta *}_{\tilde{B},\mu}$, the symbol of the first line of [3, (2.8)], in place of $D^\eta_{\tilde{B},\mu}$, so the print most likely dropped an asterisk. This is harmless for Proposition 1, which takes absolute values of every factor [3, p. 426], the graphical rule [3, (2.9)] recording only where the derivative sits.

Which vertex, and why the identity holds graph by graph. The vertex of [3, (2.4)] is [3, (1.8)] with $n + n' = 1$: $-e(L^k\varepsilon)\sum_b \eta^d[(D^\eta_{\tilde{B}}\phi')(b) \cdot q\phi'(b_-)](g_k A'_b)^n(\tilde{A}_b)^{n'}$; after the localisation of [3, p. 420], g carries the partition-of-unity factor of the vector leg, so $h_\mu = gA'_\mu$ ($n = 1$) or $h_\mu = \tilde{A}_\mu$ ($n' = 1$). The supports are at depth $\geq 2\eta$ in Ω: $\operatorname{supp} g \subset \Omega_1$ at distance $\geq r(t) - 2$ from Ω^c, and $\operatorname{dist}(\operatorname{supp}\tilde{A}, \partial\Omega) > 2r(t)$. $E(G')$ is a sum over pairings; for a fixed pairing the value is obtained by substituting into each vertex form a function for each leg (a slice $G^\eta_{(j)}(\cdot, y)$ on Ω for a scalar leg, a vector slice for an A'-leg, or an external field). The vertex form is multilinear in its legs and lemma 1.26 is an identity of multilinear forms, so it holds after substitution, graph by graph, for every fixed ordering and index set $J(\tilde{l})$, even though $\phi'(x) \cdot q\phi'(x) = 0$ as a polynomial (the two legs are distinct in each graph).

Lemma 1.28 (degrees after [3, (2.8)])**.** *Use the degree count* [3, (2.1)–(2.2)]*:* $D_G(v) = (\text{number of factors } \eta \text{ in } v) \frac{2-d}{2}(\text{legs of } v \text{ on internal lines of } G) - (\text{differentiations in } v \text{ acting on internal lines of } G) + (\text{contour credit})$*,* $D(G) = \sum_v D_G(v) - d$*.* (a) $D([3, (2.4)]) = 0$*.* (b) *Let* v *be a vertex of a* [3, (2.4)]*-subgraph* $G_0 = \{l_0\}$ *and apply lemma 1.26 at* v*. In each of the three terms the graph has the same vertices and lines; only the place of the derivative at* v *changes, from* l_0 *to the* A'*-leg (or* $\tilde{A}$*-leg), to* g*, or to the* ϕ_2*-leg. For every connected* $H \ni l_0$*:* $D_{\text{new}}(H) = D_{\text{old}}(H) + 1$ *if the new place is not an internal line of* H*, and* $= D_{\text{old}}(H)$ *otherwise; in particular* $D_{\text{new}}(G_0) = 1$*.* (c) *For a connected* H *containing* v *but not* l_0*, and containing the line of the new place:* $D_{\text{new}}(H) = D_{\text{old}}(H) - 1$*.* (d) *Fix an ordering* $\tilde{l}$ *and call the component* $G_i^{(i)}$ *created by* $l(i)$ fresh *if* $l(i)$ *joins two vertices not met by* $l(1), \dots, l(i-1)$ *and these with* $l(i)$ *form a* [3, (2.4)] *graph. Apply lemma 1.26 at one vertex of each fresh component. If every component* $G_i^{(\alpha)}$ *of* G' *along* $\tilde{l}$ *has* $D > 0$ *except the fresh ones, then in each resulting* G'^* *every component* $G_i^{(\alpha)}$ *along* $\tilde{l}$ *has* $D > 0$*.*

Proof. (a) Each vertex of [3, (2.4)] has η-count d, one internal scalar leg and one derivative on it, so $D(v) = d + \frac{2-d}{2} - 1 = \frac{d}{2}$ and $D = 2 \cdot \frac{d}{2} - d = 0$. (b) The three terms keep the η-count and the legs; only "differentiations acting on internal lines" changes: for $H \ni l_0$ the old derivative was on an internal line, the new one is on an internal line exactly when H contains the A'-line or the ϕ_2-line at v (the g-term never is). (c) For $H \not\ni l_0$ the old derivative was not on an internal line of H and the new one may be. (d) A vertex [3, (1.8)] has exactly one derivative leg; a fresh component uses the derivative legs of both its vertices, so each vertex lies in at most one fresh component, the fresh components are vertex-disjoint and the transformations commute. If v lies in the fresh component created by $l_0 = l(i_0)$, then v is met by no earlier line, so every component $G_i^{(\alpha)}$ with $i \geq i_0$ containing v contains l_0, and none with $i < i_0$ contains v. By (b) every component containing a transformed vertex has $D_{\text{new}} \geq D_{\text{old}} > 0$ or is fresh with $D_{\text{new}} = 1$; the others are unchanged. □

Remark 1.29. [3, p. 425] states "the degrees of these graphs are $\geq D(G_0)$" for a graph G_0 containing the [3, (2.4)] subgraph, where it holds by (b), and claims positivity only for the subgraphs $G_1, \dots, G_m = G'^*$ along the ordering, which is (d). By (c), a connected subgraph through the transformed vertex avoiding l_0 can lose one unit, so G'^* need not have every connected subgraph of positive degree; the induction of S5 uses only (d). The transformation depends on the ordering, so the sum over orderings of [3, (2.7)] is taken first, as printed. The new vertices are admissible for the bounds of S3 and S5: a derivative on A' is a derivative at one end of a vector slice (lemma 1.8(a), one extra s^{-1}); $|d^*g| \leq c$ since g_k is bounded with its derivatives up to second order [3, p. 412]; a derivative on ϕ_2 is a derivative at one end of a scalar slice (lemma 1.7) or of the external field (bounded by $\|h\Phi\|_{1,\alpha}$); the shifts by ηe_μ keep every point within η of the vertex localisation; the η-count is unchanged, and no line carries two derivatives at one end, the ϕ_2- and A'-legs of [3, (1.8)] carrying none.

1.7. **S5: the induction** [3, (2.16)]**.** After [3, (2.13)–(2.14)] the quantity to bound is, for a graph G with lines l, slice indices $j \in J(\tilde{l})$ and cubes $\Delta(v)$,

$$\tilde{E}(G(j), \{\Delta(v)\}) = \sum_{\Delta(v)} \prod_v s_{j(v)}^{d+e_v} \prod_l s_{j_l}^{d_l} \prod_l \exp\big[-\delta\, s_{j_l}^{-1} \operatorname{dist}_\infty(\Delta(v_l), \Delta(v_l'))\big], \tag{3}$$

with $\delta := \delta_1/2$, $s_j = L^j\eta$, $j(v) := \min_{l \ni v} j_l$, $e_v \geq 0$ the explicit η-power of a vertex [3, (1.8)–(1.11)] $(n+n'-1, n+n'-2, \dots$ from [3, p. 413]; 0 for the other vertices), the bound $\eta^{e_v} \leq s_{j(v)}^{e_v}$ being the "proper power" of [3, (2.14)], and $d_l := -(d-2) - (\text{derivatives on } l) + (\text{contour legs of } l)$, each line having two legs. Put $D(H) := \sum_{l \in H} d_l + \sum_{v \in V(H)} e_v + d(|V(H)| - 1)$ for a connected subgraph H.

Lemma 1.30 (dictionary)**.** $D(H)$ *is Bałaban's degree* [3, (2.2)]*,* $\sum_{v \in H} D_H(v) - d$*, with the contour credit going to legs of lines internal to* H *at the line's own scale (lemma 1.8(b)).* $D(H)$ *is an integer, so* $D(H) > 0$ *implies* $D(H) \geq 1$*.*

Proof. Sum [3, (2.1)] over $v \in H$: each internal line contributes two legs, $-(d-2)$ in total, its derivatives and its contour legs; each vertex contributes d (the η^d of its sum) and e_v. □

Theorem 1.31 ([3, (2.16)] for all i, and [3, (2.15)]). *Fix an ordering $\tilde{l} = (l(1), \dots, l(m))$ of the m lines of a connected graph G; for $j \in J(\tilde{l})$, $j_{l(1)} \le \cdots \le j_{l(m)} \le k-1$. Let $G_i := \{l(1), \dots, l(i)\}$ with endpoints, $G_i^{(\alpha)}$ its components, $W_i := \sum_\alpha D(G_i^{(\alpha)})$, and assume $D(G_i^{(\alpha)}) > 0$ for all i, α, so $W_i \ge 1$. At stage i (Bałaban's G/G_i) every component of G_i is shrunk to one vertex u carrying only its cube factor $s^d_{j(u)}$, every vertex not yet met keeps $s^{d+e_v}_{j(v)}$, $j(u)$ is the minimum index of the remaining lines at u, $\Delta(u)$ has side $s_{j(u)}$, and the remaining lines keep $s^{d_l}_{j_l}$ with rates $\delta_i > 0$; call the resulting expression $\tilde{E}_i(G/G_i)$. Then for every fixed $j_{l(i+1)} \le \cdots \le j_{l(m)}$,*

$$\sum_{j_{l(1)} \le \cdots \le j_{l(i)} \le j_{l(i+1)}} \sum_{\Delta \text{'s of } G} \tilde{E} \;\le\; C_i \, s^{W_i}_{j_{l(i+1)}} \sum_{\Delta \text{'s of } G/G_i} \tilde{E}_i(G/G_i), \tag{4}$$

which is [3, (2.16)]: the single factor $(L^{j_{l(i+1)}}\eta)^{\sum_\alpha D(G_i^{(\alpha)})}$ and a constant depending on the number of lines (hence on $\bar{n}$), on δ_1 and on L. The constants are $C_0 = 1$, $C_{i+1} := C_i e^{\delta_i' n_i} S_d(\delta_i')(1 - L^{-1})^{-1}$, $\delta_{i+1} := \delta_i' := \delta_i/(n_i+1)$, n_i the number of other remaining lines at the far endpoint of $l(i+1)$, $S_d(\delta) := \sum_{n \in \mathbb{Z}^d} e^{-\delta(|n|_\infty - 1)_+}$; at $L = 5$, $(1 - L^{-1})^{-1} = \frac{5}{4}$. At $i = m$: $\sum_{j \in J(\tilde{l})} \sum_\Delta \tilde{E} \le C_m 5^d$, which is [3, (2.15)].

Proof. Induction on i; at $i = 0$ the statement is the definition. Assume $(4)_i$, let $l := l(i+1)$, $j' := j_l$, and u, u' its endpoints in G/G_i. (1) l is the lowest remaining line, so $j(u) = j(u') = j'$ and $\Delta(u)$, $\Delta(u')$ have side $s_{j'}$. (2) *Case $u \ne u'$.* Split $\exp[-\delta_i s_{j'}^{-1} \operatorname{dist}(\Delta(u), \Delta(u'))]$ into $n_i + 1$ equal factors of rate $\delta' := \delta_i/(n_i+1)$. For another line l' at u' with far end u'' and $j_{l'} \ge j'$: $s_{j'}^{-1} \ge s_{j_{l'}}^{-1}$ and $\operatorname{dist}(\Delta(u), \Delta(u'')) \le \operatorname{dist}(\Delta(u), \Delta(u')) + \operatorname{diam}_\infty \Delta(u') + \operatorname{dist}(\Delta(u'), \Delta(u''))$ with $\operatorname{diam}_\infty \Delta(u') \le s_{j'} \le s_{j_{l'}}$, so one share times l''s factor is $\le e^{\delta'} \exp[-\delta' s_{j_{l'}}^{-1} \operatorname{dist}(\Delta(u), \Delta(u''))]$: the line l' now ends at u with rate δ'. The last share sums $\Delta(u')$ over the paving of side $s_{j'}$: $\sum_{\Delta(u')} \exp[-\delta' s_{j'}^{-1} \operatorname{dist}(\Delta(u), \Delta(u'))] \le S_d(\delta')$, scale-free since the cubes are labelled by $s_{j'}\mathbb{Z}^d$ and $\operatorname{dist} \ge s_{j'}(|n|_\infty - 1)_+$. The factor of u' stays ($s^d_{j'}$ for a shrunk component, $s^{d+e_{u'}}_{j'}$ for a vertex not yet met). (3) *Shrinking.* Merge u, u' into $\hat{u}$ with $j(\hat{u})$ the minimum index of the remaining lines at $\hat{u}$, $\ge j_{l(i+2)} \ge j'$ (if no line remains, $\hat{u}$ is the last vertex). Fix $\Delta(\hat{u})$ of side $s_{j(\hat{u})}$ in the nested paving and sum $\Delta(u) \subset \Delta(\hat{u})$: every remaining line at $\hat{u}$ has $\exp[-\cdots \operatorname{dist}(\Delta(u), \cdot)] \le \exp[-\cdots \operatorname{dist}(\Delta(\hat{u}), \cdot)]$, independent of $\Delta(u)$; there are $(s_{j(\hat{u})}/s_{j'})^d$ cubes $\Delta(u)$ in $\Delta(\hat{u})$, which with u's factor $s^d_{j'}$ (times $s^{e_u}_{j'}$ if u was not yet met) gives $\hat{u}$'s cube factor $s^d_{j(\hat{u})}$. What is left at scale j' is $s^{d_l + d + E}_{j'}$, E the sum of e_v over the endpoints of l not met before. (4) *Case $u = u'$* (l closes a loop or is a tadpole). There is no $\Delta(u')$ sum; what is left at scale j' is $s^{d_l}_{j'}$ if u was met before and $s^{d_l + e_u}_{j'}$ for a tadpole at a vertex not met before; shrinking is as in (3). (5) *Degrees.* With the stage-i factor $s^{W_i}_{j'}$, the power of $s_{j'}$ is W_{i+1}: for $u \ne u'$, $D(\text{new}) = D(\text{comp } u) + D(\text{comp } u') + d_l + d$ with $D(\{v\}) := e_v$ for a vertex not yet met (the vertex count adds $|V| + |V'| - 1$); for $u = u'$, $D(\text{new}) = D(\text{old}) + d_l$; all other components are unchanged. (6) *The scale sum.* The only dependence on j' is now $s^{W_{i+1}}_{j'}$ with $W_{i+1} \ge 1$ (the $j(\cdot)$ of the vertices involve only $l(i+2), \dots$, independent variables $\ge j'$), so $\sum_{j' \le j_{l(i+2)}} s^{W_{i+1}}_{j'} \le (1 - L^{-1})^{-1} s^{W_{i+1}}_{j_{l(i+2)}}$; if no line remains, the sum over $j' \le k-1$ is $\le (1 - L^{-1})^{-1}$. This is $(4)_{i+1}$. (7) *The end.* At $i = m$ one vertex remains with its cube factor; its cube sum, over cubes of side s meeting a region of sup-diameter ≤ 4 at weight s^d, is $\le 5^d$. The rates obey $\delta_m \ge \delta / \prod_i (n_i + 1) > 0$, the n_i being bounded by the number of legs, a function of $\bar{n}$. □

Remark 1.32. (i) The induction uses only $W_{i+1} \ge 1$ at each stage, which is weaker than the positivity of each component that lemma 1.28(d) supplies. (ii) Booking each component's degree at its own next scale, as a vertex weight D at $j(u) \ge j_{l(i+1)}$, gives a larger right side (since $s_{j_{l(i+1)}} \le s_{j(u)}$ and $D > 0$); the printed common factor $s^{W_i}_{j_{l(i+1)}}$ is the sharper bookkeeping, and [3, (2.16)] is correct as stated. (iii) The theorem is stated for general line dimensions d_l and vertex powers e_v, which covers the lines of [3, Prop. 2.2] as far as they are of the slice type; the

generalised graphs themselves are the next step (part II of this supplement). (iv) The hypothesis "every component along the ordering has positive degree" is exactly what S4 gives for the graphs G'^*; it is not the hypothesis that every connected subgraph has nonnegative degree, which the G'^* need not satisfy (remark 1.29): e.g. two [3, (1.8)] vertices v, v' with $n = 1$ joined by a scalar line l' carrying both derivative legs, with v's A'-leg contracted by a vector line l_A to a third [3, (1.8)] vertex, all other legs external, have $D(\{l'\}) = 0$, $D(\{l_A\}) = 2$, $D(\{l', l_A\}) = 2$; in the ordering (l_A, l') the line l' is not fresh, is not transformed, and $G'^* = G$ keeps the zero-degree connected subgraph $\{l'\}$, while the components along this ordering have degrees 2, 2 and the theorem gives $O(1)$.

2. The sketched steps of Proposition 1, II: generalised graphs, power counting, Ward–Takahashi identities and the estimate (3.1)

This part contains the next four sketched steps of the proof of [3, Prop. 1, (1.33)], in the order in which the printed proof invokes them, and the degree facts about cut lines on which the regrouping of [3, p. 432], proved in the main paper, rests:

step	[3] page	printed object
S6	428	[3, Prop. 2.2]: the induction [3, (2.16)] for "generalized expressions and graphs"
S7	428–429	the estimate [3, (2.17)] and [3, Cor. 2.3]: which graphs have $D(G) \leq 0$
S8	430–431	the Ward–Takahashi identities [3, (2.23)–(2.28)] "for free boundary conditions ... by taking a
S9	432	the estimate [3, (3.1)] for a line between separated localisations
—	432	degrees of a graph after a line is cut (section 2.5)

The order is S6 (section 2.1), S7 (section 2.2), S8 (section 2.3), S9 (section 2.4) and the cut-line degrees (section 2.5). S6 and S9 are implications from the slice bounds of section 1.2 (and, for the subgraphs [3, (2.4)], the summation by parts of section 1.6); S7, S8 and the cut-line degrees are self-contained.

Conventions for this part.
Scales. As in section 1.1: $d = 3$ unless a statement says otherwise (the lemmas of section 2.1 and section 2.3 hold for general d with the same proofs), $L = 5$, $\eta = \eta_k = L^{-k}$, $k \geq 1$, and the slice scales are $s_j := L^j \eta \in (0, \frac{1}{L}]$, $0 \leq j \leq k-1$. The running charge is e_k, and $U(a) := \exp(e_k \eta q a)$ with q real antisymmetric, $\|q\| \leq 1$; all U are orthogonal and commute.
Vertices and lines. The vertices are those of [3, (1.6)–(1.15)]: the quartic vertex [3, (1.6)]; the mass vertex [3, (1.7)], with a finite counterterm (δm^2_{fin} or $\delta m^2_{K,k}$) or with a counterterm of δm^2_k carrying its own graph G_0; the expansion vertices [3, (1.8)–(1.11)] with n legs A' and n' legs $\tilde{A}$ (the $\tilde{A}$ are external fields inside the vertex, not legs), the R-vertices [3, (1.9), (1.11)] having $n' = \bar{n} + 1$; and the block vertices [3, (1.13)–(1.15)], each counted as a vertex of its own with one scalar leg ϕ' and n vector legs ([3, p. 414]: "we will treat (1.13)–(1.15) as the vertices"; [3, p. 423 (v)]). A graph is connected and has at least one line [3, p. 415]. We write V, I for its numbers of vertices and lines, $\ell := I - V + 1 \geq 0$ for its number of independent loops, E for its number of external legs among the ϕ' and A' legs, E_s, E_v for the scalar and vector parts of E, Δ_{ext} for the number of differentiations acting on external legs, and l_v for the number of ϕ', A' legs of the vertex v (so $\sum_v (l_v - 2) = 2I + E - 2V$).
Degrees. The degree is that of [3, (2.1)–(2.3)]: a scalar or vector leg lying on a line of the graph has dimension $-\frac{d-2}{2}$; a vector leg of [3, (1.14)] or [3, (1.15)] lying on a line has $-\frac{d-2}{2} + 1$ ([3, p. 422] prints "(1.4) and (1.15)" for (1.14) and (1.15); [3, (2.1)] itself has (1.14) and (1.15)); a differentiation acting on a line of the graph has -1; each factor η has $+1$; external fields have 0. For a connected subgraph H (a set of lines with their end vertices) $D(H) := \sum_{v \in H} D_H(v) - d$, where $D_H(v)$ counts only the legs of v which lie on lines of H. The degree $D(v)$ of a vertex is $D_H(v)$ with all its legs on lines. A mass vertex with a counterterm of δm^2_k is counted by the convention [3, (2.3)], through its attached graph.

2.1. **S6: generalised graphs, [3, Prop. 2.2].** After the proof of [3, Prop. 2.1] the text says ([3, p. 428]): "It is easily seen from the proof that the theorem can be generalized to a much wider

class of graphs and expressions. We can have vertices with an arbitrary number of legs and arbitrary power of η. We can have lines with the same exponential factors but with arbitrary dimensions instead of $-d+2$. The only thing which matters is that propagators have representations corresponding to (2.6) with the estimates corresponding to (2.10)–(2.12), so that we have the inequality (2.13) with the proper generalization of (2.14)." The induction step of [3, Prop. 2.1] itself is "We can prove (2.16) for $i+1$ repeating the same reasoning as in the proof of (2.15)." Both are written out here once, for the generalised class. What is used later is: the sum over indices and localisations of the bound function is $O(1)$, with $O(1)$ depending on the number of lines, the decay rate and the minimal positive degree only, and, with the top index held fixed, the bound carries a factor $(L^{j_*}\eta)^{D(G)}$ without a sign condition on $D(G)$ ([3, (3.4)–(3.5)]).

Definition 2.1 (generalised bound function)**.** The data are: a finite connected multigraph G (self-loops allowed) with vertex set $\mathcal{V}$, line set $\mathcal{L}$, $m := |\mathcal{L}| \geq 1$, and $N_G :=$ the largest number of line-ends at a vertex of any graph obtained from G by contracting lines ($N_G \leq 2m$); reals ω_l ($l \in \mathcal{L}$, the line dimension) and p_v ($v \in \mathcal{V}$, the vertex power); localisation boxes $\square(v)$, each a union of unit cubes of the integer grid with $|\square(v)| \leq 2^d$ ([3, p. 420]: unit cubes, or cubes of side 2 for the partition-of-unity localisations; a side-2 support which is not grid-aligned is replaced by the at most 3^d unit cubes meeting it, which changes constants only); an index assignment $j = (j_l)$, $j_l \in \{0, \dots, k-1\}$, with $s_l := L^{j_l}\eta$, $j(v) := \min\{j_l : l \text{ has an end at } v\}$ and $s(v) := L^{j(v)}\eta$; a localisation $\Delta = (\Delta(v))_v$, $\Delta(v)$ a cube of the grid $s(v)\mathbb{Z}^d$ contained in $\square(v)$ (possible since $s(v) = L^{j(v)-k}$ divides 1); and a rate $\delta_0 > 0$ ([3, (2.14)] has $\delta_0 = \delta_1/2$). The bound function is

$$\tilde{E}(G(j), \Delta) := \prod_{v \in \mathcal{V}} s(v)^{d+p_v} \prod_{l \in \mathcal{L}} s_l^{\omega_l} \exp\big[-\delta_0 s_l^{-1} \operatorname{dist}(\Delta(v_l), \Delta(v_l'))\big],$$

v_l, v_l' the ends of l and dist the sup-distance of sets. The degree of a subgraph H (a set $\mathcal{L}_H$ of lines with their end vertices $\mathcal{V}_H$) is $D(H) := \sum_{v \in \mathcal{V}_H}(d + p_v) + \sum_{l \in \mathcal{L}_H} \omega_l - d$.

Lemma 2.2 (Bałaban's graphs are instances)**.** (a) *For a graph of* [3, Sect. 1] *with lines* $G^{\eta}_{(j)}$ *(scalar or vector) put, for each line* l, $\omega_l := -(d-2) - (\text{number of differentiations acting on } l) + (\text{number of ends of } l \text{ which are vector legs of (1.14)/(1.15) vertices})$, *and for each vertex* $p_v :=$ *the number of factors* η *in* v *other than the* η^d *of its summation. Then the bound* [3, (2.14)] *is* $\tilde{E}(G(j), \Delta)$ *with* $\delta_0 = \delta_1/2$, *and* $D(H)$ *of definition 2.1 is the degree* [3, (2.1)–(2.2)]. (b) $p_v \geq 0$ *for every vertex* [3, (1.6)–(1.15)]*:* [3, (1.8)] *has* $\eta^{n+n'-1}$ *with* $n + n' \geq 1$, [3, (1.9)] $\eta^{n+\bar{n}}$, [3, (1.10)] $\eta^{n+n'-2}$ *with* $n + n' \geq 2$, [3, (1.11)] $\eta^{n+\bar{n}-1}$, *and* [3, (1.6), (1.7), (1.13)–(1.15)] *none.*

Proof. (a) In [3, (2.14)] each vertex carries $(L^{j(v)}\eta)^d$ and "a proper power of $L^{j(v)}\eta$"; the latter is the bound $\eta^{p_v} \leq s(v)^{p_v}$, valid because $p_v \geq 0$ by (b) and $\eta \leq s(v)$. Each leg of l carries $s_l^{-(d-2)/2}$, each differentiation on l carries s_l^{-1}, each vector leg of l at a (1.14)/(1.15) vertex carries s_l ([3, (2.12)]), and each line carries $\exp[-\frac{1}{2}\delta_1 s_l^{-1} \operatorname{dist}(\Delta(v), \Delta(v'))]$; collecting the factors of l gives $s_l^{\omega_l}$. For the degree, sum [3, (2.1)] over $v \in \mathcal{V}_H$: the η-factors give $\sum(d + p_v)$, each line of H contributes its two leg dimensions, minus its differentiations, plus the (1.14)/(1.15) bonus, that is ω_l, and legs on lines not in H do not enter. (b) is read off [3, (1.8)–(1.15)]. □

Lemma 2.3 (one shrinking step)**.** *Fix an ordering* $l(1), \dots, l(m)$ *of* $\mathcal{L}$ *and let* $J(l)$ *be the set of index assignments with* $j_{l(1)} \leq \cdots \leq j_{l(m)}$. *For* $0 \leq i \leq m$ *let* G_i *be the subgraph formed by* $l(1), \dots, l(i)$, *with connected components* $G_i^{(\alpha)}$, *and* G/G_i *the graph* G *with each* $G_i^{(\alpha)}$ *contracted to one vertex. For a vertex* u *of* G/G_i *put* $p_u := p_v$ *if* u *is an original vertex* $v \notin G_i$ *and* $p_u := 0$ *if* u *is a contracted component;* $\square(u) :=$ *the box of a representative original vertex of* u *(when two vertices are merged below, the merged vertex keeps the representative of the end which is not summed);* $j(u)$, $s(u)$, $\Delta(u)$ *as in definition 2.1 with the lines of* G/G_i *only. For* $0 \leq i < m$ *and* $j \in J(l)$ *let*

$$F_i(j, \Delta) := \prod_{\alpha} s_{l(i+1)}^{D(G_i^{(\alpha)})} \prod_{u} s(u)^{d+p_u} \prod_{l \notin G_i} s_l^{\omega_l} \exp\big[-\delta_i s_l^{-1} \operatorname{dist}(\Delta(u_l), \Delta(u_l'))\big], \qquad \delta_i := \delta_0 (2N_G)^{-i},$$

Δ ranging over the localisations of the vertices of G/G_i (at $i=0$ there are no components and $F_0=\tilde{E}$). Assume

$$\text{(PD)} \qquad D(G_i^{(\alpha)}) \geq \theta > 0 \qquad \text{for every } 1 \leq i \leq m-1 \text{ and every } \alpha.$$

Then for $0 \leq i \leq m-2$ and fixed $j_{l(i+2)}, \dots, j_{l(m)}$,

$$\sum_{j_{l(i+1)} \leq j_{l(i+2)}} \sum_{\Delta} F_i(j,\Delta) \;\leq\; c_{i+1} \sum_{\Delta'} F_{i+1}(j,\Delta'), \qquad c_{i+1} := c_d(\delta_i/2)\, e^{d\delta_i}\, (1-L^{-\theta})^{-1}, \quad c_d(r) := \sum_{n\in\mathbb{Z}^d} e^{-r(|n|_\infty - 1)_+}$$

Proof. Write $l := l(i+1)$, $j := j_l$, $s := s_j$. Since j is the smallest index among the lines of G/G_i, every end u of l has $j(u) = j$ and $s(u) = s$.

Step 1: l joins two distinct vertices $u \neq u'$ of G/G_i. (i) *Moving the other lines of u'.* Let $l' \neq l$ be a line of G/G_i with an end at u' and other end u'', and put $a := \operatorname{dist}(\Delta(u), \Delta(u'))$, $b := \operatorname{dist}(\Delta(u'), \Delta(u''))$, $c := \operatorname{dist}(\Delta(u), \Delta(u''))$. The sup-diameter of $\Delta(u')$ is s, so $c \leq a+s+b$. First lower the rate, $e^{-\delta_i b/s_{l'}} \leq e^{-\delta_{i+1} b/s_{l'}}$ $(b \geq 0)$, then insert $b \geq c-a-s$ and use $s_{l'} \geq s$:

$$e^{-\delta_i b/s_{l'}} \leq e^{-\delta_{i+1} c/s_{l'}}\, e^{\delta_{i+1} a/s}\, e^{\delta_{i+1}}.$$

(The order matters: the triangle inequality at the rate δ_i would cost $e^{\delta_i a/s}$ per moved line, which half of the exponential of l does not pay.) There are at most $N_G - 1$ such l', and their total cost $e^{(N_G-1)\delta_{i+1} a/s} \leq e^{(\delta_i/2)a/s}$ is paid by half of the exponential of l; this is Bałaban's "we represent the exponential factor for the line $l(1)$ as a product of as many equal factors as there are legs in v" [3, p. 427]. If $u'' = u$, the line l' becomes a self-loop after the merge and its exponential is bounded by 1. (ii) *Summing $\Delta(u')$.* The remaining factor $e^{-(\delta_i/2)a/s}$ gives $\sum_{\Delta(u')} e^{-(\delta_i/2)\operatorname{dist}(\Delta(u),\Delta(u'))/s} \leq c_d(\delta_i/2)$, uniformly in s, because the cubes of the grid $s\mathbb{Z}^d$ are at sup-distance $s(|n|_\infty - 1)_+$ from $\Delta(u)$. (iii) *Relocalising the merged vertex.* Let β be the component of G_{i+1} containing l. Every line of G/G_{i+1} at β has index $\geq j_{l(i+2)} \geq j$, so $s(\beta) \geq s$, and the cube $\Delta(\beta)$ of side $s(\beta)$ containing $\Delta(u)$ lies in $\square(u)$. For the lines at β, $\exp[-\delta s_{l''}^{-1} \operatorname{dist}(\Delta(u), \cdot)] \leq \exp[-\delta s_{l''}^{-1} \operatorname{dist}(\Delta(\beta), \cdot)]$ because $\Delta(u) \subset \Delta(\beta)$. The summand then depends on $\Delta(u)$ only through $s(u)^d = s^d$, and $\sum_{\Delta(u)\subset\Delta(\beta)} s^d = s(\beta)^d$ is the volume factor of β in F_{i+1}. (iv) *Powers of s.* Not yet used are $s^{d+p_{u'}}$ ($p_{u'} = 0$ if u' is contracted), s^{p_u}, s^{ω_l} and the prefactors $s^{D(\alpha)}$ of the components merged into β. Their product is $s^{D(\beta)}$, because $D(\beta) = D(\alpha_u) + D(\alpha_{u'}) + d + \omega_l$, where $D(\alpha_w)$ is the degree of the component w if w is contracted and $D(\alpha_w) := p_w$ if w is an original vertex (a one-vertex "component" without lines has degree $(d+p_w) - d$): this is the additivity of the degree of definition 2.1 under adding one line.

Step 2: l is a self-loop at a vertex u of G/G_i. There is no summation; only (iii) (with $\Delta(u)$ itself) and (iv) apply, now with $D(\beta) = D(\alpha_u) + \omega_l$, the factor s^{d+p_u} supplying the volume $s(\beta)^d$ after relocalisation and s^{p_u} going into the degree.

Step 3: untouched components. A component α of G_i not touched by l is a component of G_{i+1}; its prefactor satisfies $s_{l(i+1)}^{D(\alpha)} \leq s_{l(i+2)}^{D(\alpha)}$, since $j_{l(i+1)} \leq j_{l(i+2)}$ and $D(\alpha) > 0$ by (PD).

Step 4: the index sum. After Steps 1–3 the only dependence on $j_{l(i+1)}$ is $s^{D(\beta)}$ with $D(\beta) \geq \theta$ (β is a component of G_{i+1}, $i+1 \leq m-1$), and $\sum_{j=0}^{j_{l(i+2)}} (L^j \eta)^{D(\beta)} \leq (1-L^{-\theta})^{-1} (L^{j_{l(i+2)}} \eta)^{D(\beta)}$, the prefactor of β in F_{i+1}. Lowering all other rates from δ_i to δ_{i+1} only increases the bound. The factors collected are $c_d(\delta_i/2)$, at most $e^{N_G \delta_{i+1}} \leq e^{\delta_i}$ from Step 1(i), and the geometric sum; $e^{d\delta_i} \geq e^{\delta_i}$ covers the second. □

Theorem 2.4 ([3, Prop. 2.2], with the top index free or fixed)**.** *Assume* (PD). (i) *For each fixed $j_* \in \{0, \dots, k-1\}$,*

$$\sum_{j \in J(l),\, j_{l(m)} = j_*} \sum_{\Delta} \tilde{E}(G(j), \Delta) \leq C_A\, (L^{j_*}\eta)^{D(G)},$$

with no sign condition on $D(G)$. (ii) *If in addition $D(G) \geq \theta$, then $\sum_{j\in J(l)} \sum_\Delta \tilde{E}(G(j),\Delta) \leq C_A (1-L^{-\theta})^{-1}$.* (iii) *Summed over the $m!$ orderings (the sets $J(l)$ cover all index assignments) the same bounds hold with an extra factor $m!$. Here $C_A := 2^d \prod_{i=0}^{m-2} c_{i+1} \cdot c_d(\delta_{m-1}/2) e^{d\delta_{m-1}}$*

depends only on d, L, δ_0, θ and m (through $N_G \le 2m$); it is independent of η, k, the boxes $\square(v)$, the torus and the domain.

Proof. Apply lemma 2.3 for $i = 0, \dots, m-2$. At the last line $l(m)$ (index j_*, scale s_*) do Steps 1–2 of lemma 2.3 once more: afterwards G/G_m is a single vertex $\beta = G$ without lines, with the factor $s_*^{D(G)}$ (Step 1(iv) with $\beta = G$) and the volume s_*^d of the representative vertex, whose sum over Δ inside $\square$ is $|\square| \le 2^d$. That is (i). For (ii), sum (i) over $j_* \le k-1$: $\sum_{j_*} (L^{j_*}\eta)^{D(G)} \le (1-L^{-\theta})^{-1}(L^{k-1}\eta)^{D(G)} \le (1-L^{-\theta})^{-1}$, since $L^{k-1}\eta = L^{-1} < 1$. (iii) is the finite union over orderings. □

Remark 2.5 (scope of theorem 2.4). (a) For Bałaban's own lines the hypothesis of [3, Prop. 2.2], the generalised [3, (2.13)–(2.14)], is the slice bounds of section 1.2; for subgraphs [3, (2.4)] the summation by parts of section 1.6 is applied first, as on [3, p. 426]. The induction [3, (2.16)] itself (S5) is the instance $\omega_l = -(d-2)$, p_v as in lemma 2.2. (b) The proof uses lines only through $s_l^{\omega_l}$ and the exponential, and vertices only through $s(v)^{d+p_v}$; so "arbitrary dimensions" (ω_l real) and "arbitrary number of legs" are free. A line of dimension 0 whose kernel is bounded by $e^{\delta L\sqrt{d}} e^{-\delta L \operatorname{dist}}$ (the contact joining the two factors of one product [3, (1.18)] at the top scale $s = 1/L$, convention (CV) of the main paper) is such a line. A Hölder factor $|x-x'|^\alpha$ between two vertex positions enters as $\omega_l \to \omega_l + \alpha$ with δ_0 halved, by $|x-x'|^\alpha e^{-\delta|x-x'|/s} \le (2\alpha/(e\delta))^\alpha s^\alpha e^{-\delta|x-x'|/(2s)}$. (c) "Arbitrary power of η" means $p \ge 0$: the bound $\eta^p \le s(v)^p$ which turns a vertex factor into [3, (2.14)] needs $p \ge 0$, and for $p < 0$ the conclusion of theorem 2.4 does not control η^p (which can be as large as $L^{k|p|}$). Vertex factors which are powers of the vertex scale $s(v)$ (any real exponent) are also covered, since they enter $s(v)^{d+p_v}$ directly. This covers every use in [3]: lemma 2.2(b) for [3, (1.6)–(1.15)], and in [3, Sect. 3] the new vertices carry powers of the lowest external index $(L^{j_0}\eta)^{\cdots}$ (for instance [3, (3.30), p. 442]), that is of $s(v)$. (d) A mass vertex [3, (1.7)] with a counterterm of δm_k^2 from a graph G_0 has degree $2 + D(G_0)$ by [3, (2.3)], and $D(G_0)$ can be -1; its bound is not of the form η^p with $p \ge 0$. Such a vertex is estimated through its attached graph, whose lines have their own indices [3, (1.27)–(1.28)]; that bookkeeping is part of the renormalisation step (Step 3(D_4) of the main paper), not of theorem 2.4. Thus graphs containing δm_k^2 vertices are not instances of theorem 2.4 as single graphs (their attached G_0 is divergent, so (PD) fails for it); only their classes are bounded.

2.2. **S7: power counting, [3, (2.17)] and [3, Cor. 2.3].** [3, pp. 428–429] states: a case analysis for the vertices [3, (1.13)–(1.15)] ("analyzing more carefully the possibilities we arrive at the conclusion $D(G) > 0$ again"); "if a graph G has an R-vertex ... then it has positive degree"; for graphs without [3, (1.13)–(1.15)] vertices the estimate

$$D(G) \;\ge\; -2\tfrac{d-2}{2} - 1 + (V-1)\tfrac{4-d}{2} + (E-1)\tfrac{d-2}{2}, \tag{2.17}$$

"from which it follows that the graphs with more than four external legs have positive degree in $d = 3$"; and [3, Cor. 2.3]: "If a graph G has a vertex of the form (1.13)–(1.15), or an R-vertex, or a vertex (1.7) with finite counterterms, or it has more than four external legs, then $D(G) > 0$." What is used afterwards ([3, pp. 429–432]) is the classification of the graphs with $D(G) \le 0$, corollary 2.11 below.

Lemma 2.6 (vertex degrees)**.** (a) *$D(v) = 4 - d$ for [3, (1.6)]; $D(v) = 2$ for [3, (1.7)] with a finite counterterm (whose coefficient has degree 0 by the convention of [3, p. 423]) and $D(v) = 2 + D(G_0)$ for [3, (1.7)] with a counterterm of δm_k^2 attached to the graph G_0 [3, (2.3)]; $D(v) = n\frac{4-d}{2} + n'$ for [3, (1.8)–(1.11)] ($n' = \bar n + 1$ for the R-vertices); $D(v) = \frac{d+2}{2} + n\frac{4-d}{2}$ for [3, (1.13)–(1.15)].* (b) *For v not of type (1.14)/(1.15), $D_G(v) = D(v) + e_v\frac{d-2}{2} + \delta_{\text{ext}}(v) \ge D(v)$, with e_v the number of external legs of v and $\delta_{\text{ext}}(v)$ the number of its differentiations acting on them. For v of type (1.13)–(1.15), $D_G(v) = d - [\phi' \text{ on a line}]\frac{d-2}{2} + n_{\text{int}}(v)\frac{4-d}{2} \ge \frac{d+2}{2}$, with $n_{\text{int}}(v)$ the number of its A' legs on lines.* (c) *At $d = 3$: $D(v) \ge \frac12$ for every vertex not of type (1.13)–(1.15), $D_G(v) \ge \frac12$ for every vertex of every graph, and $D_G(v) \ge \frac52$ if v is of type (1.13)–(1.15). Moreover $D(G_0) \ge -1$ for every counterterm graph G_0 of δm_k^2, so a δm_k^2 vertex has $D(v) \ge 1$.*

Proof. (a), (b) follow from [3, (2.1)]: count η^d and the explicit η-powers, the legs ($-\frac{d-2}{2}$ each, $+1$ more for A' legs of (1.14)/(1.15)) and the differentiation of [3, (1.8)]/[3, (1.9)]; a leg not on a line has dimension 0 and its differentiation is not counted. For (1.13)–(1.15) a leg made external removes $-\frac{d-2}{2}$ (ϕ') or $+\frac{4-d}{2}$ (A'), whence the separate formula. (c) The first part is (a), (b) at $d=3$ with $n+n'\geq 1$ for [3, (1.8)] and $n+n'\geq 2$ for [3, (1.10)]. For $D(G_0)$: G_0 is one-particle-irreducible with two external legs [3, pp. 416–417], built of [3, (1.6), (1.7), (1.8), (1.10)] with $n'=0$; so $\ell(G_0)\geq 1$ (a tree has bridges), and lemma 2.7 below, applied to G_0 (by induction on the order for its own counterterm vertices), gives $D(G_0)=\ell(G_0)+2-4+\sum\kappa+\Delta_{\text{ext}}\geq -1$. □

The vertex (1.13)–(1.15) is read as Bałaban's one-scalar-leg vertex ([3, p. 423 (v)], $n+1$ legs). Under the alternative reading, in which the two factors of a product [3, (1.14)]·[3, (1.15)] at a common y form one bilinear vertex, every statement $D>0$ below still holds, because each factor carries its own $\sum_{x\in B^k(y)}\eta^d$ [3, p. 414], that is $D_G\geq\frac{5}{2}$ per factor; only $D>0$ is used.

Lemma 2.7 (the exact power-counting identity)**.** *Let G have no vertex of type (1.13)–(1.15), and put $\kappa_v := D(v)-(l_v-2)\frac{4-d}{2}$, so that $\kappa=0$ for [3, (1.6)], $\kappa=n'$ for [3, (1.8)–(1.11)], $\kappa=2$ for [3, (1.7)] with a finite counterterm, and $\kappa=2+D(G_0)$ (≥ 1 at $d=3$) for [3, (1.7)] with δm_k^2. Then*

$$D(G)=(4-d)\,\ell+E-4+\sum_v\kappa_v+\Delta_{\text{ext}}.$$

At $d=3$, $D(G)=\ell+E-4+\sum\kappa+\Delta_{\text{ext}}$ with every term on the right nonnegative except -4.

Proof. By lemma 2.6(b), $D(G)=\sum_v D_G(v)-d=\sum_v D(v)-d+E\frac{d-2}{2}+\Delta_{\text{ext}}$ (the counterterm degrees are inside $D(v)$ by [3, (2.3)]). Counting leg-ends, $\sum_v(l_v-2)=2I+E-2V$, so $\sum_v D(v)=(2I+E-2V)\frac{4-d}{2}+\sum\kappa$. Hence $D(G)=(I-V)(4-d)+E+\sum\kappa-d+\Delta_{\text{ext}}=(\ell-1)(4-d)+E-d+\sum\kappa+\Delta_{\text{ext}}$. The values of κ follow from lemma 2.6(a) with $l_v=4$ for (1.6), 2 for (1.7) and $n+2$ for (1.8)–(1.11). □

Lemma 2.8 ((2.17) and its use)**.** *Let G have no vertex of type (1.13)–(1.15), $d\in\{2,3\}$.* (a) *$D(G)\geq V\frac{4-d}{2}+E\frac{d-2}{2}-d$, and the right side equals the right side of (2.17) identically.* (b) *At $d=3$ and $(V,E)=(1,5)$, (2.17) gives only $D(G)\geq 0$, so the printed inference "graphs with more than four external legs have positive degree" does not follow from (2.17) in this one case; it does follow from lemma 2.7: $E\geq 5$ implies $D(G)\geq 1$ (and at $V=1$ even $D(G)\geq 2$, since then $\ell=I\geq 1$).*

Proof. (a) $D(G)=\sum D_G(v)-d\geq\sum D(v)+E\frac{d-2}{2}-d$ by lemma 2.6(b), and $D(v)\geq\frac{4-d}{2}$ for every vertex ((a) of lemma 2.6 at $d\in\{2,3\}$; for a δm_k^2 vertex $2+D(G_0)\geq 1\geq\frac{4-d}{2}$ by the identity of lemma 2.7 applied to G_0). Expanding, $-(d-2)-1+(V-1)\frac{4-d}{2}+(E-1)\frac{d-2}{2}=V\frac{4-d}{2}+E\frac{d-2}{2}-d$. (b) Insert $V=1$, $E=5$, $d=3$: $-1-1+0+2=0$. The rest is lemma 2.7 with $\kappa\geq 0$. □

Lemma 2.9 ([3, Cor. 2.3], with its exact scope)**.** *Let $d=3$ and G connected with at least one line.*

(a) *If G has a vertex of type (1.13)–(1.15), then $D(G)\geq 1$.*
(b) *If G has an R-vertex [3, (1.9)] or [3, (1.11)] and $\bar n\geq 3$, then $D(G)\geq\bar n-2\geq 1$ ([3] takes $\bar n\geq 12$, [3, p. 421]).*
(c) *If G has a vertex [3, (1.7)] with a finite counterterm, then: $E\geq 2$ implies $D(G)\geq 1$; $E=1$ implies $D(G)\geq 0$, with $D(G)=0$ exactly for one graph G_{tad}, the vertex $(1.7)_{\text{fin}}$ and a vertex [3, (1.8)] with $n=1$, $n'=0$ joined by two scalar lines, the A' leg external; $E=0$ allows $D(G)=-1$ (the single $(1.7)_{\text{fin}}$ vertex with its two legs contracted, $D=2-d$) and $D(G)=0$. So the third clause of [3, Cor. 2.3] holds for $E\geq 2$ and fails as printed for vacuum graphs and for G_{tad}.*
(d) *If $E\geq 5$, then $D(G)\geq 1$.*

Proof. Degrees lie in $\frac{1}{2}\mathbb{Z}$ at $d=3$. Let $\mathcal{V}_5$ be the set of (1.13)–(1.15) vertices and $\mathcal{V}'$ the rest.

(a) By lemma 2.6(b) at $d=3$, $D_G(v)=\frac{5}{2}+\frac{1}{2}e_v^\phi+\frac{1}{2}n_{\text{int}}(v)$ for $v\in\mathcal{V}_5$, where $e_v^\phi=1$ if the ϕ' leg of v is external and 0 otherwise, and $D_G(v')\geq\frac{1}{2}+\frac{1}{2}e_{v'}$ for $v'\in\mathcal{V}'$. Put $E':=\sum_{\mathcal{V}'}e_{v'}+\sum_{\mathcal{V}_5}e_v^\phi$.

Then $D(G) \geq -3 + \frac{5}{2}|\mathcal{V}_5| + \frac{1}{2}|\mathcal{V}'| + \frac{1}{2}E' + \frac{1}{2}\sum n_{\rm int}$. If $|\mathcal{V}_5| \geq 2$, $D(G) \geq 2$. Let $\mathcal{V}_5 = \{v\}$. Every vertex of $\mathcal{V}'$ has 2 or 4 scalar legs and every scalar line uses two, while v has one ϕ' leg; so the number of external scalar legs, which are counted in E', is odd, and $E' \geq 1$. If $|\mathcal{V}'| \geq 2$: $D(G) \geq -3 + \frac{5}{2} + 1 + \frac{1}{2} = 1$. If $\mathcal{V}' = \emptyset$: all lines are self-loops at v, hence vector self-loops (v has one scalar leg), so $n_{\rm int} \geq 2$ and the ϕ' leg is external: $D(G) \geq -3 + \frac{5}{2} + \frac{1}{2} + 1 = 1$. If $\mathcal{V}' = \{v'\}$: $D(G) = D_G(v) + D_G(v') - 3$. If $D(v') \geq 1$, then $D(G) \geq \frac{5}{2} + 1 + \frac{1}{2}E' - 3 \geq 1$. If $D(v') = \frac{1}{2}$, then v' is [3, (1.8)] with $n = 1$, $n' = 0$ (two scalar legs and one A' leg; lemma 2.6(a)). If the ϕ' leg of v is external, v and v' are joined by a vector line only, so $n_{\rm int} \geq 1$, both scalar legs of v' are external, and $D(G) \geq \frac{5}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} - 3 = 1$. If the ϕ' leg of v lies on the line to v', the other scalar leg of v' is external and its A' leg is external ($e_{v'} = 2$) or joins v ($e_{v'} = 1$, $n_{\rm int} = 1$); in both cases $D(G) \geq \frac{5}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} - 3 = 1$.

(b) If G has a vertex $w \in \mathcal{V}_5$, then with u the R-vertex $D(G) \geq D_G(w) + D_G(u) - 3 \geq \frac{5}{2} + (\bar{n}+1) - 3 = \bar{n} + \frac{1}{2}$, all other $D_G \geq \frac{1}{2}$ being positive. Otherwise lemma 2.7 with $\kappa_u \geq \bar{n}+1$ and all other terms ≥ 0: a tree has $E \geq 2$ ($\sum(l_v - 2) = E - 2 \geq 0$), so $\ell + E \geq 1$ and $D(G) \geq 1 - 4 + \bar{n} + 1 = \bar{n} - 2$.

(c) If G has a vertex $w \in \mathcal{V}_5$, then with u the $(1.7)_{\rm fin}$ vertex ($D_G(u) \geq D(u) = 2$), $D(G) \geq \frac{5}{2} + 2 - 3 = \frac{3}{2}$, which implies every clause. Otherwise lemma 2.7 gives $D(G) \geq \ell + E - 2 + \sum_{v \neq u} \kappa_v + \Delta_{\rm ext}$. $E \geq 3$ gives $D(G) \geq 1$. $E = 2$: $\ell \geq 1$ gives $D(G) \geq 1$; if $\ell = 0$, then $\sum(l_v - 2) = 0$, every vertex has two legs, every two-leg vertex has $\kappa \geq 1$ ((1.7): ≥ 1; (1.8) with $n = 0$: $\kappa = n' \geq 1$; (1.10) with $n = 0$: $\kappa \geq 2$; R-vertices: $\bar{n} + 1$), and $V \geq 2$, so $D(G) \geq 1$. $E = 1$: a tree is impossible ($\sum(l_v - 2) = -1$); $\ell \geq 2$ gives $D(G) \geq 1$; if $\ell = 1$, then $\sum(l_v - 2) = 1$, one vertex has three legs and all others two, and $D(G) = \sum_{v \neq u} \kappa_v + \Delta_{\rm ext} = 0$ forces $\kappa_v = 0$ at every other vertex; no two-leg vertex has $\kappa = 0$, and the only three-leg vertex with $\kappa = 0$ is [3, (1.8)] with $n = 1$, $n' = 0$. So $V = 2$, $G = G_{\rm tad}$, its external leg being the undifferentiated A' leg ($\Delta_{\rm ext} = 0$; E_s is even since every vertex has 2 or 4 scalar legs and every scalar line uses two, so the external leg is a vector leg). $E = 0$: the one-vertex ring has $\ell = 1$ and $D = 1 - 4 + 2 = -1$; the pair $(1.7)_{\rm fin}$, [3, (1.6)] joined by three lines has $\ell = 2$ and $D = 0$.

(d) If G has a vertex of type (1.13)–(1.15), (a) gives $D(G) \geq 1$. Otherwise lemma 2.7 with $\kappa \geq 0$ (lemma 2.6(c) for the δm_k^2 vertices) gives $D(G) \geq E - 4 \geq 1$. □

Lemma 2.10 (the two exceptions are immaterial)**.** (a) *At background $\tilde{B} = 0$ (after the gauge transformation removing B_0, [3, p. 433]) every localised value of $G_{\rm tad}$, and of every graph containing $G_{\rm tad}$ as a subgraph, is* 0. (b) *Vacuum graphs never rely on* [3, Cor. 2.3]*: by* [3, p. 432] *"all vacuum energy counterterms corresponding to these graphs are added" to every vacuum class, whatever its degree.*

Proof. (a) At $\tilde{B} = 0$ the scalar propagator is a scalar kernel times $\mathbf{1}_N$. The vertex [3, (1.7)] is $\phi' \cdot \delta m^2 \phi'$, where δm^2 is a matrix: [3, (1.22)–(1.23), pp. 416–417] contain q^2 and q^4, so δm^2 is an even polynomial $P(q^2)$ in q, real symmetric and commuting with q; this holds for $\delta m^2_{\rm fin}$ and $\delta m^2_{K,k}$ alike. The vertex [3, (1.8)] with $n + n' = 1$ is $[D\phi'(b) \cdot q\phi'(b_-)]A'_b$. Contracting the scalar loop gives a scalar kernel times $\mathrm{Tr}(qP(q^2)) = 0$, the trace of an antisymmetric matrix times a symmetric one commuting with it. The same trace occurs in every graph containing $G_{\rm tad}$. (b) is the printed rule. □

At $\tilde{B} \neq 0$, before the cube step, $G_{\rm tad}$ need not vanish; it is not used there, because the one-vector-leg classes are formed "by graphs necessary to renormalize all divergent proper subgraphs" [3, p. 431] and the $\tilde{B}'$-expansion vertices carry $\kappa \geq 1$, which makes $D > 0$. By the same count, the statement of [3, p. 431] that graphs with "one external vector field leg ... do not introduce any new divergences" is also true only after the gauge transformation, by lemma 2.10(a).

Corollary 2.11 (the classification used in [3], $d = 3$)**.** *Let $D(G) \leq 0$, G not a vacuum graph and not $G_{\rm tad}$. Then G has no vertex of type (1.13)–(1.15), no R-vertex and no finite-counterterm vertex, and $E \leq 4$; $E = 4$ only for the graph* [3, (2.4)]*; $E = 3$ only with $(E_s, E_v) \in \{(0,3), (2,1)\}$, and then $\ell \leq 1$; $E = 2$ only with $(E_s, E_v) \in \{(2,0), (0,2)\}$ and $V \leq 4$; $E = 1$ only with $E_v = 1$.*

Proof. The first statement is lemma 2.9. $E=4$: $D=\ell+\sum\kappa+\Delta_{\text{ext}}\le 0$ forces $\ell=0$, $\kappa=0$ everywhere and $\Delta_{\text{ext}}=0$. A tree with four legs has $\sum(l_v-2)=2$, and a $\kappa=0$ vertex has $l_v\ge 3$ ((1.8) with $n\ge 1$, $n'=0$: $n+2$ legs; (1.10) with $n'=0$: $n\ge 2$; (1.6): four legs); so $V\le 2$, and $V=2$ because G has a line. Both vertices are [3, (1.8)] with $n=1$, $n'=0$. A vector line between them would leave both differentiated scalar legs external ($\Delta_{\text{ext}}=2$), so they are joined by a scalar line carrying both differentiations: this is [3, (2.4)]. $E=3$: $D=\ell-1+\cdots\le 0$ gives $\ell\le 1$, and E_s is even. $E=2$: lemma 2.8(a) gives $D\ge\frac{V}{2}-2$, and E_s is even. $E=1$: E_s is even. □

2.3. **S8: the Ward–Takahashi identities, and the limit from periodic to free boundary conditions.** [3, pp. 430–431] prints "the basic identity" [3, (2.23)], its first-order differential [3, (2.24)] and the examples [3, (2.25)–(2.28)], and continues: "Taking other functions F, or differentiating (2.24) to higher order in A, we can get all necessary Ward–Takahashi identities. They hold for free boundary conditions also by taking a limit of the identities with periodic boundary conditions." ("Free boundary conditions" is the infinite lattice, as in [4, p. 584].) The identities are consumed on [3, pp. 441–442], on $\xi\mathbb{Z}^d$ with $C^\xi=(-\Delta^\xi+1)^{-1}$, after "we resum over proper orderings and j-indices, we replace the propagators … by C^ξ": the square bracket of [3, (3.29)] is [3, (2.26)] with A concentrated at one point y and $\lambda(y')=\sum_{\mu'}g'(y)A'_{\mu'}(y)(y'_{\mu'}-y_{\mu'})$, a linear function whose gradient is the constant $g'(y)A'(y)$; the same zeroth-moment identity at the next order is "apply a proper Ward–Takahashi identity" in [3, (3.31)–(3.32)]. So the consumed form is: the identities on the infinite lattice, for full propagator lines, with a constant gauge gradient, for sums over all connected graphs of a fixed order (the vertex positions and the internal photon lines summed). This section proves exactly that.
Setting. X is the infinite lattice $\eta\mathbb{Z}^d$ or a torus $T_\ell:=\eta\mathbb{Z}^d/\ell\mathbb{Z}^d$, $\ell\in\eta\mathbb{N}$. Fields $\phi\colon X\to\mathbb{R}^N$; q real antisymmetric; $U(a):=\exp(e_k\eta q a)$. B is a fixed real bond field (periodic on T_ℓ), A a bond field, $\langle\phi,-\Delta_{B+A}\phi\rangle:=\sum_b\eta^{d-2}|U(B_b+A_b)\phi(b_+)-\phi(b_-)|^2$, and $H(A):=-\Delta_{B+A}+M^2$ with $M^2>0$. For $\lambda\colon X\to\mathbb{R}$ put $(d\lambda)(b):=(\lambda(b_+)-\lambda(b_-))/\eta$ and $(T_\lambda\phi)(x):=\exp(e_kq\lambda(x))\phi(x)$. A polynomial F in finitely many $\phi(x)$ is gauge invariant if $F(T_\lambda\phi)=F(\phi)$ for all λ; polynomials in $\phi(x)\cdot P(q)\phi(x)$, P a polynomial, are examples, and they include the Wick-ordered $:|\phi(x)|^4:$, $:|\phi(x)|^2:$ of [3, (2.24)–(2.28)] (the Wick matrix is a function of q, since every kernel below is built from the U) and the mass term $-\frac12\delta m^2:|\phi|^2:$ (δm^2 is an even polynomial in q, [3, (1.23)] and lemma 2.10).

Lemma 2.12 (exponential decay of the kernels)**.** *Let $H=-\Delta_B+M^2$ on X with any real bond field B, and let $C:=H^{-1}$ have kernel $C(u,u')$ with respect to the weight η^d. There are $\theta,c>0$ depending on d, η and M only (not on B, X or ℓ) such that $|C(u,u')|\le c\,e^{-\theta\,\mathrm{dist}_X(u,u')}$, dist_X the sup-distance on X.*

Proof. $H\ge M^2$ since $-\Delta_B\ge 0$. Fix u' and let $\rho(x):=\mathrm{dist}_X(x,u')$, which is 1-Lipschitz for the sup-distance. The conjugate $H_\theta:=e^{\theta\rho}He^{-\theta\rho}$ differs from H only in the hopping terms, each multiplied by $e^{\pm\theta(\rho(b_+)-\rho(b_-))}\in[e^{-\theta\eta},e^{\theta\eta}]$; hence $\|H_\theta-H\|\le 2d\eta^{-2}(e^{\theta\eta}-1)\le M^2/2$ for $\theta\le\theta_0(d,\eta,M)$, and H_θ is invertible with $\|H_\theta^{-1}\|\le 2/M^2$. Then $C(u,u')=e^{-\theta\rho(u)}(H_\theta^{-1})(u,u')e^{\theta\rho(u')}$ with $\rho(u')=0$, and a kernel entry is bounded by η^{-d} times the operator norm. □

Lemma 2.13 (gauge covariance)**.** *$H(A+d\lambda)=T_\lambda^{-1}H(A)T_\lambda$ on X, for every λ (finitely supported if $X=\eta\mathbb{Z}^d$; arbitrary on T_ℓ).*

Proof. With $\psi=T_\lambda\phi$: $U(B_b+A_b+d\lambda(b))(T_\lambda^{-1}\psi)(b_+)-(T_\lambda^{-1}\psi)(b_-)=e^{-e_kq\lambda(b_-)}[U(B_b+A_b)\psi(b_+)-\psi(b_-)]$, because $U(\eta\,d\lambda(b))=\exp(e_kq(\lambda(b_+)-\lambda(b_-)))$ and all these matrices commute. The prefactor is orthogonal, so the quadratic forms agree, and M^2 commutes with T_λ. □

Lemma 2.14 (the identities on a torus: exact, every order, connected form)**.** *On T_ℓ let $F_1,\dots,F_p$ be gauge-invariant polynomials and, for a multi-index β, $Z_\beta(A):=\int d\phi\,e^{-\frac12\langle\phi,H(A)\phi\rangle}F^\beta(\phi)$, $F^\beta:=\prod_r F_r^{\beta_r}$; let $Z(A,\tau):=\sum_\beta\tau^\beta Z_\beta(A)/\beta!$ as a formal power series in τ. Then $Z_\beta(A+d\lambda)=$*

$Z_\beta(A)$ for every β and λ. Consequently, for bond fields $A^{(1)},\dots,A^{(n)}$ and every β, the coefficient

$$W_{n,\beta}(A^{(1)},\dots,A^{(n)};\lambda) := \partial_s\partial_{t_1}\cdots\partial_{t_n}\,[\tau^\beta]\log Z\Big(\sum_i t_i A^{(i)} + s\,d\lambda,\tau\Big)\Big|_{t=s=0}$$

vanishes ($[\tau^\beta]$ denoting the coefficient of $\tau^\beta/\beta!$ in the formal logarithm). $W_{n,\beta}$ is a finite sum of connected graphs with lines $C := H(0)^{-1}$, vertices from the Taylor coefficients of $H(A)$ in A (the vertices [3, (1.8), (1.10)] with $g=1$ at background B) at the bonds of $\operatorname{supp} A^{(i)}$*, one vertex linear in $d\lambda$, and β_r vertices F_r. The identities [3, (2.25)–(2.28)] are the instances $(n,\beta)=(0,0)$, $(1,0)$, $(1,e_r)$ with $F=-\lambda_k\sum_{x\in\Delta}\eta^d:|\phi(x)|^4:$ and $B=0$.*

Proof. Finiteness. $H(A)\ge M^2$ for real A, and $\operatorname{Re}H(A)\ge M^2/2$ for A in a complex neighbourhood of the real bond fields; so each Z_β is a finite-dimensional Gaussian integral of a polynomial, analytic in A there, and $Z_0>0$ on real A. (τ is a formal parameter only; $e^{\tau F}$ is never integrated.) *Invariance.* Substitute $\phi=T_\lambda^{-1}\psi$: the Jacobian is 1 (orthogonal), the quadratic form changes by lemma 2.13, and F^β is unchanged; so $Z_\beta(A+d\lambda)=Z_\beta(A)$. *Logarithm.* Every coefficient of the formal series $\log Z(A,\tau)$ is a polynomial in the $Z_\beta(A)/Z_0(A)$, hence gauge invariant in A, and its ∂_s-derivative at $s=0$ vanishes identically in t. *Graph form.* By Wick's theorem the A-Taylor coefficients at $A=0$ of these coefficients are the connected Gaussian moments of the polynomials $\partial_A^a H(A)|_{A=0}$ and of the F_r: finite sums of connected graphs with lines C. *Instances.* [3, (2.25)] is $n=0$; [3, (2.26)–(2.27)] is $n=1$ at $B=0$ (the two bubbles, the seagull loop and the η-loop); [3, (2.28)] adds one F-vertex. □

Lemma 2.15 (free boundary conditions as the limit of the periodic ones)**.** *On $\eta\mathbb{Z}^d$ let $C_\infty := H(0)^{-1}$ (bounded on ℓ^2, $H\ge M^2$) with B bounded, let λ and the $A^{(i)}$ be finitely supported and the F_r polynomials in the $\phi(x)$, $x\in X_F$ finite. Define $W^\infty_{n,\beta}$ by the finite graph sum of lemma 2.14 with C_∞ in place of C. Then $W^\infty_{n,\beta}=0$.*

Proof. Finitely many positions. Every vertex of the graphs sits at a fixed position: the $A^{(i)}$-vertices at bonds of $\operatorname{supp}A^{(i)}$, the λ-vertex at bonds of $\operatorname{supp}d\lambda$, the F-vertices in X_F; no position is summed, because $H(A)-H(0)$ vanishes off $\operatorname{supp}A$. So W^∞ is a fixed polynomial in the finitely many numbers $C_\infty(u,u')$, $u,u'\in S$, S the union of the supports and their η-neighbours. *Embedding in a torus.* Choose $\ell>\operatorname{diam}S+2\eta$ and B_ℓ periodic of period ℓ agreeing with B on a box of side ℓ around S (the periodisation of its restriction; it changes H only at distance $\ge\ell/2-\operatorname{diam}S$ from S). The projection $\eta\mathbb{Z}^d\to T_\ell$ is injective on the convex hull of S, which contains $\operatorname{supp}\lambda$ (a finitely supported λ has its support in the hull of $\operatorname{supp}d\lambda$), so λ, the $A^{(i)}$ and the F_r lift to T_ℓ, and lemma 2.14 gives $W^{T_\ell}=0$. *Convergence of the kernels.* Let $\tilde C_\infty := (-\Delta_{B_\ell}+M^2)^{-1}$ on $\eta\mathbb{Z}^d$. The torus kernel is the image sum $C_\ell(u,u')=\sum_{n\in\mathbb{Z}^d}\tilde C_\infty(u,u'+n\ell)$: the sum converges absolutely by lemma 2.12, it is ℓ-periodic, and $(-\Delta_{B_\ell}+M^2)$ applied to it gives $\sum_n\delta_{u'+n\ell}$, so it solves the periodic equation on T_ℓ, whose solution is unique since $H_\ell\ge M^2>0$. By the second resolvent identity and lemma 2.12, $|\tilde C_\infty - C_\infty|(u,u')\le c'\exp(-\frac{\theta}{2}(\ell/2-\operatorname{diam}S))$ on $S\times S$ (one half of each exponential pays the sum over the support of $B_\ell-B$), and the images $n\ne0$ are $\le c\,e^{-\theta(\ell-\operatorname{diam}S)}$ each, summable. *Limit.* Hence $C_\ell(u,u')\to C_\infty(u,u')$ for $u,u'\in S$ as $\ell\to\infty$, and $W^\infty=\lim_\ell W^{T_\ell}=0$, W being a fixed polynomial in these entries. □

Lemma 2.16 (the constant-gradient form)**.** *In lemma 2.15 let $n\ge1$ or $\beta\ne0$, and write $W^\infty=\sum_{b'}\eta^d K(b')(d\lambda)(b')$, which is possible since W^∞ is linear in $d\lambda$. Then:* (a) $|K(b')|\le Ce^{-\theta\operatorname{dist}(b',S_0)}$*, with S_0 the union of the supports of the $A^{(i)}$ and X_F;* (b) *for every $c\in\mathbb{R}^d$, $\sum_{b'}\eta^d K(b')\,c_{\mu(b')}=0$, $\mu(b')$ the direction of b'. This is the zeroth-moment identity of* [3, (3.29)] *($c=g'(y)A'(y)$) and of* [3, (3.31)–(3.32)].

Proof. (a) $K(b')$ is the sum of the connected graphs with the λ-vertex at b'; all other positions lie in S_0 and its η-neighbourhood, and in each connected graph some line joins a neighbour of b' to S_0, so each monomial carries a factor $|C_\infty(u,u')|\le ce^{-\theta|u-u'|}$ with u near b' and u' near S_0 (lemma 2.12); the number of graphs is finite. (b) Put $\lambda(x):=\sum_\mu c_\mu x_\mu$ and $\chi_R(x):=\min\{1,\max\{0,2-|x|_\infty/R\}\}$. Then $\lambda_R:=\chi_R\lambda$ is finitely supported, $(d\lambda_R)(b')=c_{\mu(b')}$

if both ends of b' lie in $\{|x|_\infty \le R\}$, and everywhere $|d\lambda_R(b')| \le |c|_\infty + 2|c|_1 + O(\eta/R)$, since $|\lambda| \le 2R|c|_1$ on $\operatorname{supp}\chi_R$ and $|d\chi_R| \le 1/R$. Hence $|\sum_{b'} \eta^d K(b')(c_{\mu(b')} - d\lambda_R(b'))| \le C(|c|_1)\sum_{|b'|_\infty \ge R-\eta} \eta^d |K(b')| \to 0$ as $R \to \infty$ by (a), S_0 being fixed; and $\sum_{b'} \eta^d K(b')\, d\lambda_R(b') = 0$ for every R by lemma 2.15. □

Lemmas 2.15 and 2.16 are stated for finitely supported $A^{(i)}$, a finite X_F and no photon integration. The uses need more: [3, (2.28)] has $F = -\lambda_k \sum_{x\in\Delta} \eta^d : |\phi(x)|^4 :$ summed over all cubes Δ of the class, and the degree-0 two-vector-leg classes of [3, (3.31)–(3.32)] ("all possible connected graphs … of a fixed order", [3, p. 431]) have $\ell = 2$ (lemma 2.7), so they contain internal vertices and internal photon lines at all positions. The next lemma supplies this.

Lemma 2.17 (summed internal positions and internal photon lines)**.** *Work on $\xi\mathbb{Z}^d$ (so $\eta = \xi$) at $B = 0$ with the scalar covariance $C^\xi = (-\Delta^\xi + M^2)^{-1}$ and the photon covariance $C^\xi_{\mathrm{ph}} = (-\Delta^\xi + M^2)^{-1} \otimes \mathbf{1}_d$, $M^2 = 1$ ([3, p. 442]: "replace the propagators $G_{j_0}(0)$, G_{j_0} by C^ξ"). Fix finitely many external vector legs at bonds of a finite set S_0, gauge-invariant vertex densities f_r (polynomials in $\phi(x)\cdot P(q)\phi(x)$, for instance $-\lambda_k :|\phi|^4:$ and $-\frac12\delta m^2 :|\phi|^2:$), and an order in e_k and in the f_r. Let W^{sum} be the sum, over all connected graphs of that order with full lines C^ξ (scalar) and C^ξ_{ph} (photon), of the graphs with one λ-vertex (linear in $d\lambda$), the external legs at S_0, the f_r-vertices summed over all $x \in \xi\mathbb{Z}^d$ with weight ξ^d, and the A-vertices of $H(A)$ (the vertices [3, (1.8), (1.10)] at $B = 0$) summed over all bonds, their A-legs either external (at S_0) or contracted in pairs by C^ξ_{ph}. Then:* (a) *for finitely supported λ, $W^{\mathrm{sum}} = 0$;* (b) *if $S_0 \neq \emptyset$, then writing $W^{\mathrm{sum}} = \sum_{b'} \xi^d K^{\mathrm{sum}}(b')(d\lambda)(b')$ one has $|K^{\mathrm{sum}}(b')| \le Ce^{-\theta' \operatorname{dist}(b', S_0)}$ for some $\theta' > 0$, and $\sum_{b'} \xi^d K^{\mathrm{sum}}(b') c_{\mu(b')} = 0$ for every $c \in \mathbb{R}^d$.*

With $S_0 = \emptyset$ the class is translation invariant, $K^{\mathrm{sum}}(b')$ depends only on the direction of b', and (b) is not meaningful as written; (a) needs no such hypothesis. The uses on [3, p. 442], [3, (3.29), (3.31)–(3.32)], have external vector legs.

Proof. Step 1: the polynomial identity, pointwise in the photon field. On a torus T_ℓ ($\ell \in \xi\mathbb{N}$) let A be any real bond field. For real e_k near 0 the functions $e_k \mapsto Z_\beta(A; e_k)$ of lemma 2.14 are smooth ($H(A) \ge M^2$), and their Taylor coefficients $P_{r,\beta}(A) := \frac{1}{r!}\partial^r_{e_k} Z_\beta(A; e_k)|_{e_k=0}$ are polynomials in the finitely many A_b (the expansion of $U(A) = \exp(e_k \xi q A)$ gives vertices with A-legs, each e_k^r coming with A^r). The invariance $Z_\beta(A + d\lambda; e_k) = Z_\beta(A; e_k)$ for all real e_k near 0 and all A, λ (lemma 2.14, T_λ depending on e_k) gives $P_{r,\beta}(A + d\lambda) = P_{r,\beta}(A)$ as polynomial identities, for every r, β, A and λ on T_ℓ. *Step 2: integrating the photon.* Put $A = A_{\mathrm{ext}} + A_{\mathrm{int}}$ with A_{int} Gaussian of covariance C^ℓ_{ph}, the torus photon covariance at $M^2 = 1$ (a nondegenerate finite-dimensional Gaussian). The A_{int}-expectation of the identity of Step 1 (a polynomial in A_{int}, so a finite Wick sum) gives $\mathbb{E}[P_{r,\beta}(A_{\mathrm{ext}} + d\lambda + A_{\mathrm{int}})] = \mathbb{E}[P_{r,\beta}(A_{\mathrm{ext}} + A_{\mathrm{int}})]$, and by Wick's theorem $\mathbb{E}[P_{r,\beta}(\cdot)]$ is the sum over graphs with internal photon lines C^ℓ_{ph}. Every coefficient of the formal logarithm in (e_k, τ) is a polynomial in the $\mathbb{E}[P_{r,\beta}]/\mathbb{E}[P_{0,0}]$ ($P_{0,0} > 0$ is A-independent), so it is gauge invariant in A_{ext} and its mixed A_{ext}-$d\lambda$ derivatives vanish. These derivatives are the connected graphs of the class with scalar lines C^ℓ, photon lines C^ℓ_{ph} and all internal positions summed over T_ℓ (the f_r-vertices entering through $F_r = \sum_x \xi^d f_r(\phi(x))$). This is (a) on T_ℓ, with λ finitely supported and lifted as in lemma 2.15. *Step 3: absolute convergence, uniformly in ℓ.* By lemma 2.12, $|C^\xi(u,u')|, |C^\xi_{\mathrm{ph}}(u,u')| \le ce^{-\theta|u-u'|}$ and $|C^\ell(u,u')|, |C^\ell_{\mathrm{ph}}(u,u')| \le ce^{-\theta d_\ell(u,u')}$ uniformly in ℓ, d_ℓ the torus distance. Every graph of the class is connected and has a fixed vertex (an external leg at S_0, or the λ-vertex on the finite $\operatorname{supp} d\lambda$). Choose a spanning tree rooted at a fixed vertex and bound the non-tree lines by c; changing variables to the tree differences z_i (minimal representatives on T_ℓ) bounds each graph by $\prod_i ce^{-\theta|z_i|}$ times constants, summable over $(\xi\mathbb{Z}^d)^{\#\text{tree lines}}$ uniformly in ℓ. *Step 4: the limit $\ell \to \infty$.* For fixed z's all positions stay within a fixed distance of the root, so for ℓ large the torus distances equal the lattice distances and $C^\ell \to C^\xi$, $C^\ell_{\mathrm{ph}} \to C^\xi_{\mathrm{ph}}$ there (lemma 2.15). By dominated convergence (Step 3) each graph converges to its value on $\xi\mathbb{Z}^d$, with absolutely convergent position sums; the number of graphs of the fixed order is finite. So $W^{\mathrm{sum}} = \lim_\ell W^{\mathrm{sum},T_\ell} = 0$, which is (a). *Step 5: (b).* As in

lemma 2.16(a): in each connected graph the λ-vertex at b' is joined to S_0 by a path of lines; summing the intermediate positions along this path with $\sum_z e^{-\theta|b'-z|}e^{-\theta|z-s|} \le C_{\theta'}e^{-\theta'|b'-s|}$ ($\theta' < \theta$) and the other positions by Step 3 gives the decay of K^{sum}. The cut-off $\lambda_R = \chi_R\lambda$ of lemma 2.16(b) and dominated convergence then give the zeroth-moment identity exactly as there. □

Remark 2.18 (scope of the identities). (a) *Why the limit is needed.* On T_ℓ a linear λ is not a function on the torus, and a constant $d\lambda = c$ is a flat connection with holonomy, not gauge-trivial; so lemma 2.16(b) is false on a torus in general, the defect being a sum over images winding around the torus, exponentially small in ℓM. Bałaban's "by taking a limit" is essential, and it is exactly lemmas 2.15 to 2.17: the torus identity with finitely supported λ, the limit, then the extension to linear λ on $\xi\mathbb{Z}^d$. (b) *Full lines only.* Lemmas 2.14 to 2.17 are identities between sums over all graphs of a given order with every line the full propagator. They say nothing about a single scale assignment or a fixed ordering of slices; after the decomposition [3, (2.6)] the per-ordering, per-slice analogue is false. [3] applies [3, (2.26)] only after "we resum over proper orderings and j-indices, we replace the propagators $G_{j_0}(0)$, G_{j_0} by C^{ξ}" [3, p. 442], which is the full-line form; that this resummation and replacement is legitimate at the places of [3, Sect. 3] is part of the renormalisation step (Step 3(D_4) of the main paper), not of S8. (c) *Background and averaging terms.* Lemmas 2.13 to 2.16 hold at any background B; [3] uses $B = 0$, after the gauge transformation removing B_0. If $H(A)$ contains a covariant averaging term $a_kP_k(B + A)$ (the propagators G_k), lemma 2.13 still holds, since $Q_k(A+d\lambda)T_\lambda^{-1} = T_\lambda^{\mathrm{unit}}Q_k(A)$ by transport along the contours; so the identities hold with the vertices [3, (1.13)–(1.15)] added. The printed and the consumed identities use C_{M^2}, which needs no such remark. (d) The constants of lemma 2.17 depend on ξ and the order; only the limits $\ell \to \infty$ and $R \to \infty$ are taken, at fixed ξ, which is how the identities are used.

2.4. **S9: the estimate** [3, (3.1)] **for lines between separated localisations.** [3, p. 432]: "Now if two vertices v, v' have localizations satisfying $\mathrm{dist}(\square(v), \square(v')) \ge 1$, then we consider every propagator corresponding to a line connecting these vertices as an external field also. Such a possibility is assured by the following estimates

$$\|hG_k(\Omega, \tilde{B})h'\|_{1,\alpha} \le O(1)e^{-\delta_0 \mathrm{dist}(\square(v),\square(v'))}, \tag{3.1}$$

and similarly for the vector field propagator, h, h' are localization functions." No argument is printed. What is used is (3.1) with $O(1)$ and δ_0 independent of ε, k, Ω, $\tilde{B}$ and the torus: it turns every line between separated localisations into an external field of exponentially small $(1, \alpha)$-norm, which with [3, (2.5)] and [3, (3.2)] gives [3, (1.33)]. In the main paper the estimate is applied to the cluster localisation functions h_K, $h_{K'}$ (Step 3(D_3)); proposition 2.22 below covers them.

Definition 2.19 (the two-variable norm)**.** For a kernel $f(x, x')$ on η-lattice points, an operation in one variable is one of $O_0 :=$ the identity, $O_1 := D^{\eta}_{\tilde{B},\mu}$ (a covariant forward difference), $O_2 :=$ the Hölder quotient of [3, (1.32)], $H_\mu f(x_1, x_2) := |x_2 - x_1|^{-\alpha}[U(\tilde{B}(\Gamma_{x_1x_2}))(D_\mu f)(x_2) - (D_\mu f)(x_1)]$, $x_1 \ne x_2$; put $|O_0| = 0$, $|O_1| = 1$, $|O_2| = 1 + \alpha$, and

$$\|f\|_{1,\alpha} := \max_{(a,a')\in\{0,1,2\}^2,\ \mu,\mu'} \sup \big|(O_a \text{ in } x)(O_{a'} \text{ in } x')f\big|.$$

This is the strongest natural extension of [3, (1.32)] to two variables ([3, p. 420]: "extends in a natural way"), the reading of section 1.1: every operation a vertex can apply to one leg of a two-leg external field, independently at the two legs. Any weaker reading follows from it; the summed form of [3, (1.32)] in two variables is at most 9 times the maximum above.

Input: the slice bounds. [3, (2.6)] writes $G_k(\Omega, \tilde{B}) = \sum_{j=0}^{k-1} G^{\eta}_{(j)}(\Omega, \tilde{B})$, and section 1.2 proves constants $c_s, \delta_1 > 0$, independent of ε, k, j, Ω, $\tilde{B}$ and the torus, such that for every j, with $s := L^j\eta$, and all evaluation points at the depth required there (the line ends and their Hölder partners at depth $\ge R_0$ in Ω),

$$\big|(O_a \text{ in } x)(O_{a'} \text{ in } x')\, G^{\eta}_{(j)}(x, x')\big| \le c_s\, s^{-(d-2)-|O_a|-|O_{a'}|} \exp\big(-\delta_1 s^{-1}\,\mathrm{dist}(X, X')\big), \tag{SL}$$

where $X = \{x\}$ or $\{x_1, x_2\}$ is the set of points used by O_a, and similarly X'. This is [3, (2.10)–(2.11)] read as stated there ("for each differentiation, there is an additional factor $(L^j\eta)^{-1}$ ... This applies also to Hölder norms", [3, p. 426]), applied at each end; the two-ended Hölder case $(a, a') = (2, 2)$ is the mixed second difference of section 1.2.

Lemma 2.20 (summing the slices at a fixed separation)**.** *Let $\rho_0 > 0$, $P \geq 0$. For $\eta = L^{-k}$ and all $k \geq 1$, all $\rho \geq \rho_0$,*

$$\sum_{j=0}^{k-1} (L^j\eta)^{-P} \exp\left(-\delta_1\rho/(L^j\eta)\right) \leq K(P, \delta_1, \rho_0)\, e^{-(\delta_1/2)\rho}, \qquad K(P, \delta_1, \rho_0) := \Big(\frac{2P}{e\delta_1\rho_0}\Big)^P \Big(1 - e^{-\delta_1\rho_0(L-1)/2}\Big)^{-1},$$

independent of k. (Where the lemma is used with several P at once, K is read as the maximum over those P.)

Proof. Write $s_j = L^{j-k} = L^{-m}$ with $m := k - j \geq 1$, and split the exponential into two equal halves. With $u := \rho/s$, $s^{-P} e^{-\delta_1\rho/(2s)} = \rho^{-P} u^P e^{-\delta_1 u/2} \leq \rho^{-P}(2P/(e\delta_1))^P \leq (2P/(e\delta_1\rho_0))^P$. The remaining factors sum to $\sum_{m\geq 1} \exp(-\frac{\delta_1}{2}\rho L^m)$; since $L^m \geq 1 + m(L-1)$, this is $\leq e^{-\delta_1\rho/2} \sum_{m\geq 1} \exp(-\frac{\delta_1}{2}\rho_0(L-1)m)$, bounded by the geometric factor. □

Lemma 2.21 (localisation functions)**.** *Let h, h' be real functions with supports in the w-neighbourhoods of $\square(v)$, $\square(v')$, where the collar w is fixed independently of η and k with $\eta \leq w \leq \frac{1}{4}$, $h = 1$ on $\square(v)$, $h' = 1$ on $\square(v')$, and $|h|$, $|d^\eta h|$ and the Lipschitz constant of $d^\eta h$ bounded by c_h (likewise h'); for Bałaban's smooth h with derivatives up to second order bounded* [3, pp. 412, 420]*, $c_h \sim w^{-2}$. Let $\tilde{B}$ satisfy $e_k \|d^\eta \tilde{B}\|_\infty \leq 1$ (the hypothesis (E) of section 1.1 gives this). Then*

$$\|hfh'\|_{1,\alpha} \leq C(c_h, d) \max_{(a,a'),\mu,\mu'} \sup \big|(O_a)(O_{a'})f\big|,$$

the supremum over the points of $\operatorname{supp} h \times \operatorname{supp} h'$ and their η-neighbours.

Proof. Product rule. $D_{\tilde{B},\mu}(hf)(x) = (d^\eta_\mu h)(x)\, U(\tilde{B}_b) f(x + \eta e_\mu) + h(x)(D_\mu f)(x)$. *Hölder quotient of the first term.* It is controlled by the Lipschitz bound of $d^\eta h$ together with the transported difference $|U(\tilde{B}(\Gamma_{x_1x_2}))f(x_2 + \eta e_\mu) - f(x_1 + \eta e_\mu)|$. That difference is at most $|x_2 - x_1|_1 \sup|Df|$ plus the change of transport between the two shortest contours, which by Stokes' theorem is at most $c(d)\, e_k \|d\tilde{B}\|_\infty |x_2 - x_1|^2 \sup|f| \leq c(d)|x_2 - x_1| \sup|f|$ for $|x_2 - x_1| \leq 1$ (the area enclosed by the two shortest contours is at most $c(d)|x_2 - x_1|^2$). For $|x_2 - x_1| > 1$ the quotient is at most $2\sup|D(hf)|$. If one of the two points, say x_1, lies outside $\operatorname{supp} h$, the term $h(x_1)(Df(x_2) - Df(x_1))$ vanishes, and the remaining terms are bounded by $|h(x_2) - h(x_1)| \sup|Df| \leq c_h|x_1 - x_2| \sup|Df|$ and by the dh-terms; all of them use f and Df only at points within $w + \eta$ of $\square(v)$. *The second term* is h times a quotient of f. *Both variables.* The same in x'. □

Proposition 2.22 ([3, (3.1)], scalar lines)**.** *Let $L \geq 5$, $k \geq 1$, and let h, h' be as in lemma 2.21 with $r := \operatorname{dist}(\square(v), \square(v')) \geq 1$. Assume (SL) at the points used (they lie within $w + \eta$ of $\square(v)$, $\square(v')$; the depth there is supplied by the hypotheses of section 1.1). Then, with $\rho_0 := 1 - 2(w + \eta) > 0$,*

$$\|hG_k(\Omega, \tilde{B})h'\|_{1,\alpha} \leq C_9\, e^{-(\delta_1/2)(r - 2(w+\eta))}, \qquad C_9 := C(c_h, d)\, c_s\, K(d + 2\alpha, \delta_1, \rho_0),$$

with C_9 independent of ε, k, Ω, $\tilde{B}$ and the torus. So [3, (3.1)] holds with $\delta_0 := \delta_1/2$ and $O(1) = C_9 e^{\delta_1(w+\eta)}$; for $w = \frac{1}{4}$, $\eta \leq \frac{1}{5}$ one has $\rho_0 \geq \frac{1}{10}$, and the bound reads $C_9 e^{-(\delta_1/2)(r - \frac{1}{2} - 2\eta)}$, the form used for the cluster functions h_K, $h_{K'}$ of the main paper.

Proof. By [3, (2.6)] the kernel is a sum of slices. The points X, X' used lie within $w + \eta$ of $\square(v)$, $\square(v')$, so $\operatorname{dist}(X, X') \geq r - 2(w + \eta) \geq \rho_0$. (SL) for each slice, with $P := (d - 2) + |O_a| + |O_{a'}| \leq d + 2\alpha$, and lemma 2.20 with $\rho = \operatorname{dist}(X, X')$ give, for each pair (a, a'), the bound $c_s K(P, \delta_1, \rho_0) e^{-(\delta_1/2)(r - 2(w+\eta))}$ (Hölder pairs x_1, x_2 with $|x_1 - x_2| > 1$ are bounded by twice the derivative entry). Lemma 2.21 multiplies by $C(c_h, d)$. The sum over j converges because the points are separated: at separation $\geq \rho_0$ the factors $(L^j\eta)^{-P} e^{-\delta_1\rho/(L^j\eta)}$ are uniformly summable in j, and the diagonal singularity of the mixed derivative never occurs. At $k = 0$ there is no slice sum. □

Proposition 2.23 ([3, (3.1)], vector lines)**.** *The A' covariance is G_k on the torus* [3, p. 414]*, with the slice decomposition "the similar equality for the vector field propagator"* [3, p. 424]*; its slices satisfy* [3, (2.10)–(2.11)] *("the same inequalities hold for vector field propagators",* [3, p. 426]*) with no depth condition, and a leg in a vertex (1.14)/(1.15) is a contour sum $A'(\Gamma^{(k)}_{y,x})$ inside the unit block $B^k(y) \subset \square(v)$, whose slice carries the extra factor $L^j\eta$ of* [3, (2.12)] *(section 1.2). The statement and proof of proposition 2.22 then hold verbatim, with P lowered by* 1 *at a contour leg and with ordinary differences ($\tilde{B} = 0$) in definition 2.19.*

Remark 2.24 (collars and depth). (a) The vertex positions are sharp (unit cubes, side-2 supports), but the localisation functions h, h' of a cut line must be smooth for lemma 2.21: with a sharp indicator ($w = 0$) the Hölder quotient of hf jumps at the edge of the box. Any fixed collar $\eta \le w \le \frac{1}{4}$ works, since $h = 1$ on $\square(v)$, where all vertex positions lie; [3, p. 420] ("$h = 0$ outside some neighborhood of box(v)") leaves the collar unspecified, and the constants depend on it. For wider collars [3, (3.1)] holds for $r \ge 1 + 2w + 2\eta$ instead of $r \ge 1$, and lemma 2.20 holds for any $\rho_0 > 0$. (b) The only depth used is that of (SL) at points within $w + \eta \le \frac{9}{20}$ of the two boxes; the vertex supports lie within 1 of Ω_1 or of supp $\tilde{A}$ (section 1.1), so the depth hypothesis holds once the layer width satisfies $r(t) \ge R_0 + 5$, which the choice of R in [2, (2.7)] secures. The block vertex [3, (1.13)] never stands alone: at least one factor of each product [3, (1.18)] is (1.14) or (1.15) [3, p. 414], and those carry $g_k A'$ or $\tilde{A}$, so they are deep. (c) In contrast with [3, (2.5)], which bounds δG_k and [3, (1.16)] without any separation and therefore needs the two-domain theorem (S2), [3, (3.1)] needs only separated points and the slice bounds.

2.5. **Degrees after a line is cut.** [3, p. 432]: "if we replace a line in an arbitrary graph G by a pair of external fields, then the degree of the new graph is greater or equal to the degree of G, thus a convergent graph is transformed into a convergent one. From our analysis in previous chapters it follows that most of the divergent graphs are transformed into convergent ones, with the possible exception of graphs with one external vector field leg and vacuum graphs." The regrouping into classes which this sentence prepares is proved in the main paper; the three degree facts it uses are proved here. Throughout, a cut set S is a set of lines of G, and $G|_S$ denotes G with every line of S replaced by its two legs, which become external fields (the kernel of the line, an external field by propositions 2.22 and 2.23). Degrees are those of [3, (2.1)–(2.3)], $d = 3$ where stated.

Lemma 2.25 (degrees are intrinsic)**.** (a) *The connected subgraphs of $G|_S$ are exactly the connected subgraphs H of G with no line in S, and $D(H)$ is the same whether computed in $G|_S$ or in G.* (b) *Hence, if every connected subgraph of G has positive degree (the hypothesis of* [3, Prop. 2.1]*, with or without the exception* [3, (2.4)]*), so does every connected subgraph of $G|_S$; and the divergent connected subgraphs of $G|_S$ are exactly the divergent connected subgraphs of G which avoid S.*

Proof. A subgraph is given by its lines and their end vertices. [3, (2.1)] for $D_H(v)$ counts only legs on lines of H, and a leg on a line of S is on no line of H, both in G and in $G|_S$. □

Lemma 2.26 (the degree of the whole graph)**.** *Apply* [3, (2.2)] *verbatim to $G|_S$, with one $-d$ even if $G|_S$ is disconnected. Then $D(G|_S) - D(G) = \sum_{l \in S} \big[\sum_{ends\ of\ l} c_{\rm end} + \delta_l \big]$, with $c_{\rm end} = \frac{d-2}{2}$ for an ordinary leg, $c_{\rm end} = \frac{d-4}{2}$ for a vector leg of a vertex (1.14)/(1.15), and δ_l the number of differentiations acting on l. So the printed sentence "the degree of the new graph is greater or equal to the degree of G" holds whenever no line of S ends at a vector leg of a (1.14)/(1.15) vertex, and at $d = 3$ the degree drops by $\frac{1}{2}$ per such end (two (1.14) vertices with $n = 2$, ϕ' external, joined by their two vector lines have $D = 5$; cutting one line gives $D = 4$). What the regrouping uses is lemma 2.25, together with lemma 2.9(a), which keeps every component containing a (1.13)–(1.15) vertex at $D \ge 1$.*

Proof. By lemma 2.6(b), an ordinary leg leaving a line changes $D_G(v)$ by $+\frac{d-2}{2}$, a (1.14)/(1.15) vector leg by $-\frac{4-d}{2}$, and a no longer counted differentiation by $+1$. In the example, $D_G = 3 + 2 \cdot \frac{1}{2} = 4$ per vertex before and $3 + \frac{1}{2}$ after; with -3 once, 5 and 4. □

Lemma 2.27 (cutting a divergent graph; $d = 3$)**.** *Let γ be connected without (1.13)–(1.15) vertices, with $E(\gamma) \geq 2$, and let S' be a set of $t \geq 1$ lines of γ whose removal leaves $\gamma|_{S'}$ connected (for $t = 1$: a line which is not a bridge; every line, if γ is one-particle-irreducible) and such that $\gamma|_{S'}$ still has a line. Then*

$$D(\gamma|_{S'}) = D(\gamma) + t + (\textit{differentiations on } S') \;\geq\; \max\{0, D(\gamma) + t\},$$

and $D(\gamma|_{S'}) = 0$ only if $t = 1$ and $\gamma|_{S'}$ is the graph [3, (2.4)]*. If $E(\gamma) \geq 3$, then $D(\gamma|_{S'}) \geq 1$.*

Proof. Exact change. By lemma 2.7 for the connected $\gamma|_{S'}$: ℓ drops by t, E rises by $2t$, Δ_{ext} rises by the differentiations on S', and κ is unchanged. *Lower bound.* Since the removal of S' leaves a connected graph, γ has a cycle, $\ell(\gamma) \geq 1$ and $D(\gamma) \geq \ell + E - 4 \geq -1$; hence $D(\gamma|_{S'}) \geq 0$, with equality only if $t = 1$, $D(\gamma) = -1$ and S' carries no differentiation. *Equality.* Then $\gamma|_{S'}$ has $\ell = 0$, $E = 4$, all $\kappa = 0$ and $\Delta_{\text{ext}} = 0$, so it is [3, (2.4)] by corollary 2.11, which requires a line; this is where the last hypothesis enters (without it, a self-loop cut of a one-vertex γ leaves a bare vertex with four or two external legs and $D = 0$: the [3, (1.6)] tadpole and the [3, (1.10)] tadpoles with $n + n' = 2$). $E \geq 3$. $D(\gamma) \geq 1 + 3 - 4 = 0$, so $D(\gamma|_{S'}) \geq 1$. □

The cut of [3, p. 432] removes only lines joining distinct vertices whose localisations are at distance ≥ 1, never a self-loop, and a connected $\gamma|_{S'}$ with at least two vertices has a line; so the last hypothesis of lemma 2.27 holds at every use. The outcome [3, (2.4)] arises from the e^2 self-energy with its photon line cut and from the one-loop vacuum polarisation with one scalar line cut; it needs no partner, since the summation by parts of section 1.6 treats it. The other two possible outcomes of cutting a divergent subgraph, a vacuum graph and a one-vector-leg graph, are Bałaban's exception list: vacuum classes always receive their counterterms [3, p. 432], and the one-vector-leg loop through a finite-counterterm vertex vanishes after the gauge transformation by lemma 2.10(a) (its coefficient is an even polynomial in q). The regrouping itself, with these inputs, is Step 3(D_3) of the main paper.

3. The sketched steps of Proposition 1, III: the component cube, the constant field, the expansion in the new background and the old-vertex components

This part contains the steps of [3, pp. 432–433] that prepare the passage from the domain Ω to a cube $\square$ of constant side and from the background $\tilde{B}$ to a constant background B_0:

step	[3] page	printed object
S11	432	"the localizations of each connected component are contained in some cube of a linear size de a number of vertices, hence on $\bar{n}$ only"
S12	433	the cube split $G_k(\Omega, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k(\Omega, \square, \tilde{B})$ (printed; only its three properties (P-i)–(
S13	433	the constant field B_0 with $\lvert\tilde{B} - B_0\rvert \leq O(r(L^k\varepsilon)p(L^k\varepsilon))$ on $\square$
S14	433	the expansion in $\tilde{B}' = \tilde{B} - B_0$ by [1, (3.14), (3.44), (3.45)], and its decomposition into renorm
S15	433	"If a class contains an old vertex ..., then all vertices of this class are localized in $\square_1$"

The order is S11 (section 3.1), the properties of the cube split (section 3.2), S13 (section 3.3), the expansion S14 (sections 3.4 and 3.5) and S15 (section 3.6). Everything is at $L = 5$, $d = 3$, on the scale-k lattice with $k \geq 1$; the geometric lemmas hold for every d with the same proofs. The step S10 (the regrouping after the separated lines are cut) is the main paper's; its cluster form is recalled below, because S11 and S15 are statements about the components it leaves. The chains without an old vertex (S16), the gauge covariance and the gauge-out of B_0 (S17–S18), the cube propagator (S19), the norms (S20), the renormalisation (S21) and the closure lemmas of the class family are in the fourth part.

Conventions for this part.
Scales and fields. As in section 1.1: $\eta = L^{-k}$, $t = L^k\varepsilon$, $e_k = e\,t^{1/2}$; $p(t) = b_0(1 + \ln t^{-1})^{\bar{p}}$ with $\bar{p} > 2$ and $r(t) = R(1 + \ln t^{-1})^{\varrho}$ with $\varrho > 1$, $R > R_0$ [2, (2.3), (2.7)]; q is the fixed real antisymmetric $N \times N$ matrix, $U(A_b) := \exp(qe_k\eta A_b)$, and for an oriented contour Γ, $A(\Gamma) := \sum_{b\subset\Gamma} \eta A_b$ and $U(A(\Gamma)) := \exp(qe_k A(\Gamma))$; all U are orthogonal and commute, $U(A + A') = U(A)U(A')$. The covariant derivative is $(D^\eta_A\phi)(b) := \eta^{-1}(U(A_b)\phi(b_+) - \phi(b_-))$ on the bond $b = (b_-, b_+)$,

$Q_k(A)\phi(y) := \sum_{x\in B^k(y)} \eta^d U(A(\Gamma^{(k)}_{y,x}))\phi(x)$ with the contours $\Gamma^{(k)}_{y,x} \subset B^k(y)$ of η-length $\le d$ [3, (1.13)], $P_k(A) = Q_k(A)^* Q_k(A)$, and on a union X of unit blocks $H_X(A) := -\Delta^{\eta,N}_{A,X} + m^2t^2 + a_k P_k(A)$, $G_k(X,A) := H_X(A)^{-1}$, with $-\Delta^{\eta,N}_{A,X} = (D^\eta_A)^* D^\eta_A$ over the bonds of X and $m^2t^2 > 0$. The hypotheses of [3, p. 412] on $\tilde B$ are those of section 1.1: $|\partial^\eta_\nu \tilde B_\mu| \le c_1 p(t)$ on every pair of bonds of Ω, and the Hölder quotients of exponent α_0 of $\partial^\eta \tilde B$ are $\le c_1 p(t)$ on pairs of points at distance ≤ 1 at depth $\ge R_{\mathrm{Ba}}+2$ in Ω, where $R_{\mathrm{Ba}} \ge R_0$ is the depth parameter of the background regularity of the main paper. The constant c_1 is independent of k (this is the k-uniformity of the background bounds proved in the main paper's Step 3(C)); nothing else about $\tilde B$ is used.

Grid boxes and label boxes. $|\cdot|_\infty$ is the sup norm; "side", "diameter" and dist are sup-norm notions unless said otherwise. A *grid box* is a closed set $\prod_\mu [\alpha_\mu, \beta_\mu] \subset \mathbb{R}^d$ with all $\alpha_\mu \le \beta_\mu$ in the block-corner grid (a translate of $\mathbb{Z}^d$). A *label box* is a grid box with all sides ≤ 2. The localisation of [3, p. 420] is read as in the main paper (its convention (Loc)): a label is attached to each leg of a vector field (the support box of its partition function $H_y \subset y + [-1,1]^d$, the partition $\{H_y\}_{y\in\mathrm{grid}}$ being smooth with $\operatorname{supp} H_y \subset y + [-\frac34, \frac34]^d$ and derivatives bounded independently of η; a bond b with $H_y(m_b) \ne 0$ at its midpoint m_b has both ends in $y + [-1,1]^d$), to each [3, (1.6)] or [3, (1.7)] vertex (the unit grid cube of its block), and to each block factor [3, (1.12)–(1.15)] (the unit grid cube containing $B^k(y)$, which contains the contours $\Gamma^{(k)}_{y,x}$). The *localisation box* Q_v of a vertex v is the intersection of the boxes of its labels; it is a label box (possibly degenerate) containing every summation point of the vertex. A counterterm vertex carries the labels of the anchor vertex of the subgraph it replaces (the main paper's reading (Agr)), so its localisation box contains its point. The distance "$\mathrm{dist}(\square(v), \square(v'))$" of [3, pp. 432–433] is $\mathrm{dist}(Q_v, Q_{v'})$; the point-set distance of the summation points is at least this, so the rule "distance $\ge 1 \Rightarrow$ external field" is applied only where the separation that [3, (3.1)] needs holds for the actual points.

Classes and the cut rule. The classes are the main paper's per-label classes: all graphs of a type and order with a given multiset $\mathcal{B}$ of label boxes, with the counterterm graphs of the divergent subgraphs; every member of a class has the same $\mathcal{B}$. The *clusters* of $\mathcal{B}$ are the connected components of the graph on $\mathcal{B}$ in which two boxes are joined when they intersect, and U_K is the union of the boxes of the cluster K. The cut rule of [3, p. 432] in the main paper's form: a line is *cut* (made an external field by [3, (3.1)], section 2.4) when its two end vertices lie in different clusters, and *kept* otherwise. Every vertex lies in the intersection of its label boxes, so its labels lie in one cluster and "the cluster of a vertex" is well defined. A *component* is a connected component of a graph with respect to its kept lines.

Vertex counts. Three counts occur. The *merged* count V_{mer} counts a product [3, (1.18)] of two block factors at a common y as one vertex; Bałaban's count $V_{\mathrm{Bal}} \le 2V_{\mathrm{mer}}$ counts each of [3, (1.13)–(1.15)] separately ([3, p. 414]; this is the count in which degrees are computed, section 2); the *label* count $V_{\mathrm{lab}} \le (2\bar n + 1)V_{\mathrm{Bal}}$ is the number of label boxes, since a vertex has at most $2\bar n + 1$ labels (at most $n + n' \le 2\bar n + 1$ vector legs for [3, (1.8)–(1.11)], the R-vertices having $n' = \bar n + 1$ and $n \le \bar n$; one unit cube for [3, (1.6)–(1.7)]; one unit cube and $n \le \bar n$ legs for a block factor). Lemma 3.3 bounds V_{mer} by a function of $\bar n$; with it $V_{\mathrm{lab}}(\bar n) := (2\bar n + 1)\cdot 2\bar n(4N_A + 5)$ and

$$l_N := 2V_{\mathrm{lab}}(\bar n)$$

are the constants of the main paper. Here N_A is the fixed maximal number of insertions in the expansion of a scalar line or external scalar field in the fluctuation of the background, [1, (3.45)] to order $\bar n$ on a line ($N_A = \bar n$) and the fixed odd order $n_{\mathrm{loop}} \ge \bar n$ of [1, (3.47)] on a closed scalar loop; $N_A := \max\{\bar n, n_{\mathrm{loop}}\}$ depends on $\bar n$ only.

The cube step. $r_\square$ denotes the side parameter of the cube step: $r_\square = r^\flat$, the constant of the main paper's block-factor-5 reading (its cube step replaces the cubes of sides $r(t)$, $3r(t)$ by concentric cubes of constant sides $r^\flat$, $3r^\flat$), and $r_\square = r(t)$ wherever the printed step is applied as printed. M is the side of the large blocks. Rounding a set outward to large blocks enlarges each side by at most $2M$.

3.1. **S11: a connected component lies in a cube of side l_N.** [3, p. 432]: "each line of each connected component of G'_{ren} is localized in cubes $\square(v)$, $\square(v')$ with $\mathrm{dist}(\square(v),\square(v')) = 0$ (by the definition of new graphs it should be < 1, but by the construction of localizations it is then $= 0$). From this it follows that the localizations of each connected component are contained in some cube of a linear size depending on a number of vertices, hence on $\bar{n}$ only." The consumed form is corollary 3.4: after the cut, every component of every graph of a class has all its label boxes in one grid cube Q_0 of side l_N, and Q_0 can be chosen common to the class component. Three facts enter: the dichotomy for grid boxes (the parenthesis of the quotation), the diameter bound for a chain of intersecting boxes, and the bound on the number of vertices in terms of $\bar{n}$ ([3, p. 414]: "The maximal number of vertices depends in a simple way on $\bar{n}$, but it is unessential here").

Lemma 3.1 (the dichotomy "$< 1 \Rightarrow = 0$")**.** *Let Q, Q' be grid boxes. Then for every ℓ^p distance, $1 \le p \le \infty$, either $Q \cap Q' \neq \emptyset$ (so $\mathrm{dist}(Q, Q') = 0$) or $\mathrm{dist}(Q, Q') \ge 1$. The same holds on a torus whose side is an integer number of unit blocks, with the torus distance.*

Proof. Write $Q = \prod_\mu I_\mu$, $Q' = \prod_\mu I'_\mu$ with $I_\mu = [\alpha_\mu, \beta_\mu]$, $I'_\mu = [\alpha'_\mu, \beta'_\mu]$, all endpoints in one translate of $\mathbb{Z}$. If $I_\mu \cap I'_\mu \neq \emptyset$ for every μ, pick $x_\mu \in I_\mu \cap I'_\mu$; then $(x_\mu)_\mu \in Q \cap Q'$. Otherwise there is a μ with, say, $\alpha'_\mu > \beta_\mu$; the gap $g := \alpha'_\mu - \beta_\mu$ is a difference of two points of a translate of $\mathbb{Z}$, so $g \in \mathbb{Z}$ and $g \ge 1$, and for $x \in Q$, $x' \in Q'$, $|x - x'|_p \ge |x_\mu - x'_\mu| \ge g \ge 1$. On the torus the distance is the minimum over the lifts of Q' by the period lattice; integer periods map grid boxes to grid boxes, so the argument applies to each pair of lifts. □

With the labels of (Loc) this is the printed parenthesis: every line kept by the threshold 1 of [3, p. 432] joins two vertices whose localisation boxes intersect, and under the cluster form of the rule every kept line joins two boxes of one cluster.

Lemma 3.2 (chains of intersecting boxes)**.** *Let $Q_1, \dots, Q_V$ be label boxes (sides ≤ 2), and let the graph on $\{1, \dots, V\}$ in which i and j are joined when $Q_i \cap Q_j \neq \emptyset$ be connected.*

(a) *If i and j are joined by a path of m edges, then $|x - z|_\infty \le 2(m+1)$ for all $x \in Q_i$, $z \in Q_j$, and $\mathrm{dist}_\infty(z, Q_i) \le 2m$ for all $z \in Q_j$.*
(b) *$\bigcup_i Q_i \subset Q$ for a grid cube Q of side $2V$.*
(c) *On a torus of side $\ell_T > 2V$ (in unit blocks) the same holds, with Q a cube of the torus that does not wrap.*

Proof. (a) Let $i = i_0, i_1, \dots, i_m = j$ be the path and pick $p_s \in Q_{i_{s-1}} \cap Q_{i_s}$, $s = 1, \dots, m$. Consecutive points of the sequence $x, p_1, \dots, p_m, z$ lie in a common box ($x, p_1 \in Q_{i_0}$; $p_s, p_{s+1} \in Q_{i_s}$; $p_m, z \in Q_{i_m}$), each of sup-diameter ≤ 2, so $|x - z|_\infty \le 2(m+1)$ (≤ 2 when $m = 0$). Dropping the first segment, $\mathrm{dist}_\infty(z, Q_i) \le |z - p_1|_\infty \le 2m$.

(b) Any two indices are joined by a path of $\le V - 1$ edges, so by (a) the union has sup-diameter $\le 2V$. For each μ the projection of the union onto the μ-axis is a union of intervals with grid endpoints; its hull $[a_\mu, b_\mu]$ has grid endpoints and $b_\mu - a_\mu \le 2V$. Put $Q := \prod_\mu [a_\mu, a_\mu + 2V]$.

(c) Choose a spanning tree of the intersection graph and a lift of Q_{i_0} to $\mathbb{R}^d$. Lift the boxes along the tree, each to the lift that meets the lift of its parent (it exists because the boxes meet on the torus). Tree paths have $\le V - 1$ edges, so (a) bounds the sup-diameter of the lifted union by $2V$; every box is lifted once, so the projection of the lifted union is $\bigcup_i Q_i$. It lies in the projection of a cube of side $2V$ of $\mathbb{R}^d$, which is injective and does not wrap when $\ell_T > 2V$. □

Lemma 3.3 (the number of vertices depends on $\bar{n}$ only)**.** *Every graph of the expansion* [3, (1.5)]*, and every graph obtained from one by cutting lines, by passing to the graphs of its renormalised class, or by localisation, has at most*

$$V_{\mathrm{mer}}(\bar{n}) := \bar{n}(4N_A + 5)$$

vertices in the merged count, hence at most $2V_{\mathrm{mer}}(\bar{n})$ in Bałaban's count and at most $V_{\mathrm{lab}}(\bar{n}) = (2\bar{n} + 1) \cdot 2V_{\mathrm{mer}}(\bar{n})$ label boxes.

Proof. The facts about the expansion that are used, all from [3, pp. 412–415] and [1, (3.44)–(3.47)]:

(E1) The expressions are the derivatives [3, (1.5)], $\sum_{1\le\alpha+\beta\le\bar n}(\alpha!\beta!)^{-1}\partial_{e'}^{\alpha}\partial_{\lambda'}^{\beta}E_k$ at $e'=\lambda'=0$, of the connected functional $E_k=-\log[\dots]$ [3, (1.4)]. So each graph is connected, has at least one internal line [3, p. 415], and has total order $\alpha+\beta\in[1,\bar n]$ in (e',λ').

(E2) e' multiplies only the fluctuation field g_kA' in [3, (1.4)], λ' the quartic term and the counterterms ([3, p. 419]: the dependence of δm^2, E_1 on e',λ' "is obtained by replacing e,λ by $e'e,\lambda'\lambda$"). So a vertex with $n\ge1$ legs A' carries e'^n, a [3, (1.6)] vertex carries λ', and a [3, (1.7)] vertex carries a counterterm all of whose Taylor terms have order ≥1 (the counterterms are sums of graphs). Call these vertices *order-carrying*. Every other vertex has $n=0$ and no A'-leg; it is two-legged (the types [3, (1.8)–(1.11)] at $n=0$ and the products [3, (1.18)] of two block factors at $n=0$, merged) or one-legged (the product of [3, (1.12)] with a block factor at $n=0$).

(E3) The $n=0$ vertices arise only from the expansion [1, (3.45)] of the propagator $G_k(\Omega,\tilde A+\tilde B)$ and of the external scalar field $a_kG_k(\Omega,\tilde A+\tilde B)Q_k^*(\tilde A+\tilde B)\phi$ in $\tilde A$ ([3, p. 414]: the vertices of [3, (1.5)] "are given by (1.6)–(1.8), (1.10), (1.13), and (1.14) with $n'=0$"), taken to the fixed number N_A of insertions with the remainder [3, (1.16)] treated as an external field. So the scalar path that replaces one propagator, or one external scalar field, carries at most N_A two-legged $n=0$ vertices and at most one one-legged one. On a closed scalar loop the expansion [1, (3.46)–(3.47)] is taken to the fixed odd order $n_{\rm loop}\ge\bar n$, which N_A dominates.

(E4) Every vertex has at most four scalar legs: [3, (1.6)] has four, all others at most two.

(E5) The expressions of [3, p. 419] "obtained by an integration and a cumulant expansion in kth step, where some external legs are contracted by propagators $C^{(k)}_{\Lambda_5^{(k)}}$" are handled there with "external fields in the form of functions $\Phi_{\rm ext}(x,x',x'',\dots)$, $A_{\rm ext}(y,y',\dots)$ of many variables". With this device the contractions are external fields, not internal lines, and (E1) holds for each factor of the cumulant separately; the count below applies to each factor.

Order-carrying vertices. Each consumes at least one unit of the total order $\alpha+\beta\le\bar n$ by (E1)–(E2), so there are $V_c\le\bar n$ of them.

Chains. A *chain* is a maximal path or cycle, along scalar lines, of $n=0$ vertices whose interior vertices are two-legged. Every $n=0$ vertex lies in exactly one chain: a two-legged one has both its lines in the chain, a one-legged one ends its chain. A cycle of $n=0$ vertices only would be a whole component of order 0, excluded by (E1); a cycle through an order-carrying vertex uses two of that vertex's legs and is counted below like a chain that uses those legs.

Every chain ends at an order-carrying vertex. $n=0$ vertices have no A'-leg (E2), so a chain meets the rest of the graph only through the scalar lines at its ends. If neither end were a leg of an order-carrying vertex, the chain would be a connected component of order 0, contradicting (E1).

Number of chains. Each scalar leg of an order-carrying vertex ends at most one line, so distinct chains use distinct legs; by (E4) there are at most $4V_c\le4\bar n$ chains.

Vertices per chain. Each chain replaces one propagator or one external scalar field of the unexpanded expression, so by (E3) it has at most N_A+1 vertices.

Total. $V\le V_c+4V_c(N_A+1)\le\bar n(4N_A+5)$.

Stability. Cutting lines can only disconnect, and does not change the vertex set; the counterterm graphs of a class replace a divergent subgraph by one counterterm vertex, which lowers the count; localisation does not change the vertex set. Bałaban's count at most doubles the merged one, since only the products [3, (1.18)] are merged, and the label count is at most $2\bar n+1$ per vertex (conventions of this part). □

Corollary 3.4 (S11, consumed form)**.** *Let $\mathcal C$ be a class (every member carries the same multiset $\mathcal B$ of label boxes), cut by the cluster rule. Then:*

(a) *every cluster K of $\mathcal B$ has* $\operatorname{diam}_\infty U_K\le2|K|\le l_N$*, and U_K lies in a grid cube Q_K of side l_N;*

(b) *every component of every $G\in\mathcal C$ has all its label boxes in one cluster K, hence in Q_K;*

(c) *a connected component of the class (the members of $\mathcal{C}$ restricted to the components lying in a fixed cluster K, including the counterterm graphs) has all its label boxes, for all its graphs, in the one cube $Q_0 := Q_K$ of side l_N.*

On a torus the same holds when its side exceeds l_N. The side l_N depends on $\bar{n}$ only, not on ε, k, Ω, Ω_1, Ω_2, $\tilde{B}$ or the torus.

Proof. (a) A cluster is a family of $|K| \le V_{\mathrm{lab}}(\bar{n})$ label boxes (lemma 3.3) whose intersection graph is connected, so lemma 3.2(b), (c) give the cube of side $2|K| \le l_N$ (enlarge the cube of side $2|K|$ to one of side l_N on the grid). (b) A kept line joins two vertices of one cluster, and every vertex lies in the intersection of its own label boxes, so all boxes along a kept-connected component lie in one cluster. (c) $\mathcal{B}$, hence the clusters and the cubes Q_K, are common to all members of the class; a counterterm vertex carries the labels of its anchor, which lie in the cluster of the subgraph it replaces. □

Remark 3.5. (a) The cube of (c) is what the cube step centres $\square_1$ on ([3, p. 433]: "The localization cubes of each component are contained in some cube of a size $O(\bar{n})$. We take a cube $\square_1$ ... containing the above cube in the center"). It must be common to the class component, since the cancellations inside a class (a graph against its counterterm graph, the leg-exchange and Ward partners) are between different graphs of the class, and the split $G_k(\Omega, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k$ must be the same for all of them; with per-label classes this is automatic, as (c) shows. (b) For a single graph cut by the threshold 1 on single lines, lemmas 3.1 and 3.2 give the per-graph cube of side $2V_{\mathrm{Bal}}(\bar{n}) \le l_N$ directly: a kept line joins intersecting boxes, and a component has at most $V_{\mathrm{Bal}}(\bar{n})$ vertices. (c) The parenthesis "it is then = 0" of [3, p. 432] is true because the label boxes lie on a common grid (lemma 3.1); for half-open blocks of the η-lattice adjacent blocks are at distance η. Bałaban's freedom in choosing the partitions of unity allows the grid choice, and it is made in (Loc).

3.2. **The cube split: the three properties used.** [3, p. 433]: "We take a cube $\square_1$ of size $r(L^k\varepsilon)$ containing the above cube in the center, and next we take a cube $\square$ of size $3r(L^k\varepsilon)$ and with $\square_1$ in the center. We assume that $\square_1$, $\square$ are sums of big blocks of the unit lattice. Now we replace each scalar field propagator $G_k(\Omega, \tilde{B})$ [or $G_k(\Omega_2, \tilde{B})$] in the expression by $G_k(\square, \tilde{B})$, more exactly we represent $G_k(\Omega, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k(\Omega, \square, \tilde{B})$, and we treat $\delta G_k(\Omega, \square, \tilde{B})$ as an external field. Thus again we drop the lines corresponding to these operators and decompose the obtained set of graphs into renormalized classes." This step is printed. The main paper performs it with the constant side $r^\flat$ in place of $r(t)$ (the containment $\square \subset \Omega$ fails at $L = 5$ for the printed side) and proves the bound on $\delta G_k(\Omega, \square, \tilde{B})$ there. The steps S13–S15 use only the following three properties, which hold for $r_\square = r^\flat$ under the main paper's cube step and for $r_\square = r(t)$ wherever the printed step applies.

(P-i) $\square_1 \subset \square$ are unions of large blocks, obtained by rounding outward the cubes of sides $r_\square$ and $3r_\square$ concentric with the class-component cube Q_0 of corollary 3.4(c). So $\square_1$ contains the cube of side $r_\square$ concentric with Q_0, $\square$ contains the cube of side $3r_\square$ concentric with Q_0, and $\mathrm{side}(\square) \le 3r_\square + 2M$.

(P-ii) $\square \subset \Omega$ (resp. $\square \subset \Omega_2 \subset \Omega$), so $\tilde{B}$ is defined, and satisfies the hypotheses of [3, p. 412], on every bond of $\square$; and $\square$ does not wrap around the torus.

(P-iii) After the split every scalar line is $G_k(\square, \tilde{B})$; $\delta G_k(\Omega, \square, \tilde{B})$, $\delta G_k(\Omega, \Omega_2, \tilde{B})$ and the kernels [3, (1.16)] are external fields.

3.3. **S13: the constant field B_0.** [3, p. 433]: "The field $\tilde{B}$ is regular in the sense that $\|\partial_\mu^\eta \tilde{B}\|_{0,\alpha} \le O(p(L^k\varepsilon))$. Hence there exists a constant field B_0 such that $|\tilde{B} - B_0| \le O(r(L^k\varepsilon)p(L^k\varepsilon)) \le O(p(L^k\varepsilon)^2)$ on the cube $\square$." The consumed form: a constant vector field B_0 with $|\tilde{B} - B_0| \le Cr_\square p(t)$ on the bonds of $\square$, such that $\tilde{B}' := \tilde{B} - B_0$ has the derivative and Hölder bounds of $\tilde{B}$ and is an external vector field of norm [3, (1.32)] $O((r_\square + 1)p(t))$ on $\square$: $O(p(t))$ at the constant side $r^\flat$.

Lemma 3.6 (the constant field)**.** *Assume (P-i), (P-ii), and let c be a site of $\square$ with $c + \eta e_\mu \in \square$ for every μ and $|x - c|_\infty \le \frac{1}{2}\,\mathrm{side}(\square) + \eta$ for all $x \in \square$ (a site next to the centre). Put*

$B_{0,\mu} := \tilde B_\mu(c)$ for every μ (a constant field) and $\tilde B' := \tilde B - B_0$ on the bonds of $\square$. Then for every bond $(x, x+\eta e_\mu) \subset \square$ and every contour $\Gamma \subset \square$:

(a) $|\tilde B'_\mu(x)| \le c_1 p(t)\,|x-c|_1 \le C_{13}\,p(t)$, *with* $C_{13} := c_1 d\big(\frac{3}{2} r_\square + M + 1\big)$;

(b) $\partial^\eta \tilde B' = \partial^\eta \tilde B$ *on* $\square$, *so* $|\partial^\eta_\nu \tilde B'_\mu| \le c_1 p(t)$, *and the Hölder quotients of* $\partial^\eta \tilde B'$ *are those of* $\partial^\eta \tilde B$;

(c) $|\tilde B'(\Gamma)| \le |\Gamma|_\eta\, C_{13}\, p(t)$, *with* $|\Gamma|_\eta$ *the length of* Γ *($\le d$ for the contours $\Gamma^{(k)}_{y,x}$)*;

(d) *the norm* [3, (1.32)] *of* $\tilde B'$ *as an external vector field ("with* $\tilde B = 0$", [3, p. 420]*) over* $\square$ *satisfies* $\|\tilde B'\|_{1,\alpha_0;\square} \le (C_{13} + 3c_1)p(t)$ *wherever the Hölder clause on* $\tilde B$ *holds, i.e. on pairs of points at depth* $\ge R_{\mathrm{Ba}} + 2$ *in* Ω;

(e) *at* $r_\square = r^\flat$, C_{13} *is a constant and* $\tilde B' = O(p(t))$, *the size class of* $\tilde A$; *at the printed* $r_\square = r(t)$, $|\tilde B'| \le c_1 d(\frac{3}{2} R(1 + \ln t^{-1})^\varrho + M + 1)p(t)$, *which is* $\le C(R/b_0 + 1)p(t)^2$ *uniformly in* $t \le 1$ *if and only if* $\varrho \le \bar p$.

Proof. A path inside $\square$. Fix a bond $(x, x+\eta e_\mu) \subset \square$. Go from c to x by changing one coordinate at a time, $\nu = 1, \dots, d$, in steps of $\pm\eta$; every intermediate site z has z_ν between c_ν and x_ν for the coordinates already moved and equal to c_ν or x_ν otherwise. $\square$ is a product of intervals of sites (a cube that does not wrap, (P-ii)); both $c + \eta e_\mu$ and $x + \eta e_\mu$ lie in $\square$, so $\max(c_\mu, x_\mu) + \eta$ is at most the last site coordinate of $\square$ in direction μ, and $z + \eta e_\mu \in \square$ for every intermediate z. Hence every pair of bonds $(z, z+\eta e_\mu)$, $(z + \eta e_\nu, z + \eta e_\nu + \eta e_\mu)$ used below lies in $\square \subset \Omega$, where the hypothesis on $\partial^\eta \tilde B$ holds.

(a) Telescoping. $\tilde B_\mu(x) - \tilde B_\mu(c) = \sum_{\text{steps}} \pm\eta\, \partial^\eta_\nu \tilde B_\mu(z)$ over $|x-c|_1/\eta$ steps, each bounded by $\eta c_1 p(t)$. So $|\tilde B'_\mu(x)| \le c_1 p(t)|x-c|_1 \le c_1 p(t)\, d(\frac{1}{2}\,\mathrm{side}(\square) + \eta) \le C_{13} p(t)$, using $\frac{1}{2}\,\mathrm{side}(\square) \le \frac{3}{2} r_\square + M$ (P-i) and $\eta \le 1$.

(b) B_0 is constant, so $\partial^\eta B_0 = 0$.

(c) $\tilde B'(\Gamma) = \sum_{b \subset \Gamma} \eta \tilde B'_b$ has $|\Gamma|_\eta/\eta$ terms, each bounded by $\eta C_{13} p(t)$ by (a).

(d) The norm is the sum of the sup (a), the derivative bound (b) and the Hölder bound: for $|x - x'| \le 1$ the quotient is $\le c_1 p(t)$ by (b) and the hypothesis; for $|x - x'| > 1$ it is $\le 2 \sup |\partial^\eta \tilde B'| \le 2 c_1 p(t)$.

(e) Insert $r_\square$. At the printed cube, $r(t)p(t) = (R/b_0)(1 + \ln t^{-1})^{\varrho - \bar p} p(t)^2$, which is $O(p(t)^2)$ uniformly in $t \le 1$ exactly when $\varrho \le \bar p$. $\square$

Remark 3.7. (a) *The inequality "$O(rp) \le O(p^2)$".* [2, (2.3), (2.7)] fix only $\bar p > 2$ and $\varrho > 1$; the printed inequality needs $\varrho \le \bar p$, which holds for the values used in the main paper ($\varrho = 2$, $\bar p = 6$) and is not needed at the constant side, where $\tilde B' = O(p)$ by (e).

(b) *Where the Hölder clause is used.* The norm (d) is needed at the vertices of the $\tilde B'$-expansion (section 3.5). At the vertices of the components containing an old vertex the depth is $\ge r_\square$ (lemma 3.14), and the clause holds there when $r^\flat \ge R_{\mathrm{Ba}} + 2$. At the vertices of the chains without an old vertex (S16, the fourth part), which may lie anywhere in $\square$, a point $z \in \square$ has depth $\ge \rho - D_\square$, with ρ the depth of the old vertices and $D_\square := \frac{3}{2} r_\square + 2M + l_N$; the main paper's cube step places Q_0 at depth $> D^\flat = \frac{3}{2} r^\flat + 2M + l^\sharp$, so $\mathrm{depth}(z) > l^\sharp - l_N = 4V^\sharp_{\mathrm{lab}}$ (section 3.6). The parameter row $R \ge R_{\mathrm{flat}} + R_{\mathrm{Ba}} + 2$ of the main paper (R_{flat} the depth parameter of its cube step) makes every point of $\square$ lie at depth $\ge R_{\mathrm{Ba}} + 2$ in Ω, so (d) holds on all of $\square$; it is a lower bound on R of Bałaban's kind, fixed after R_{Ba} and $l^\sharp$. The chain step of the fourth part in fact uses only $\sup|\tilde B'|$ and $\sup|\partial^\eta \tilde B'|$ at the chain vertices, so the row is a convenience there.

(c) *On a torus* the same proof works for a cube $\square$ that does not wrap (P-ii), so that the coordinate path is defined.

(d) Nothing in the lemma depends on k except through c_1, whose k-uniformity is the input named in the conventions.

3.4. **S14, I: the expansion of the vertices and of the propagator in $\tilde B'$.** [3, p. 433]: "We have $\tilde B = B_0 + \tilde B'$, and we expand in $\tilde B'$ the expressions connected with each renormalized class of graphs using the formulas (I.3.14), (I.3.44), and (I.3.45). After this expansion we get a sum of expressions corresponding to some new big set of graphs, which can be decomposed into a sum of renormalized classes." The consumed form has three parts: (i) the expression of a class after the cube split equals a finite sum of expressions of graphs with scalar lines $G_k(\square, B_0)$, old

vertices of the same types and localisations with B_0 in place of $\tilde B$ and extra external $\tilde B'$-legs, new vertices of the types [3, (1.8)–(1.11)] and [3, (1.13)–(1.15)] "with external vector field $\tilde B'$ only", at most N_B of them on each old line, each with two scalar legs, no A'-leg and a factor e_k per $\tilde B'$-leg, and remainder operators treated as external fields; (ii) the number of terms is bounded in terms of $\bar n$, N_B, d only, and the new factors obey lemma 3.6; (iii) the terms can be regrouped into renormalised classes. This subsection proves the two identities behind (i); section 3.5 proves (i)–(ii) and (iii).

The Taylor remainder. For z in the Lie algebra spanned by q (real multiples of q, antisymmetric) put

$$R_{\bar n+1}(z) := (\bar n+1)\int_0^1 (1-s)^{\bar n} e^{sz}\,ds, \qquad \text{so that} \qquad e^z = \sum_{j=0}^{\bar n}\frac{z^j}{j!} + \frac{z^{\bar n+1}}{(\bar n+1)!}R_{\bar n+1}(z)$$

(Taylor's formula with integral remainder); e^{sz} is orthogonal, so $\|R_{\bar n+1}(z)\| \le (\bar n+1)\int_0^1(1-s)^{\bar n}ds = 1$. Define, for a bond value A and a contour value $A(\Gamma)$,

$$F_{1,k}(A) := \eta^{-1}\big(U(A)-1\big) = \sum_{j=1}^{\bar n}\eta^{j-1}\frac{(qe_kA)^j}{j!} + \eta^{\bar n}\frac{(qe_kA)^{\bar n+1}}{(\bar n+1)!}R_{\bar n+1}(\eta q e_k A), \qquad U(A(\Gamma))-1 = \sum_{j=1}^{\bar n}\frac{(qe_kA(\Gamma))^j}{j!}+$$

These are [1, (3.14)–(3.15)] (with the sign convention of this supplement; every identity below holds for either sign of the exponent). $F_{1,k}(A)$ is a sum of $\bar n+1$ terms, the jth homogeneous of degree j in e_kA with the factor η^{j-1}, and the last carrying the matrix $R_{\bar n+1}$ of norm ≤ 1.

Lemma 3.8 (the vertex factors)**.** *On $\square$, with $\tilde B = B_0 + \tilde B'$ as in lemma 3.6, for every bond $b \subset \square$ and contour $\Gamma \subset \square$:*

(a) $U(\tilde B_b) = U(B_{0,b})U(\tilde B'_b)$ *and* $U(\tilde B(\Gamma)) = U(B_0(\Gamma))U(\tilde B'(\Gamma))$*;*

(b) $U(\tilde B'_b) = 1 + \eta F_{1,k}(\tilde B'_b)$ *and* $U(\tilde B'(\Gamma)) = 1 + [U(\tilde B'(\Gamma)) - 1]$*, with the finite expansions displayed above;*

(c) $D^\eta_{\tilde B}\phi(b) = D^\eta_{B_0}\phi(b) + F_{1,k}(\tilde B'_b)U(B_{0,b})\phi(b_+)$*;*

(c′) *equivalently, in the form of* [1, (3.44)] *with* $(A,B) = (\tilde B', B_0)$*:* $D^\eta_{\tilde B}\phi(b) = U(\tilde B'_b)\big[D^\eta_{B_0}\phi(b) - F_{1,k}(-\tilde B'_b)\phi(b_-)\big]$*;*

(d) $Q_k(\tilde B)\phi(y) = Q_k(B_0)\phi(y) + F_{2,k}(\tilde B', B_0)\phi(y)$*, with* $F_{2,k}(A,B)\phi(y) := \sum_{x\in B^k(y)}\eta^d\,[U(A(\Gamma^{(k)}_{y,x}))-1]\,U(B(\Gamma^{(k)}_{y,x}))\phi(x)$*.*

Proof. (a) The U are exponentials of real multiples of the fixed matrix q; they commute and multiply, and $A \mapsto A(\Gamma)$ is additive. (b) is the definition of $F_{1,k}$ and the Taylor formula. (c) Insert (a)–(b) into $D^\eta_{\tilde B}\phi(b) = \eta^{-1}(U(\tilde B_b)\phi(b_+) - \phi(b_-))$: $U(\tilde B_b)\phi(b_+) = U(B_{0,b})\phi(b_+) + \eta F_{1,k}(\tilde B'_b)U(B_{0,b})\phi(b_+)$. (c′) Write $U(\tilde B_b) = U(\tilde B'_b)U(B_{0,b})$ and $\phi(b_-) = U(\tilde B'_b)U(-\tilde B'_b)\phi(b_-)$ with $U(-\tilde B'_b) = 1 + \eta F_{1,k}(-\tilde B'_b)$; then $\eta^{-1}(U(\tilde B_b)\phi(b_+) - \phi(b_-)) = U(\tilde B'_b)\eta^{-1}\big(U(B_{0,b})\phi(b_+) - \phi(b_-) - \eta F_{1,k}(-\tilde B'_b)\phi(b_-)\big)$. (d) Insert (a) into the definition of $Q_k(\tilde B)$. □

Lemma 3.9 (the propagator expansion and the vertex operators)**.** *Let $H_\square(A) = -\Delta^{\eta,N}_{A,\square} + m^2t^2 + a_kP_k(A)$ be the Neumann operator on the sites of $\square$ and $G_k(\square,A) = H_\square(A)^{-1}$; both $H_\square(\tilde B)$ and $H_\square(B_0)$ are invertible since $m^2t^2 > 0$ (the identities below need nothing more; the uniform lower bound* [4, (1.8)] *enters only the estimates of the fourth part). Put $G_0 := G_k(\square, B_0)$, $D_0 := D^\eta_{B_0}$, $Q_0 := Q_k(B_0)$, $(F\phi)(b) := F_{1,k}(-\tilde B'_b)\phi(b_-)$, $F_2 := F_{2,k}(\tilde B', B_0)$, and $V_k := H_\square(B_0) - H_\square(\tilde B)$. Then:*

(a) *(*[1, (3.44)] *with $(A,B) = (\tilde B', B_0)$.) $V_k = F^*D_0 + D_0^*F - F^*F - a_k\big[F_2^*Q_0 + Q_0^*F_2 + F_2^*F_2\big]$, adjoints with respect to the η^d-weighted inner products on sites and bonds of $\square$.*

(b) *(*[1, (3.45)]*.) For every $N \ge 0$, $G_k(\square,\tilde B) = \sum_{m=0}^N G_0(V_kG_0)^m + \mathcal{R}_N$ with $\mathcal{R}_N := G_0(V_kG_0)^N V_kG_k(\square,\tilde B)$.*

(b′) *The two-sided form, for $n_1, n_2 \ge 1$:*

$$G_k(\square,\tilde B) = \sum_{m<n_1} G_0(V_kG_0)^m + \sum_{m'<n_2}(G_0V_k)^{n_1}G_0(V_kG_0)^{m'} + (G_0V_k)^{n_1}G_k(\square,\tilde B)(V_kG_0)^{n_2}.$$

Its last term is the analogue on $\square$ *of the kernel* [3, (1.16)]*; its main terms have at most* $N_B := n_1 + n_2 - 1$ *insertions of* V_k*.*

(c) *Inserting the finite expansions of* $F_{1,k}$ *and* $U(\tilde B'(\Gamma)) - 1$ *into (a) writes* V_k *as a sum of at most* $6(\bar n + 1)^2$ vertex operators*, each a bilinear form in* (ϕ, ψ) *with exactly two scalar legs and no* A'*-leg, every factor* $\tilde B'$ *accompanied by* e_k*,* B_0 *standing where* $\tilde B$ *stood in* [3, (1.8)–(1.15)] *and* $\tilde B'$ *where* $\tilde A$ *stood:*

- *the terms of* F^*D_0*,* D_0^*F *are* $\sum_b \eta^d \big[D_0\psi(b)\cdot \eta^{j-1}\frac{(-qe_k\tilde B'_b)^j}{j!}\phi(b_-)\big]$ *and their adjoints,* $1 \le j \le \bar n$*, with the R-term for* $j = \bar n + 1$*: the forms* [3, (1.8)] *at* $n = 0$*,* $n' = j$*, and* [3, (1.9)]*;*
- *the terms of* F^*F *are* $\sum_b \eta^d\eta^{j+j'-2}(e_k\tilde B'_b)^{j+j'}\big[\psi(b_-)\cdot c_{jj'}q^{j+j'}\phi(b_-)\big]$ *with scalar coefficients* $c_{jj'}$*: the forms* [3, (1.10)]*, and* [3, (1.11)] *when an R-factor occurs.* The terms with $j + j'$ odd cancel as operators*: the* (j, j') *and* (j', j) *terms are* $(-1)^j\frac{(z)^{j+j'}}{j!j'!}$ *and* $(-1)^{j'}\frac{(z)^{j+j'}}{j!j'!}$ *with* $z = -\eta qe_k\tilde B'_b$*, of opposite signs when* $j + j'$ *is odd. Equivalently* $F_{1,k}(A)^TF_{1,k}(A) = \eta^{-2}(2 - e^z - e^{-z})$ *with* $z = \eta qe_kA$*, which is even in* z*. This is the parity condition "*$n + n'$ *even" of* [3, (1.10)]*; the terms with an R-factor carry no parity, as* [3, (1.11)] *carries none. A single odd* (j, j') *term is a nonzero operator, so the cancellation is between the two orderings, not termwise;*
- *the terms with* F_2 *are block sums* $\sum_y$ *of products of a block factor* [3, (1.13)] *at* B_0 *with a factor* [3, (1.14)] *or* [3, (1.15)] *at* $n = 0$*, and of two such factors: the products* [3, (1.18)].

The η*-powers per leg are those of* [3, (1.8)–(1.15)]*:* η^{j-1} *at a term of type* [3, (1.8)] *and* $\eta^{j+j'-2}$ *at a term of type* [3, (1.10)]*.*

Proof. (a) By lemma 3.8(c′) and (d), as operators on the fields of $\square$, $D^\eta_{\tilde B} = U(\tilde B')(D_0 - F)$ with $U(\tilde B')$ the bondwise orthogonal multiplication by $U(\tilde B'_b)$, and $Q_k(\tilde B) = Q_0 + F_2$. Hence $(D^\eta_{\tilde B})^*D^\eta_{\tilde B} = (D_0-F)^*(D_0-F) = D_0^*D_0 - D_0^*F - F^*D_0 + F^*F$ and $P_k(\tilde B) = (Q_0+F_2)^*(Q_0+F_2)$. Subtracting $H_\square(\tilde B)$ from $H_\square(B_0)$, the mass terms cancel and V_k is the stated bracket. (The factorisation of lemma 3.8(c), $D^\eta_{\tilde B} = D_0 + \mathfrak F$ with $(\mathfrak F\phi)(b) := F_{1,k}(\tilde B'_b)U(B_{0,b})\phi(b_+)$, gives the different exact form $V_k = -[D_0^*\mathfrak F + \mathfrak F^*D_0 + \mathfrak F^*\mathfrak F] - a_k[F_2^*Q_0 + Q_0^*F_2 + F_2^*F_2]$; the two brackets are equal as operators but not term by term. The form (a) is the one whose terms are the printed vertices.)

(b) $G_0^{-1} - G_k(\square, \tilde B)^{-1} = V_k$ gives the second resolvent identity $G_k(\square, \tilde B) = G_0 + G_0V_kG_k(\square, \tilde B)$; iterate it N times.

(b′) Apply the identity on the other side, $G_k(\square, \tilde B) = G_0 + G_k(\square, \tilde B)V_kG_0$, n_2 times to the factor $G_k(\square, \tilde B)$ of the remainder of (b) at $N = n_1 - 1$, i.e. to $(G_0V_k)^{n_1}G_k(\square, \tilde B)$.

(c) Distribute (a) over the $\bar n+1$ terms of F and of F_2: at most $2(\bar n+1)+(\bar n+1)^2$ terms from the three D-terms and the same from the three Q-terms, at most $6(\bar n+1)^2$ in all. Each term has one ϕ and one ψ leg (two scalar legs) and no A'. The forms are read off: $(F\phi)(b) = F_{1,k}(-\tilde B'_b)\phi(b_-)$ has the undifferentiated leg at b_- and no $U(B_0)$ factor, so F^*D_0 pairs $D_0\psi(b)$ with $q^j\phi(b_-)$, the printed shape "$[(D^\eta_{\tilde B}\phi')(b)\cdot q^{n+n'}\phi'(b_-)]$" of [3, p. 413]; F^*F pairs $\psi(b_-)$ with $\phi(b_-)$ through $F_{1,k}(-\tilde B'_b)^TF_{1,k}(-\tilde B'_b)$, whose (j, j') term is $\eta^{j+j'-2}\frac{((-z)^T)^j(-z)^{j'}}{j!j'!}$ with $z = \eta qe_k\tilde B'_b$, and $(q^T)^j = (-1)^jq^j$ gives the parity statement. The F_2 terms are the definition of $F_{2,k}$ expanded. $\square$

Remark 3.10. The identities of this subsection are exact finite-dimensional identities, valid for any invertible $H_\square(\tilde B)$, $H_\square(B_0)$ and any bond field $\tilde B'$; the regularity of lemma 3.6 is used only in lemma 3.11(v) and in the estimates of the fourth part. [13, (3.47)–(3.53)] prints the same expansion on the torus.

3.5. **S14, II: the expanded expression and its classes.**

Lemma 3.11 (the expanded expression)**.** *Let* $\mathcal C$ *be a connected component of a class after the cube split, with all scalar lines* $G_k(\square, \tilde B)$ *(P-iii) and all old vertices localised in the cube* $Q_0 \subset \square_1$ *of corollary 3.4(c). Expand every scalar line by lemma 3.9(*b′*) with fixed* $n_1, n_2 \ge 2$ *(the orders*

for which the last term of (b′) is a regular kernel of both variables, as in the fourth part), so with at most $N_B = n_1 + n_2 - 1$ insertions in a main term, and every old vertex by lemma 3.8: its dependence on $\tilde{B}$ is through $D^{\eta}_{\tilde{B}}$ in [3, (1.8)–(1.9)] *and through $U(\tilde{B}(\Gamma))$ in* [3, (1.13)–(1.15)], *while* [3, (1.6), (1.7), (1.10), (1.11)] *and the counterterm vertices do not depend on $\tilde{B}$. Then*

$$E(\mathcal{C}) = \sum_{G\in\mathcal{C}} \sum_{\tau\in T(G)} E(G_\tau),$$

a finite sum, where a pattern *τ chooses, for each scalar line, one main term of (b′) together with one vertex operator of lemma 3.9(c) per insertion, or the remainder term, and for each old vertex one term of its expansion. Each G_τ has:*

(i) *scalar lines $G_k(\square, B_0)$ and vector lines unchanged (they do not depend on $\tilde{B}$); the external fields unchanged; each line with the remainder choice replaced by a pair of external scalar legs carrying the two-variable field $(G_0V_k)^{n_1}G_k(\square,\tilde{B})(V_kG_0)^{n_2}$;*

(ii) *old vertices of the same type and the same localisation box Q_v, with B_0 in place of $\tilde{B}$, and extra $\tilde{B}'$-legs at the vertex's own bond or contour, inside Q_v, so that no relocalisation is needed. "Same type" is meant collectively, as* [3, p. 433] *says it ("Vertices are now of the same type (1.6)–(1.11) and (1.13)–(1.15), but with B_0 instead of $\tilde{B}$ and with the new external vector field $\tilde{B}'$ beside the old one"): the term $F_{1,k}(\tilde{B}'_b)U(B_{0,b})\phi(b_+)\cdot q^m\phi(b_-)$ of an expanded* [3, (1.8)] *or* [3, (1.9)] *vertex has no derivative leg and $n'+j$ vector legs, i.e. the shape of* [3, (1.10)] *or* [3, (1.11)]*; its degree is the old one plus j;*

(iii) *new vertices, at most N_B on each old scalar line, each of the form of lemma 3.9(c): exactly two internal lines, both on the chain of that old line, no A'-leg, lying on bonds and blocks of $\square$ (their label boxes may leave $\square$ by at most 1). An old scalar line may have one old end and one external scalar leg; its chain then has a single old end;*

(iv) *$|T(G)| \le \big[(N_B+2)(6(\bar{n}+1)^2)^{N_B}\big]^{L(\bar{n})}\,(2(\bar{n}+1))^{2dV_{\mathrm{mer}}(\bar{n})}$, with $L(\bar{n}) \le 2V_{\mathrm{mer}}(\bar{n})$ the maximal number of scalar lines: a bound in terms of $\bar{n}$, N_B, d only;*

(v) *every $\tilde{B}'$-factor is $e_k\tilde{B}'_b$ or $e_k\tilde{B}'(\Gamma)$, with $|e_k\tilde{B}'_b| \le e_kC_{13}p(t)$ and $|e_k\tilde{B}'(\Gamma)| \le e_kdC_{13}p(t)$ (lemma 3.6); the R-factors have norm ≤ 1;*

(vi) *$V_{\mathrm{mer}}(G_\tau) \le V_{\mathrm{mer}}(\bar{n}) + N_BL(\bar{n}) \le V_{\mathrm{mer}}(\bar{n})(1+2N_B)$; the label count after the expansion is bounded by a function $V^{\sharp}_{\mathrm{lab}}(\bar{n})$ of $\bar{n}$ (and the fixed N_B) only, since each new vertex carries at most $\bar{n}+1$ labels.*

Proof. $E(G)$ is multilinear in the factors that depend on $\tilde{B}$: each scalar line, and each $D^{\eta}_{\tilde{B}}$ or $U(\tilde{B}(\Gamma))$ at an old vertex. Replace each by its finite expansion (lemmas 3.8 and 3.9) and distribute; this is the sum over τ. (i)–(iii) read off the terms: an inserted vertex operator of lemma 3.9(c) lies on a bond or block of $\square$ (the operators act on fields of $\square$), its two scalar legs are joined to its neighbours on the chain, and its label boxes, of side ≤ 2 around a point of $\square$, leave $\square$ by at most 1. (iv) Per scalar line there are $n_1+n_2 = N_B+1$ main terms and one remainder, and each insertion has at most $6(\bar{n}+1)^2$ vertex operators; per old vertex at most $2d$ bond or contour factors of $\tilde{B}$-type, each with at most $\bar{n}+2 \le 2(\bar{n}+1)$ terms; a graph has at most $4V_{\mathrm{mer}}/2$ scalar lines. (v) is lemma 3.6(a), (c) with $|\Gamma|_\eta \le d$. (vi) Each line gains at most N_B vertices. □

The regrouping into renormalised classes. The class rules of [3, Sect. 2] (section 2.2): divergent graphs ($D \le 0$) are built of [3, (1.6)–(1.8), (1.10)] only, and a graph with an R-vertex, a block vertex [3, (1.13)–(1.15)], a [3, (1.7)] vertex with finite counterterm or more than four external legs has $D > 0$ [3, Cor. 2.3]; the two-scalar-leg and $\phi\phi A$ classes are "the smallest sets of graphs, symmetric with respect to permutation of external scalar field legs and with each divergent subgraph renormalized" [3, p. 429], two-scalar-leg classes with their mass counterterm [3, p. 430]; the $2A$ classes contain "all graphs connected with these graphs by Ward–Takahashi identities", or, "if we refrain from taking the smallest classes", "all possible connected graphs of the considered type … of a fixed order" [3, p. 431]; $1A$ and $3A$ graphs and differentiated two-scalar-leg graphs "do not introduce any new divergences" [3, p. 431]; vacuum classes carry their vacuum counterterms [3, p. 432]; and "for an arbitrary graph G we form a class G_{ren} containing

this graph adding all graphs necessary to renormalize all divergent proper subgraphs. Of course localizations in these graphs should be in agreement" [3, p. 432]. For $G \in \mathcal{C}$, a pattern τ and a subgraph S of G, say τ *touches* S if τ has an insertion on an internal scalar line of S or a nonzero $\tilde{B}'$-order at a vertex of S; S_τ is the corresponding subgraph of G_τ.

Lemma 3.12 (the expanded class is a sum of renormalised classes)**.** *Let $\mathcal{C}$ be a per-label class (every member has the label multiset $\mathcal{B}$) at background $\tilde{B}$ on $\square$.*

(a) Untouched subgraphs. *Let S be a divergent proper subgraph of $G \in \mathcal{C}$ and $G' \in \mathcal{C}$ a graph that differs from G only inside S (the counterterm graph G/S, or a leg-exchange or Ward partner inside S). The patterns of G not touching S correspond bijectively to the patterns of G' trivial inside S, and $(G_\tau)' = (G')_\tau$. So inside G_τ the subgraph $S_\tau = S$ is renormalised by terms of the expansion of $\mathcal{C}$.*

(b) Touched subgraphs change type. *If τ touches S, then S_τ has $j \geq 1$ more external vector legs than S and $D(S_\tau) \geq D(S) + j$; S_τ is convergent if τ puts an R-term or an F_2-term inside S. The only partner sets a touched divergent S_τ can require are: (b1) S of two-scalar-leg type with $j = 1$: the partner with the two external scalar legs of S_τ exchanged; (b2) S of type $1A$ with $j = 1$, or a vacuum component with $j = 2$: the Ward class of type $2A$; (b3) the four-leg graph* [3, (2.4)]*, handled by the summation by parts of section 1.6. Touched $1A$ and $3A$ subgraphs need nothing.*

(c) Closure. *The per-label class $\mathcal{C}$ is closed under the exchange of the two external scalar legs of any two-scalar-leg or $\phi\phi A$ subgraph, the whole graph included (so a touched two-scalar-leg graph, now of type $\phi\phi A$ and degree 0, finds its mirror partner in the expansion of $\mathcal{C}$), and the first (resp. second) $\tilde{B}'$-Taylor coefficient of the per-label $1A$ (resp. vacuum) class of an order is the per-label $2A$ class of that order with one (resp. two) external legs equal to $\tilde{B}'$, with the weights of the Taylor coefficients. Hence the partners of (b1)–(b2) are terms of the expansion of $\mathcal{C}$, and $\sum_{G,\tau} E(G_\tau)$ is a sum of per-label renormalised classes at background B_0, with $\tilde{B}'$ among the external fields.*

Proof. (a) Counterterm vertices [3, (1.26)–(1.31)] and the vertices [3, (1.6), (1.7), (1.10), (1.11)] do not depend on $\tilde{B}$. A pattern not touching S consists of choices on the lines and vertices outside S, which are the same data for G and G'; the expansion acts factor by factor (lemma 3.11), so the two expansions agree outside S, and the labels of the new vertices are fixed by their own bonds and blocks, which also agree.

(b) The vertex degrees of [3, p. 423] with the definition [3, (2.2)]: a vertex of type [3, (1.8)] or [3, (1.10)] has degree $n(4-d)/2 + n'$, so each added $\tilde{B}'$-leg raises the degree of the subgraph containing it by 1, and an inserted vertex of lemma 3.9(c) has $n = 0$, $n' = j \geq 1$ and adds j to the degree of the line it sits on (two legs of dimension $-\frac{d-2}{2}$ each and one new line, net j); an R-term or a block factor inside S makes $D(S_\tau) > 0$ by [3, Cor. 2.3] (lemma 2.9). The list of divergent types with at most four external legs and their partner sets is [3, pp. 429–431] (corollary 2.11): a two-scalar-leg subgraph of degree -1 touched once becomes $\phi\phi A$ of degree 0, whose class is its symmetrisation; a $1A$ subgraph of degree -1 (resp. a vacuum component of degree -2) touched once (resp. twice) becomes $2A$ of degree 0, whose class is the Ward class; a four-scalar-leg subgraph touched becomes [3, (2.4)] with a vector leg, of degree ≥ 1, and is handled by lemma 1.26; everything else touched has $D > 0$.

(c) The two closure statements are the two lemmas on the class family in the fourth part of this supplement (the exchange closure of per-label classes, which is a bijection of Wick pairings preserving orders and labels, and the identity of generating functions $\partial/\partial\tilde{B}' = \partial/\partial X$ for the Taylor coefficients of log of the Gaussian integral at background B_0, the vertices and lines depending on $\tilde{B}'$ and the other vector legs X only through $B_0 + \tilde{B}' + X$). Given them: for (b1), the exchange acts on the external connections of S and the insertion on its internal lines, so $(S_\tau)^\pi = (S^\pi)_\tau$ with $S^\pi \subset G^\pi \in \mathcal{C}$; for (b2), the $2A$ Ward class at the order of S_τ is the set of all single (resp. double) $\tilde{B}'$-attachments, by line insertion or vertex expansion, to all graphs of S's type and labels, which are exactly the touched terms of the expansion of $\mathcal{C}$. Together with (a) every divergent subgraph of every G_τ is renormalised, or completed to its Ward class, inside

$\sum_{G,\tau} E(G_\tau)$, and the labels of the partners agree, so the sum is a sum of per-label renormalised classes. □

Remark 3.13. (a) The sentence "which can be decomposed into a sum of renormalized classes" of [3, p. 433] is a closure property of the class family under differentiation in the background; it holds for per-label classes because these are complete at fixed labels ("if we refrain from taking the smallest classes", [3, p. 431]), and [3, Prop. 1] for these classes is all that its consumers use, since they sum over all graphs of each order and over the localisations ([3, p. 421]: "Proposition I.3.1 is completely obvious if we sum over localizations"). (b) The regrouping and the renormalisation of [3, pp. 434–445] are applied in the fourth part to per-label classes, so the classes produced here are the ones they take.

3.6. **S15: old-vertex components lie in $\square_1$.** [3, p. 433]: "We repeat some of the operations done before, i.e. we localize new vertices in the way described in Chap. 1 and we include the operators (1.16) and propagators with localizations $\square(v)$, $\square(v')$ satisfying the condition $\mathrm{dist}(\square(v), \square(v')) \geq 1$ into the external fields. These operations give us again a sum of renormalized classes. If a class contains an old vertex coming from the expression before the expansion, then all vertices of this class are localized in $\square_1$." The consumed form: after the expansion (lemma 3.11), the localisation of the new vertices by (Loc), and the cut on $\square$ by the cluster rule on the label boxes of the expanded class (the estimate [3, (3.1)] on $\square$ at the background B_0 is the repetition of section 2.4 proved in the fourth part), every component that contains an old vertex has all its vertices in $\square_1$, at distance $\geq r_\square$ from $\square^c$; the components without an old vertex are the chains of S16.

Lemma 3.14 (old-vertex components)**.** *Assume (P-i), with the old vertices of all graphs of the class component in the cube Q_0 of side l_N concentric with $\square_1$ (corollary 3.4(c)); let the lines of every G_τ of lemma 3.11 be cut by the cluster rule on the label multiset $\mathcal{B}^\sharp$ of the expanded class (common to the expanded class by lemma 3.12), with $|\mathcal{B}^\sharp| \leq V^\sharp_{\mathrm{lab}}(\bar n)$. Put*

$$l^\sharp := l_N + 4V^\sharp_{\mathrm{lab}}(\bar n), \qquad Q_0^+ := \{x : \mathrm{dist}_\infty(x, Q_0) \leq 2V^\sharp_{\mathrm{lab}}(\bar n)\},$$

the cube of side $l^\sharp$ concentric with Q_0. If $r_\square \geq l^\sharp$, then for every G and every pattern τ:

(a) *every component of G_τ containing an old vertex has all its label boxes in Q_0^+; hence it lies in $\square_1$, at sup-distance $\geq \frac{1}{2}(3r_\square - l^\sharp) \geq r_\square$ from $\square^c$;*

(b) *every component of G_τ without an old vertex consists of new vertices of a single old scalar line: a sub-chain of at most N_B consecutive new vertices of that line, whose label boxes lie in one cluster, so of sup-diameter $\leq 2V^\sharp_{\mathrm{lab}}(\bar n)$; with the threshold applied to single lines (lemma 3.1) its diameter is also $\leq 2N_B$.*

Proof. Old vertices. They are the vertices of the pre-expansion component with their boxes unchanged (lemma 3.11(ii)), so their boxes lie in Q_0.

New vertices. By lemma 3.11(iii) a new vertex lies on the chain $u = n_0, n_1, \dots, n_m, n_{m+1} = w$ ($m \leq N_B$) of exactly one old scalar line $\ell = (u, w)$ with a main-term choice; its only internal lines are its two chain lines, its $\tilde B'$-legs are external and it has no A'-leg. If ℓ has one old end u and an external scalar leg, the chain is $u = n_0, \dots, n_m$ with u its only old end.

(a) Let K be a component containing an old vertex v. Every kept line joins two vertices of one cluster of $\mathcal{B}^\sharp$, and every vertex lies in the intersection of its label boxes, so all label boxes of K lie in the cluster K_v of v. The intersection graph of K_v is connected with $|K_v| \leq V^\sharp_{\mathrm{lab}}(\bar n)$ boxes, and one of them (a box of v) meets Q_0; by lemma 3.2(a), every point z of every box of K_v has $\mathrm{dist}_\infty(z, Q_v) \leq 2(|K_v| - 1)$, hence $\mathrm{dist}_\infty(z, Q_0) \leq 2V^\sharp_{\mathrm{lab}}(\bar n)$, i.e. $z \in Q_0^+$. If $r_\square \geq l^\sharp$, the cube Q_0^+ of side $l^\sharp$ concentric with Q_0 lies in the cube of side $r_\square$ concentric with Q_0, which lies in $\square_1$ (P-i; rounding outward only enlarges). For the distance to $\square^c$: $\square$ contains the cube of side $3r_\square$ concentric with Q_0 (P-i), so every point of Q_0^+ is at sup-distance $\geq \frac{1}{2}(3r_\square - l^\sharp) \geq r_\square$ from $\square^c$.

(b) A component without an old vertex consists of new vertices only; by the structure of the new vertices its lines are chain lines of one old line, so it is a sub-chain $n_i, \dots, n_j$ with at

most N_B vertices. Its label boxes lie in one cluster (all its lines are kept), so lemma 3.2(b) with $|K| \le V^\sharp_{\rm lab}(\bar n)$ boxes bounds its diameter by $2V^\sharp_{\rm lab}(\bar n)$. Under the single-line threshold, consecutive boxes of the sub-chain intersect (lemma 3.1), and lemma 3.2(b) on the at most N_B boxes gives $2N_B$. □

Remark 3.15. (a) *Per component.* The sentence "If a class contains an old vertex ..., then all vertices of this class are localized in $\square_1$" is read per connected component: a graph of the expanded class can have an old-vertex component and, beside it, a chain component anywhere in $\square$ (part (b)). Bałaban treats the components separately ([3, p. 433], before (3.3): "Considering each connected component of $G_{\rm ren}$ separately we can assume additionally that $G_{\rm ren}$ is connected"), so the reading costs nothing. Part (a) is also what places the old-vertex components at distance $\ge r_\square$ from $\square^c$, where the bounds on $\delta G_k(\Omega, \square, \tilde B)$ and on the cube propagator of the fourth part are evaluated.

(b) *The parameter condition.* The side $l^\sharp = l_N + 4V^\sharp_{\rm lab}(\bar n)$ is "the side of the localisation cube of a connected renormalised class after the expansion" with which the main paper's cube step defines its constant, $r^\flat \ge 3l^\sharp + R_0 + 12$; so $r_\square = r^\flat \ge l^\sharp$ and the hypothesis of lemma 3.14 holds. It depends on $\bar n$ only: a chain of new vertices advances by at most 2 per vertex in the sup metric (the side of a label box), not by less than 1, which is why the side is $l_N + 4V^\sharp_{\rm lab}$ and not l_N plus the number of new vertices. At the printed cube, $r(t) \ge R$ and the condition is $R \ge l^\sharp$.

(c) *Counterterm graphs.* The counterterm graphs of the regrouped classes are members of the expanded class (lemma 3.12(a)), and lemma 3.14 holds per (G, τ) for every member; with the class-common cube Q_0 their counterterm vertices, which carry the labels of their anchors, lie in $Q_0 \subset Q_0^+$.

(d) *"These operations give us again a sum of renormalized classes."* This is the regrouping after the cut (the main paper's S10, in its cluster form), applied on $\square$ to the per-label classes of lemma 3.12: the clusters of $\mathcal{B}^\sharp$ are common to the expanded class, a touched two-scalar-leg subgraph and its mirror, and a touched $1A$ subgraph and its $2A$ completion, have the same labels, so partners are kept or cut together.

(e) *A second printed source.* [13, Sect. 3.4] reaches the same conclusion on the torus by a different device, a cut-off θ supported within $\frac{2}{3}p(L^k\varepsilon)$ of the inner cube ("The result is a sum of new graphs, all with vertices inside $\square'$"); his mechanism does not use the distance rule and is not needed here.

4. The sketched steps of Proposition 1, IV: the cube at zero field, the norms, the identities of the renormalisation, the further places and the classes after the expansion

This part contains the sketched steps of the proof of [3, Prop. 1, (1.33)] that follow the cube split of [3, p. 433] and the $\tilde B'$-expansion (S13–S15 of part III), the identities and the one-loop classes of the renormalisation step of [3, pp. 434–445] on which the renormalisation theorem of the main paper (Step 3(D_4) there) rests, the seven further sketched places F1–F7, and the closure of the classes under the $\tilde B'$-expansion (S14.4):

step	[3] page	printed object
S16	433	the classes without old vertex (chains): "the estimate (3.2) for them is a consequer
S17	433	gauge covariance of the expressions on the cube $\square$
S18	433	the gauge transformation removing the constant field B_0
S19	433	$G_k(\square,0) = G_k(0) + \delta G_k(\square, \eta\mathbb{Z}^d, 0)$, with δG_k an external field
S20	434	the norms [3, (1.32)] at $\tilde B = B_0$ and at $\tilde B$: "we get (3.2) exactly"
S21	434–445	the identities and the one-loop classes of the renormalisation (sections 4.6 and 4.7)
F1	433	the Ω_2 branch of the cube split: "[or $G_k(\Omega_2, \tilde B)$]"
F2	418–419, 433–445	torus propagators in the counterterms against $\eta\mathbb{Z}^d$ propagators in the classes
F3	433	"We repeat some of the operations done before": [3, (1.16)] and [3, (3.1)] on $\square$
F4	430–431	"we can get all necessary Ward–Takahashi identities"
F5	427	"the remaining powers of $L^j\eta$ … give us $(L^j\eta)^{D(G_1)}$"
F6	428	"In the case $v = v'$ we make the same operations"
F7	433	components separately; "We can extract also all coupling constants"
S14.4	433	"which can be decomposed into a sum of renormalized classes": the closure of the

The order is: the zero-field kernel bounds at every point of a box, the common tool of S16, S19, F2, F3 and S21 (sections 4.1 and 4.2); S16 (section 4.3); S17–S19 (section 4.4); S20 (section 4.5); the identities and one-loop classes of S21 (sections 4.6 and 4.7); F1–F7 (section 4.8); S14.4 (section 4.9). The renormalisation of the divergent classes beyond these identities (the two-loop remainder terms, the transport of legs across the subgraphs [3, (2.4)] and the induction of [3, pp. 444–445]) is proved in the main paper and is not repeated here.

Conventions for this part.

Scales, fields, propagators. As in section 1.1: $d = 3$ (the kernel bounds of section 4.1 hold for general d), $L = 5$, $\eta = L^{-k}$, $k \geq 1$, $t = L^k\varepsilon$, $e_k = e\,t^{1/2}$, and $s_i := L^i\eta$. Scalar fields take values in $\mathbb{R}^N$; $U(A_b) = \exp(e_k\eta q A_b)$ with q a fixed real antisymmetric $N \times N$ matrix, $\|q\| \leq 1$; all transporters commute and are orthogonal. $(D_A\phi)(b) = \eta^{-1}(U(A_b)\phi(b_+) - \phi(b_-))$; on a union X of unit blocks with Neumann conditions $\langle\phi, -\Delta^{\eta,N}_{A,X}\phi\rangle = \sum_{b\subset X}\eta^d|D_A\phi(b)|^2$, and $G_k(X,A) := (-\Delta^{\eta,N}_{A,X} + m_k^2 + a_kP_k(A))^{-1}$ with $P_k = Q_k^*Q_k$, $Q_k(A)$ the covariant unit-block average along the contours of [1, (2.1)–(2.2)], $0 \leq m_k^2 = m^2t^2 \leq c_m$ and $a_k \in [a(1-L^{-2}), a]$. Vertices are those of [3, (1.6)–(1.15)]; the vertices [3, (1.8)–(1.11)] carry $e_k^{n+n'}$ from the expansion of U, which fixes the factor e_k inside U. Degrees are those of [3, (2.1)–(2.3)] (section 2).

The cube. $\square \subset \Omega$ is the cube of the cube split of [3, p. 433] in the constant-side form of the main paper: a non-wrapping parallelepiped of the lift of the torus, a union of large blocks (side M), with $3r^\flat \leq \mathrm{side}(\square) \leq 3r^\flat + 2M$, containing the concentric cube $\square_1$ of side $r^\flat$ in which every old vertex is localised; so $\operatorname{diam}\square \leq \sqrt{3}(3r^\flat + 2M)$ and every point of $\square_1$ has depth $\geq r^\flat$ in $\square$. On $\square$ the background is $\tilde B = B_0 + \tilde B'$ with B_0 constant, and

$$\sigma_0 := \sup_{b\subset\square}|\tilde B'_b| \leq c\,(r^\flat + M)\,p(t), \qquad \sigma_1 := \sup_{b\subset\square}|\partial^\eta\tilde B'_b| = \sup_{\square}|\partial^\eta\tilde B| \leq c_1 p(t),$$

by the choice of B_0 in the cube step. Throughout we assume the cap

$$\text{(cap)} \qquad e_k(\sigma_0+\sigma_1)\,\big(1 + \operatorname{diam}\square\big) \leq 1,$$

which holds for $t \leq t_0$ since $e_k(\sigma_0+\sigma_1)(1+\sqrt{3}(3r^\flat+2M)) \leq C\,e\,t^{1/2}(1+\ln t^{-1})^{\bar p} \to 0$; it is a condition of the kind of [3, p. 412], fixed after L, $\bar n$, R_0 and $r^\flat$ and before ε_0. Only $e_k(\sigma_0+\sigma_1) \leq 1$ is used outside section 4.5.

Kernel notation. For a kernel $Y(x_1,x_2)$ on a lattice of spacing $\xi \in (0,1]$ and $S,T \subset \{1,2\}$, D_S applies one forward difference (any direction, bonds of the domain) in each variable x_i, $i \in S$, and Δ_T applies in each variable x_i, $i \in T \subset S$, a Hölder difference $Y(\dots,\tilde x_i,\dots) - Y(\dots,x_i,\dots)$ at a partner with $0 < |x_i - \tilde x_i| \leq 1$ (at zero field no transport is needed). Put $p(S,T) := d - 2 + |S| + \alpha|T|$ and $\rho := \min\{|u-w| : u \in \{x_1\}\cup\{\tilde x_1 \text{ if } 1\in T\},\ w \in \{x_2\}\cup\{\tilde x_2 \text{ if } 2\in T\}\}$; $h_\gamma(r) := 1$ for $\gamma < 0$, $1 + |\log r|$ for $\gamma = 0$, $r^{-\gamma}$ for $\gamma > 0$. In sections 4.1 and 4.2 distances are ℓ^1; elsewhere they are the sup-distances of section 1.1 ($|\cdot|_\infty \leq |\cdot| \leq |\cdot|_1 \leq d\,|\cdot|_\infty$, so the

choice affects constants only). A quotient over two points at distance > 1 is at most the sum of the two values, so every bound below is needed only at $|x_i - \tilde{x}_i| \le 1$; this is how the Hölder supremum of [3, (1.32)], which runs over all pairs of points, is reached from bounds at partners within distance 1: for $|x - x'| > 1$ the quotient is $\le 2 \sup |D \cdot|$, so the all-pairs supremum is $\le \max\{\text{local supremum}, 2 \sup |D \cdot|\}$.
The norm [3, (1.32)]. $\|f\|_{1,\alpha;A} := \sup|f| + \sup|D_A f| + \sup_{x \neq x'} |x - x'|^{-\alpha} |U(A(\Gamma_{x,x'}))D_A f(x') - D_A f(x)|$ for a function of one scalar slot, $\Gamma_{x,x'}$ a shortest contour; for kernels of several variables the maximal reading of section 1.1, slot by slot. Vector fields carry the norm at $A = 0$ [3, p. 420].
Inputs. Besides the inputs (P), (C), (H0) of section 1.1 and the lemmas of parts I–III, this part uses:

(i) **(A5)** the scale-uniform form of [4, Lemma 2.4] proved in the main paper: for $\xi = L^{-j}$, $j \ge 0$, $m^2 \ge 0$, $a \in [a_-, a_+]$, and X a rectangular parallelepiped of $\xi\mathbb{Z}^d$ with Neumann conditions, a periodic box or $\xi\mathbb{Z}^d$, each a union of unit blocks, the columns $G_j \mathbf{1}_{B(y)}$ of $G_j := (-\Delta_X^\xi + m^2 + aQ_j^* Q_j)^{-1}$ obey $|(G_j \mathbf{1}_{B(y)})(x)|$, $|D(G_j \mathbf{1}_{B(y)})(x)| \le c_0 e^{-\delta_0 |x-y|}$ and $|x - \tilde{x}|^{-\alpha} |D(G_j \mathbf{1}_{B(y)})(\tilde{x}) - D(G_j \mathbf{1}_{B(y)})(x)| \le c_1(\alpha) e^{-\delta_0 \min(|x-y|, |\tilde{x}-y|)}$ for $|x - \tilde{x}| \le 1$, with $c_0, c_1(\alpha), \delta_0$ depending only on $d, a_\pm, \alpha$. Its proof (reflection and periodisation of the full-lattice kernel) also gives the image identities used in lemma 4.15 and lemma 4.35: for a Neumann box X that is a union of unit blocks, $G_j(X, 0)(x, x') = \sum_\gamma G_j(\xi\mathbb{Z}^d, 0)(x, \gamma x')$ over the reflection group of the box, and for a periodic box the sum over its period lattice, both absolutely convergent.

(ii) **(C′)** the bound (C) of section 1.1 at every scale: for the unit-lattice block operators $C^{(j)}$ of lemma 4.1, $|C^{(j)}(y, y')| \le c_2 e^{-\theta_0 |y - y'|_\infty}$ with c_2, θ_0 depending on $d, L, a_\pm, a'$ only. The operators at scale j are the rescaled operators at scale k with the same unit-block structure, so (C) applies verbatim.

(iii) **(cube)** the cube split of the main paper (the replacement (a) of its block-factor-5 theorem): the properties of $\square$, $\square_1$, B_0, $\tilde{B}'$ listed above, and the fact that after the $\tilde{B}'$-expansion and the cut of separated lines every class component containing an old vertex is localised in $\square_1$ (S15).

(iv) **(expansion)** the $\tilde{B}'$-expansion of part III (S14): $G_k(\square, \tilde{B}) = \sum_{m=0}^{\bar{n}} (-1)^m (G_0 V)^m G_0 +$ remainders of the type [3, (1.16)], $G_0 := G_k(\square, B_0)$, $V := G_k(\square, \tilde{B})^{-1} - G_0^{-1}$ expanded in $\tilde{B}'$ into the vertices [3, (1.8)–(1.11)] ($n = 0$, $\tilde{A} \to \tilde{B}'$, $\tilde{B} \to B_0$) and the block pairs [3, (1.13)–(1.15)] ($A' \to 0$, $\tilde{A} \to \tilde{B}'$); the new vertices are localised by a partition of unity per $\tilde{B}'$-leg with supports in cubes of side 2 (block vertices by y), and the remainders and the lines between localisations at distance ≥ 1 become external fields.

4.1. **Zero-field kernel bounds at every point of a box, I: one scale and the cumulative propagators.** After the gauge-out of B_0 (S18) every scalar line of a class on the cube is a zero-field propagator on a Neumann box, on $\eta\mathbb{Z}^d$ or on the torus. The following bounds hold at *every* pair of points of the box, with no depth condition, which is what S16, S19, F2, F3 and the renormalisation step need. Distances in this and the next subsection are ℓ^1; $\alpha \in (0, 1)$; S, T, $p(S, T)$, ρ are as in the conventions of section 4.

Lemma 4.1 (one scale, all points)**.** *Fix $L \ge 3$ odd, $0 < a_- \le a_+$, $a' > 0$ and $\alpha \in (0, 1)$. For $j \ge 0$ let $\xi := L^{-j}$ and let X be* (N) *a Neumann parallelepiped of $\xi\mathbb{Z}^d$ that is a union of L-blocks of the unit lattice,* (P) *a periodic box $\xi\mathbb{Z}^d / K\mathbb{Z}^d$ with $K \in (L\mathbb{Z}_{\ge 1})^d$, or* (Z) *$\xi\mathbb{Z}^d$. Let $m^2 \ge 0$ and $a \in [a_-, a_+]$.*

- *For $j \ge 1$ put $\mathsf{T} := -\Delta_X^{\xi, N} + m^2 + aQ_j^* Q_j$ on $\ell^2(X, \xi^d)$, $G_j := \mathsf{T}^{-1}$, $\Delta^{(j)} := a - a^2 Q_j G_j Q_j^*$, $C^{(j)} := (\Delta^{(j)} + a' L^{-2} P)^{-1}$ on the unit lattice $X^{(j)}$ (P the projection onto functions constant on L-blocks), and $S_j := a^2 G_j Q_j^* C^{(j)} Q_j G_j$.*
- *For $j = 0$ ($\xi = 1$) put $S_0 := (-\Delta_X^{1, N} + m^2 + a' L^{-2} P)^{-1}$.*

Then there are $C_Z, \delta_Z > 0$, depending only on $(d, L, a_\pm, a', \alpha)$, such that for all j, X, m^2, a, x_1, x_2, S, T:

$$|\Delta_T D_S S_j(x_1, x_2)| \le C_Z e^{-\delta_Z \rho} \prod_{i \in T} |x_i - \tilde{x}_i|^\alpha. \tag{5}$$

Proof. Step 1 (the kernel, $j \ge 1$). $Q_j^* \delta_y = \mathbf{1}_{B(y)}$ and G_j is real symmetric, so $(Q_j G_j f)(y') = \langle G_j \mathbf{1}_{B(y')}, f \rangle$. Hence, with respect to the weight ξ^d,

$$S_j(x_1, x_2) = a^2 \sum_{y, y' \in X^{(j)}} (G_j \mathbf{1}_{B(y)})(x_1)\, C^{(j)}(y, y')\, (G_j \mathbf{1}_{B(y')})(x_2).$$

Each variable sits in its own factor, so D_S and Δ_T act factor by factor.

Step 2 (the factors). By (A5), on all three kinds of X, $|(G_j \mathbf{1}_{B(y)})(x)|$, $|D(G_j \mathbf{1}_{B(y)})(x)| \le c_0 e^{-\delta_0 |x-y|}$ and $|x - \tilde{x}|^{-\alpha} |D(G_j \mathbf{1}_{B(y)})(\tilde{x}) - D(G_j \mathbf{1}_{B(y)})(x)| \le c_1 e^{-\delta_0 \min(|x-y|, |\tilde{x}-y|)}$. By (C′), $|C^{(j)}(y, y')| \le c_2 e^{-\theta_0 |y-y'|_\infty} \le c_2 e^{-(\theta_0/d)|y-y'|}$.

Step 3 (the sum). Put $\delta := \min\{\delta_0, \theta_0/d\}$. For the points u, w that carry the factors (a partner where the Hölder clause is used), $|u - w| \le |u - y| + |y - y'| + |y' - w|$, so the product of the three exponentials is $\le e^{-(\delta/2)\rho} e^{-(\delta/2)(|u-y| + |y-y'| + |y'-w|)}$. The unit-lattice sums are bounded by $\sum_{y \in \mathbb{Z}^d} e^{-c|x-y|} \le (2/(1 - e^{-c}))^d$ for every real x (a minimum over two points costs a factor 2). This gives (5) with $\delta_Z := \delta/2$.

Step 4 ($j = 0$). On the unit lattice, with $\mathsf{T}_0 := -\Delta_X^{1,N} + m^2 + a' L^{-2} P$: for a single L-block Q with Neumann conditions the spectral gap of $-\Delta_Q^{1,N}$ above the constants is $4 \sin^2(\pi/2L) \ge 4L^{-2}$ (the one-dimensional free-end path on L sites has eigenvalues $2 - 2\cos(\pi m/L)$, $m = 0, \dots, L-1$), so $-\Delta_Q^{1,N} + a' L^{-2} P_Q \ge L^{-2} \min\{4, a'\}$; dropping the bonds between blocks only lowers the quadratic form, so $\mathsf{T}_0 \ge \gamma' := L^{-2} \min\{4, a'\}$ on X. The argument of lemma 1.10 with $Y = \mathsf{T}_0$, $v = m^2$ and P of range L (so that $|\rho(x) - \rho(x')| \le L$ on a block and $c(\beta) = 3\beta^2 \cosh \beta + a' L^{-2}(e^{\beta L} - 1)$) gives, for $\theta' > 0$ with $c(\theta') \le \gamma'/2$, $\|e^{\theta' \rho} \mathsf{T}_0^{-1} e^{-\theta' \rho}\| \le 2/\gamma'$ for every 1-Lipschitz ρ in the sup-metric; with $\rho = |\cdot - x'|_\infty$ and $\|\delta_x\| = 1$ this is $|S_0(x, x')| \le 2\gamma'^{-1} e^{-\theta' |x - x'|_\infty}$. A forward difference on the unit lattice is a difference of two neighbouring values, and two distinct partners are at distance ≥ 1, so every quotient is at most twice a supremum. This gives (5) at $j = 0$ with $\delta_Z \le \theta'/d$. □

Lemma 4.2 (geometric sums)**.** *For $p > 0$, $\delta > 0$, $\eta \in (0, 1]$, $s_i := L^i \eta \le 1$ and $\rho \ge 0$: $\sum_i s_i^{-p} e^{-\delta \rho / s_i} \le C_g(p, \delta, L)(\eta + \rho)^{-p}$.*

Proof. If $\rho \le \eta$ the sum is $\le \eta^{-p}/(1 - L^{-p}) \le 2^p (\eta + \rho)^{-p}/(1 - L^{-p})$. If $\rho > \eta$, the terms with $s_i \ge \rho$ sum to $\le \rho^{-p}/(1 - L^{-p})$. The terms with $s_i < \rho$ equal $\rho^{-p} u_i^p e^{-\delta u_i}$, where the $u_i = \rho/s_i > 1$ form a geometric sequence of ratio L; at most one u_i lies in each interval $[L^n, L^{n+1})$, $n \ge 0$, and $t^p e^{-\delta t} \le L^{(n+1)p} e^{-\delta L^n}$ there, so these terms sum to $\le \rho^{-p} \sum_{n \ge 0} L^{(n+1)p} e^{-\delta L^n}$. Finally $\rho^{-p} \le 2^p (\eta + \rho)^{-p}$. □

Corollary 4.3 (cumulative propagators at zero field, all points)**.** *Let $k \ge 1$, $\eta = L^{-k}$, $s_i := L^i \eta$, and let $X \subset \eta \mathbb{Z}^d$ be of type* (N), (P) *or* (Z) *and a union of unit blocks. Let $m^2 \ge 0$, $a_1 := a'$, $a_{i+1} := a' a_i / (a' L^{-2} + a_i)$ (so $a_i \in [a'(1 - L^{-2}), a']$), and $\mathsf{T}_i := -\Delta_X^{\eta,N} + m^2 + a_i s_i^{-2} P^{(i)}$, $P^{(i)}$ the projection onto functions constant on the blocks of side s_i, for $1 \le i \le k$, and also for $i = k + 1$ when X is a union of L-blocks of the unit lattice. Then:*

(i) *for $1 \le J \le k$, and for $J = k + 1$ under that hypothesis, $\mathsf{T}_J^{-1} = \sum_{i=0}^{J-1} G_{(i)}$, where $G_{(0)} := \mathsf{T}_1^{-1}$, $G_{(i)} := \mathsf{T}_{i+1}^{-1} - \mathsf{T}_i^{-1}$, and $G_{(i)}(x, x') = s_i^{2-d} S_i(x/s_i, x'/s_i)$ with S_i the kernel of lemma 4.1 at $j = i$ on X/s_i with mass $m^2 s_i^2$ and $a = a_i$;*

(ii) *for the same J and all S, T:*

$$|\Delta_T D_S \mathsf{T}_J^{-1}(x_1, x_2)| \le C_1 (\eta + \rho)^{-p(S,T)} e^{-\delta_Z \rho/2} \prod_{i \in T} |x_i - \tilde{x}_i|^\alpha, \tag{6}$$

with C_1 depending only on (d, L, a', α) (for $d = 3$, $p(S, T) > 0$).

In particular $G_k(X,0) = \mathsf{T}_k^{-1}$ ($s_k = 1$) at $a' = a$, and (i) is the zero-field case of the slice decomposition [3, (2.6)]. *The constants do not depend on k, J, X, $m^2 \geq 0$, nor on a' within a fixed range.*

Proof. Step 1 (rescaling). For $s > 0$ let $(U_s f)(y) := f(sy)$. If $A = U_s^{-1}\hat{A}U_s$, then $A(x,x') = s^{-d}\hat{A}(x/s, x'/s)$ for the kernels with respect to the respective lattice weights. Since $U_s(-\Delta^\eta)U_s^{-1} = s^{-2}(-\Delta^{\eta/s})$, $\mathsf{T}_i = U_{s_i}^{-1}s_i^{-2}\hat{\mathsf{T}}_i U_{s_i}$ with $\hat{\mathsf{T}}_i = -\Delta^{\xi_i} + m^2 s_i^2 + a_i\hat{P}$ on $\xi_i\mathbb{Z}^d$, $\xi_i = L^{-i}$, $\hat{P}$ the unit-block projection. Hence $\mathsf{T}_i^{-1}(x,x') = s_i^{2-d}\hat{\mathsf{T}}_i^{-1}(x/s_i, x'/s_i)$, and the same for T_{i+1} at the same rescaling, whose block term is $a_{i+1}L^{-2}\hat{P}_{(L)}$.

Step 2 (composition). In the units $s_i = 1$, the block inversion of the operator $\hat{\mathsf{T}}_i + a_{i+1}L^{-2}\hat{P}_{(L)}$, read in its $\phi\phi$-block (the inverse of the Schur complement in the block variable), gives $\hat{\mathsf{T}}_{i+1}^{-1} = \hat{\mathsf{T}}_i^{-1} + a_i^2\hat{\mathsf{T}}_i^{-1}Q^*C^{(i)}Q\hat{\mathsf{T}}_i^{-1}$ with $C^{(i)} = (a_i - a_i^2 Q\hat{\mathsf{T}}_i^{-1}Q^* + a'L^{-2}P)^{-1}$: this is the identity $(M^{-1})_{\phi\phi} = D^{-1} + D^{-1}B^*(A_\psi - BD^{-1}B^*)^{-1}BD^{-1}$ for $M = \begin{pmatrix} A_\psi & -B \\ -B^* & D \end{pmatrix}$ with $D = \hat{\mathsf{T}}_i$, $B = a_iQ$, $A_\psi = a_i + a'L^{-2}P$ (the quadratic form $\langle\phi, (-\Delta^{\xi_i} + m^2 s_i^2)\phi\rangle + a_i\|Q\phi - \psi\|^2 + a'L^{-2}\langle\psi, P\psi\rangle$ in (ϕ,ψ)). Its ϕ-block Schur complement is $D - B^*A_\psi^{-1}B = \hat{\mathsf{T}}_i - a_i^2 Q^*(a_i + a'L^{-2}P)^{-1}Q = -\Delta^{\xi_i} + m^2 s_i^2 + \frac{a_i a'L^{-2}}{a_i + a'L^{-2}}Q^*PQ$, since $Q^*(a_i + bP)^{-1}Q = a_i^{-1}Q^*(1-P)Q + (a_i+b)^{-1}Q^*PQ$ and $Q^*Q - Q^*(1-P)Q = Q^*PQ$; this is $\hat{\mathsf{T}}_{i+1}$ (block term $a_{i+1}L^{-2}\hat{P}_{(L)}$, $\hat{P}_{(L)} = Q^*PQ$) exactly when $a_{i+1} = a'a_i/(a'L^{-2} + a_i)$, the recursion which reproduces $a_k = a(1-L^{-2})/(1-L^{-2k})$ of [1, (2.15)] and is the zero-field form of [3, (1.27)]. So $G_{(i)} = s_i^{2-d}S_i(\cdot/s_i, \cdot/s_i)$, and the sum telescopes to (i). X/s_i is a union of unit blocks and of L-blocks of its unit lattice since $Ls_i = s_{i+1} \leq 1$, so lemma 4.1 applies. For $J = k+1$ the last slice is $i = k$ ($s_k = 1$, $X/s_k = X$), and lemma 4.1 at $j = k$ needs X to be a union of L-blocks of the unit lattice; this holds for $\xi\mathbb{Z}^d$, for periodic boxes with sides in $L\mathbb{Z}$, and for Neumann boxes that are unions of L-blocks. (Rescaling the statement at $J \leq k$ is not enough for this case: its exponential $e^{-\delta\rho_\eta/2}$ would become $e^{-\delta s\rho_\xi/2}$.)

Step 3 (bounds). Under the rescaling a difference costs s_i^{-1} and $|x-\tilde{x}|^\alpha = s_i^\alpha|y-\tilde{y}|^\alpha$. For $|x-\tilde{x}| \leq s_i$, (5) applies; for $s_i < |x-\tilde{x}| \leq 1$ bound the difference by two values, which is $\leq 2(|x-\tilde{x}|/s_i)^\alpha$ times a value. So $|\Delta_T D_S G_{(i)}| \leq 2^{|T|}C_Z s_i^{-p(S,T)}e^{-\delta_Z\rho/s_i}\prod|x_i - \tilde{x}_i|^\alpha$. Write $e^{-\delta_Z\rho/s_i} \leq e^{-\delta_Z\rho/2}e^{-\delta_Z\rho/(2s_i)}$ and apply lemma 4.2. □

4.2. **Zero-field kernel bounds at every point of a box, II: the free propagator and the remainder $\breve{G} - C^\xi$.**

Lemma 4.4 (free massive propagator smeared over a block)**.** *Let $\xi = L^{-J}$, $J \geq 0$, L odd, $\mu^2 \in [\frac{1}{2}, 2]$, and $C_\mu := (-\Delta^\xi + \mu^2)^{-1}$ on $\xi\mathbb{Z}^d$. Put $\delta_C := (88d)^{-1/2}$. Then for every unit block $B(y)$, every x, $\tilde{x}$ with $|x-\tilde{x}| \leq 1$ and every direction,*

$$|(C_\mu \mathbf{1}_{B(y)})(x)|,\ |D(C_\mu \mathbf{1}_{B(y)})(x)| \leq c_C e^{-\delta_C|x-y|}, \qquad |x-\tilde{x}|^{-\alpha}|D(C_\mu\mathbf{1}_{B(y)})(\tilde{x}) - D(C_\mu\mathbf{1}_{B(y)})(x)| \leq c_C(\alpha)e^{-\delta_C \min($$

with c_C, $c_C(\alpha)$ depending only on (d,α).

Proof. Let $N := 1/\xi = L^J$ (odd) and $B(y) = y + ([0,1)\cap\xi\mathbb{Z})^d$. With $\hat{f}(p) := \sum_x \xi^d f(x)e^{-ip\cdot x}$ on $B_\xi := [-\pi/\xi, \pi/\xi]^d$, the geometric sum $\sum_{n=0}^{N-1}\xi e^{-it\xi n} = e^{-i(1-\xi)t/2}g(t)$, $g(t) := \xi\sin(t/2)/\sin(\xi t/2)$, gives $\hat{\mathbf{1}}_{B(y)}(p) = e^{-ip\cdot y}u(p)$ with $u(p) := e^{-i(1-\xi)(p_1+\cdots+p_d)/2}\prod_\nu g(p_\nu)$. Hence, for $M \in \{1, \hat{\partial}_\nu, \hat{\partial}_\nu(e^{ip\cdot v'} - 1)/|v'|^\alpha\}$ ($\hat{\partial}_\nu(p) = (e^{i\xi p_\nu} - 1)/\xi$, $v' := \tilde{x} - x \in \xi\mathbb{Z}^d$, $0 < |v'| \leq 1$), the three quantities of the statement are

$$(2\pi)^{-d}\int_{B_\xi} e^{ip\cdot(x-y)}\,\frac{M(p)\,u(p)}{\Delta^\xi(p) + \mu^2}\,dp, \qquad \Delta^\xi(p) = \sum_\nu \frac{4}{\xi^2}\sin^2\frac{\xi p_\nu}{2}.$$

Step 1 (periodicity). Every factor is $(2\pi/\xi)$-periodic in each p_ν: $x - y$, $v' \in \xi\mathbb{Z}^d$; Δ^ξ and $\hat{\partial}$ are trigonometric in ξp; and u is periodic because N is odd: $g(t + 2\pi/\xi) = \xi(-1)^N\sin(t/2)/(-\sin(\xi t/2)) = g(t)$, and the phase gains $e^{-i\pi(N-1)} = 1$. The integrand is entire in each p_ν (g is entire).

Step 2 (the denominator on the strip). Let $p = q + i\sigma$, $q \in B_\xi$, $|\sigma|_\infty \leq \delta_C \leq \frac{1}{4}$. From $\sin^2(a+ib) - \sin^2 a = \sinh^2 b\,(\sin^2 a - \cos^2 a) + \frac{i}{2}\sin 2a \sinh 2b$ with $a = \xi q_\nu/2$, $b = \xi\sigma_\nu/2$, and

$\sinh x \le 1.13x$, $\sinh 2x \le 3.1x$ for $0 \le x \le \frac12$: $|\Delta_\nu^\xi(p) - \Delta_\nu^\xi(q)| \le 1.28\sigma_\nu^2 + 3.1|\sigma_\nu|\Delta_\nu^\xi(q)^{1/2}$. Summing over ν and using $\sum_\nu \Delta_\nu^{1/2} \le \sqrt{d}\,\Delta^{1/2}$ and $3.1\delta\sqrt{d}\,\Delta^{1/2} \le \frac14\Delta + 9.7d\delta^2$: $|\Delta^\xi(p) - \Delta^\xi(q)| \le \frac14\Delta^\xi(q) + 11d\delta_C^2 \le \frac14\Delta^\xi(q) + \frac18 \le \frac14(\Delta^\xi(q)+\mu^2)$. Hence $|\Delta^\xi(p)+\mu^2| \ge \frac34(\Delta^\xi(q)+\mu^2) \ge c\,(1+|q|)^2$, since $\Delta^\xi(q) \ge (4/\pi^2)|q|^2$ on B_ξ.

Step 3 (the numerator). For $z = t+i\sigma$, $|t| \le \pi/\xi$, $|\sigma| \le \frac14$: $|\sin(z/2)|^2 = \sin^2(t/2) + \sinh^2(\sigma/2)$ and $|\sin(\xi z/2)|^2 = \sin^2(\xi t/2) + \sinh^2(\xi\sigma/2) \ge (\xi t/\pi)^2 + (\xi\sigma/2)^2$. For $|t| \le \pi$ this gives $|g(z)|^2 \le \max\{(\pi/2)^2, 1.03\} \le 2.5$; for $|t| > \pi$, $|\sin(z/2)| \le \cosh\frac18 \le 1.01$ gives $|g(z)| \le 3.2/|t|$. In both cases $|g(z)| \le 10/(1+|t|)$. The phase of u has modulus $\le e^{d\delta_C/2} \le 1.2$, $|\hat\partial_\nu(p)| \le 1.3|p_\nu| \le 1.3(1+|q|)$, and $|e^{ip\cdot v'} - 1|/|v'|^\alpha \le C\min\{1, |q||v'|\}^\alpha |v'|^{-\alpha} + C|v'|^{1-\alpha} \le C(1+|q|)^\alpha$ on the strip. So $|M(p)| \le C(1+|q|)^{1+\alpha}$.

Step 4 (integrability). By Steps 2–3 the modulus of the integrand on the shifted contour is $\le e^{-\sigma\cdot(x-y)}\,C(d)\,(1+|q|)^{\alpha-1}\prod_\nu(1+|q_\nu|)^{-1} \le e^{-\sigma\cdot(x-y)}C(d)\prod_\nu(1+|q_\nu|)^{-1-(1-\alpha)/d}$, using $(1+|q|)^d \ge \prod_\nu(1+|q_\nu|)$; this is integrable over $\mathbb{R}^d$ uniformly in ξ.

Step 5 (the shift). Shift each p_ν to $p_\nu + i\delta_C\,\mathrm{sgn}(x-y)_\nu$; the two vertical sides of each rectangle cancel by Step 1, and the shifted integral carries $|e^{ip\cdot(x-y)}| = e^{-\delta_C|x-y|_1}$. For the Hölder quotient, $|x-y| \ge \min(|x-y|, |\tilde x - y|) - 1$ costs a factor e^{δ_C}. □

Lemma 4.5 (the free propagator $C^\xi := (-\Delta^\xi + 1)^{-1}$ on $\xi\mathbb{Z}^d$, $\xi = L^{-J}$)**.** *For all S, T: $|\Delta_T D_S C^\xi(x_1,x_2)| \le C'_C(\xi+\rho)^{-p(S,T)}e^{-\delta'\rho}\prod_{i\in T}|x_i - \tilde x_i|^\alpha$, with C'_C, δ' depending only on (d, L, α).*

Proof. Let $G := \mathsf{T}_J^{-1}$ of corollary 4.3 on $\xi\mathbb{Z}^d$ ($\eta \to \xi$, $k \to J$) with $m^2 = 1$; its block term is $a_J Q^*Q$ on unit blocks. Then $(C^\xi)^{-1} = G^{-1} - a_J Q^*Q$, so $C^\xi = G + a_J G Q^*Q C^\xi$, and the kernel of the second term is $a_J\sum_y (G\mathbf{1}_{B(y)})(x_1)(C^\xi\mathbf{1}_{B(y)})(x_2)$, since $(QC^\xi f)(y) = \langle C^\xi\mathbf{1}_{B(y)}, f\rangle$. G obeys (6). By (A5) for $G\mathbf{1}_{B(y)}$ and lemma 4.4 for $C^\xi\mathbf{1}_{B(y)}$, and the unit-lattice sum as in lemma 4.1, Step 3, the second term obeys $|\Delta_T D_S \cdot| \le Ce^{-\delta''\rho}\prod|\cdot|^\alpha \le C'(\xi+\rho)^{-p}e^{-\delta''\rho/2}\prod|\cdot|^\alpha$. □

Lemma 4.6 (lattice convolution, $d = 3$)**.** *For $p_1, p_2 \in [0,3)$, $\delta > 0$, $\xi \in (0,1]$ and $x, w \in \xi\mathbb{Z}^3$:*

$$\sum_{z\in\xi\mathbb{Z}^3} \xi^3(\xi+|x-z|)^{-p_1}(\xi+|z-w|)^{-p_2}e^{-\delta(|x-z|+|z-w|)} \le C_K\, h_{p_1+p_2-3}(\xi+|x-w|)\,e^{-\delta|x-w|/2}.$$

Proof. Factor out $e^{-\delta|x-w|/2}$ by the triangle inequality and let $r := \xi + |x-w|$. On $\{|x-z| \le |x-w|/2\}$ we have $\xi + |z-w| \ge r/2$, and $\sum_{|u|\le R}\xi^3(\xi+|u|)^{-p_1} \le C(R+\xi)^{3-p_1}$ since $p_1 < 3$; this region gives $\le Cr^{3-p_1-p_2}$. The region $\{|z-w| \le |x-w|/2\}$ is symmetric. On the rest, $|x-z| \le 3|z-w|$ and the sum is $\le C\sum_{|u|\ge|x-w|/2}\xi^3(\xi+|u|)^{-p_1-p_2}e^{-\delta|u|/2} \le Ch_{p_1+p_2-3}(r)$. For $r \ge 1$ the powers are absorbed by the remaining exponential. □

Lemma 4.7 (cumulative minus free: the gain of the resolvent comparison, $d = 3$)**.** *Let $\breve G := (-\Delta^\xi + \mu^2 + bP_Y)^{-1}$ on $\xi\mathbb{Z}^3$, with $\mu^2 \in [0,1]$, $b \in [b_-, b_+] \subset (0,\infty)$, P_Y the projection onto constants on the blocks Y of side $s_Y \in \{1, L\}$, and assume that $\breve G$ obeys (6) (with ξ in place of η). This holds for the rescaled cumulative propagators T_J^{-1} ($s_Y = 1$) and T_{J+1}^{-1} ($s_Y = L$; the case $J = k'+1$ of corollary 4.3 on the ξ-lattice, $\xi\mathbb{Z}^3$ being a union of L-blocks) of corollary 4.3, Step 1. Put $R := \breve G - C^\xi$. Then*

$$|\Delta_T D_S R(x_1,x_2)| \le C_R\, h_{|S|+\alpha|T|-1}(\xi+\rho)\,e^{-\delta_R\rho}\prod_{i\in T}|x_i - \tilde x_i|^\alpha. \tag{7}$$

So R and its Hölder quotients are bounded, one derivative costs at most a logarithm, and two derivatives (one per variable) cost $(\xi+\rho)^{-1}$: a gain of 2 (up to the logarithm) over C^ξ.

Proof. $\breve G - C^\xi = \breve G\big((C^\xi)^{-1} - \breve G^{-1}\big)C^\xi = (1-\mu^2)\breve G C^\xi - b\,\breve G P_Y C^\xi$; this is the identity of [3, p. 437].

Block part. $P_Y = \sum_Y |Y|^{-1}\mathbf{1}_Y\langle\mathbf{1}_Y, \cdot\rangle$, so the kernel of $\breve G P_Y C^\xi$ is $\sum_Y |Y|^{-1}(\breve G\mathbf{1}_Y)(x_1)(C^\xi\mathbf{1}_Y)(x_2)$. $\breve G\mathbf{1}_Y$ obeys the bounds of (A5): for $s_Y = 1$ directly, and for $s_Y = L$ after rescaling by L (the blocks become unit blocks), with constants depending on L. $C^\xi\mathbf{1}_Y$ is a sum of s_Y^3

unit-block terms $C^\xi \mathbf{1}_B$, $B \subset Y$, each bounded by lemma 4.4. So this part is bounded by $Ce^{-\delta''\rho}\prod|\cdot|^\alpha \le Ch_\gamma(\xi+\rho)e^{-\delta''\rho}\prod|\cdot|^\alpha$ for every γ.

Convolution part. The kernel is $\sum_z \xi^3 \breve{G}(x_1,z)C^\xi(z,x_2)$; differences and quotients in x_1 fall on $\breve{G}$ and those in x_2 on C^ξ. By (6) and lemma 4.5 the factors are bounded by $(\xi+|\cdot|)^{-p_i}e^{-\delta|\cdot|}$ with $p_1 = 1+[1\in S]+\alpha[1\in T] < 3$ and likewise p_2 (a quotient gives a minimum over two points, bounded by the sum of the two terms). Lemma 4.6 gives $h_{p_1+p_2-3} = h_{|S|+\alpha|T|-1}$. □

4.3. **S16: the classes without old vertex (chains),** [3, p. 433]**.** [3, p. 433]: after the $\tilde{B}'$-expansion "we get classes without old vertices. Such a class is a chain of propagators $G_k(\square, B_0)$ with new vertices in between … For such chains every subgraph has positive degree or contains (2.4), so the estimate (3.2) for them is a consequence of Proposition 2.1." (The subscript of the chain propagator is printed as a Q-like glyph; the expansion $G_k(\square, B_0+\tilde{B}') = \sum (G_0V)^m G_0 + \dots$ forces B_0.) Setting: (expansion) of section 4, $G_0 := G_k(\square, B_0)$, $e_k(\sigma_0+\sigma_1) \le 1$.

Lemma 4.8 (structure of a chain)**.** *A connected component that contains no old vertex is a path $v_1 - v_2 - \dots - v_m$ of new vertices of one product $(G_0V)^m G_0$, joined by lines G_0. Its end vertices v_1, v_m each carry one external scalar field (a cut line), and otherwise the component has only external $\tilde{B}'$-legs; if $m = 1$ it has no internal line and is a product of external fields. Chains contain no field* [3, (1.12)]*: their vertices are of the forms* [3, (1.8)–(1.11), (1.13)–(1.15)] *"with external vector field $\tilde{B}'$ only"* [3, p. 433]*, which is what makes "the end field is a cut line" exact.*

Proof. Lines between new vertices arise only inside one product $(G_0V)^m G_0$, where consecutive V's are joined by one G_0; new vertices of different products are joined only through old vertices. So a component without old vertices lies in one product. Cutting lines of a path leaves sub-paths. Each end of a sub-path was joined in the product either to an old vertex or to another new vertex, and in both cases that line was cut (otherwise the component would be larger), so it is an external scalar field. □

Lemma 4.9 (degrees of chains)**.** *Let $d = 3$. Every connected subgraph H of a chain component is a sub-path with $m_H \ge 2$ vertices, and $D(H) > 0$, except when $m_H = 2$ and H is the graph* [3, (2.4)]*: two* [3, (1.8)]*-vertices with $n' = 1$ whose common line carries the derivative at both ends. Every sub-path properly containing a* [3, (2.4)] *has $D \ge 1$. In particular "every subgraph has positive degree or contains (2.4)" holds in the sharper form "or is (2.4)".*

Proof. By [3, (2.1)], with external $\tilde{B}'$-legs of dimension 0 that still carry their factor η, $s \in \{1,2\}$ the number of internal scalar legs at v, and $\delta \in \{0,1\}$ with $\delta = 1$ only if the differentiated scalar leg is internal: [3, (1.8)] ($n = 0$, $n' \ge 1$; η-power $3+n'-1$) has $D_H(v) = 2 + n' - s/2 - \delta$; [3, (1.9)] ($n' \to \bar{n}+1$) has $D_H(v) \ge \bar{n}+1$; [3, (1.10)] ($n' \ge 2$; η-power $3+n'-2$) has $D_H(v) = 1+n'-s/2 \ge 2$; [3, (1.11)] has $\ge \bar{n}$; a block pair w ([3, (1.18)], one vertex made of two factors, each with η^d and one scalar leg, the $\tilde{B}'$-contour legs external) has $D_H(w) = 6 - s/2 \ge 5$, which is [3, p. 423 (v)] summed over the two factors (the reading under which the proof of [3, Cor. 2.3] on [3, p. 428], "if both endpoints of l belong to v, then $D_G(v) - d > 0$", holds). Hence an interior vertex ($s = 2$) has $D_H(v) \ge 1$, and an end vertex ($s = 1$) has $D_H(v) \ge \frac{3}{2}$, with equality only for [3, (1.8)] with $n' = 1$ and $\delta = 1$; every other end vertex has $\ge \frac{5}{2}$. With $D(H) = \sum_v D_H(v) - 3$: for $m_H = 2$, $D(H) \ge 0$ with equality iff H is [3, (2.4)], otherwise $D(H) \ge \frac{3}{2} + \frac{5}{2} - 3 = 1$; for $m_H \ge 3$, $D(H) \ge 3 + (m_H - 2) - 3 \ge 1$. □

Lemma 4.10 (the chain bound)**.** *Let H be a chain component with vertices $v_1, \dots, v_m$ ($2 \le m \le \bar{n}+1$), end fields Φ_1 at v_1 and Φ_m at v_m, and localisation cubes $\square(v)$. Assume $e_k(\sigma_0+\sigma_1) \le 1$, and let h be smooth with $h = 1$ on the 2η-neighbourhood of the end cube and $|\partial^\eta h| \le c_h$. Then for every $\delta_0 \in (0,1]$*

$$|E(H)| \le C_H \prod_v (e_k\|q\|\sigma_0)^{n'_v}\, \|h\Phi_1\|_{1,0}\|h\Phi_m\|_{1,0}\, e^{-\delta_0 d(\{\square(v)\})},$$

with C_H depending only on $(d, N, \|q\|, a, L, \bar{n}, c_h, c_m)$. This is [3, (3.2)] *for the chains, with sup-norms in place of $\|\cdot\|_{1,\alpha_0}$. No slice decomposition, no summation by parts and no depth inside $\square$ is used.*

Proof. Step 1 (coercivity). B_0 is constant, hence regular, so (H0) on $\square$ gives $G_0^{-1} \geq \gamma_0 > 0$. Put $\|\phi\|_E := \langle \phi, G_0^{-1}\phi\rangle^{1/2}$. Since $G_0^{-1} \geq D_{B_0}^* D_{B_0}$ and $G_0^{-1} \geq \gamma_0$, $\|D_{B_0}\phi\| \leq \|\phi\|_E$ and $\|\phi\| \leq \gamma_0^{-1/2}\|\phi\|_E$.

Step 2 (each vertex is a bounded form in the energy norm). Let $\beta_v(\phi,\chi)$ be the bilinear form of v in its two scalar legs. For [3, (1.8)]: $\beta = \kappa \sum_{b\subset\square} \eta^d [(D_{B_0}\phi)(b)\cdot M(b)\chi(b_-)]$ with $\kappa = e_k^{n'}\eta^{n'-1}/n'!$ and $|M(b)| \leq \|q\|^{n'}\sigma_0^{n'}$, so by Cauchy–Schwarz over the d bonds at a site, $|\beta| \leq \sqrt{d}(e_k\|q\|\sigma_0)^{n'}\gamma_0^{-1/2}\|\phi\|_E\|\chi\|_E$ ($\eta \leq 1$); the same with the derivative on χ. For [3, (1.9), (1.11)]: the same, using $|R_{\bar n+1}| \leq 1$ [3, p. 414]. For [3, (1.10)]: $\beta = \kappa' \sum_b \eta^d[\phi(b_-)\cdot M(b)\chi(b_-)]$ with $\kappa' = e_k^{n'}\eta^{n'-2}/n'!$, so $|\beta| \leq d(e_k\|q\|\sigma_0)^{n'}\gamma_0^{-1}\|\phi\|_E\|\chi\|_E$. For a block pair: a_k times a product of two factors [3, (1.13)–(1.14)] at one y; the contour factors obey $|\tilde B'(\Gamma)| \leq 2d\sigma_0$ (unit-block contours of η-length $\leq 2d$), $|U| = 1$, and $\sum_{x\in B(y)}\eta^d|\phi(x)| \leq \|\phi\|$, so $|\beta| \leq a(2de_k\|q\|\sigma_0)^{n'}\gamma_0^{-1}\|\phi\|_E\|\chi\|_E$. Altogether $|\beta_v| \leq c_v\|\cdot\|_E\|\cdot\|_E$ with $c_v \leq C(e_k\|q\|\sigma_0)^{n'_v}$.

Step 3 (sandwich). On the finite-dimensional space $\ell^2(\square,\eta^d)\otimes\mathbb{R}^N$ let K_v represent β_v ($\langle\phi, K_v\chi\rangle = \beta_v(\phi,\chi)$) and put $\hat K_v := G_0^{1/2}K_vG_0^{1/2}$, so $\|\hat K_v\| \leq c_v$ by Step 2. The component is $E(H) = \langle\bar\Phi_1, K_1G_0K_2G_0\cdots G_0K_m\bar\Phi_m\rangle$ with $\bar\Phi_i = h\Phi_i$: the end forms see their external field only within η of the end cube (transposes, for the other orientation of a leg, have the same norms). Since $K_1G_0K_2\cdots G_0K_m = G_0^{-1/2}\hat K_1\hat K_2\cdots\hat K_mG_0^{-1/2}$, we get $|E(H)| \leq \|h\Phi_1\|_E\prod_v c_v\|h\Phi_m\|_E$.

Step 4 (end fields). $h\Phi$ is supported in a cube of side ≤ 4, so $\|h\Phi\|_E^2 = \|D_{B_0}(h\Phi)\|^2 + m_k^2\|h\Phi\|^2 + a_k\|Q_k(B_0)h\Phi\|^2 \leq C(1+c_h)^2\|h\Phi\|_{1,0}^2$ by the Leibniz rule $D(h\Phi) = h(\cdot+\eta e)D\Phi + (\partial^\eta h)\Phi$; the constant uses $m_k^2 \leq c_m$.

Step 5 (exponential factor). Consecutive cubes $\square(v_i)$, $\square(v_{i+1})$ are at distance 0, because the line between them was not cut [3, pp. 432–433]. So $d(\{\square(v)\}) \leq \sum_i \operatorname{diam}\square(v_i) \leq 2\sqrt{3}m$, and $1 \leq e^{2\sqrt{3}\delta_0 m}e^{-\delta_0 d(\{\square(v)\})}$. □

Remark 4.11 (why the chains are bounded directly). (a) The operator form of Step 3 is the ℓ^2 sandwich of proposition 1.17 in its simplest case: every vertex is first order on each side and G_0 is never needed pointwise, so the subgraphs [3, (2.4)] need no summation by parts and the vertices may lie up to $\partial\square$. The printed route through [3, Prop. 2.1] would need the summation by parts [3, (2.8)] at [3, (2.4)]-subgraphs whose vertices may sit near $\partial\square$; [3, (2.8)] is printed with $\sum_{b\subset T_\eta}$, on the torus [3, p. 425], and on the Neumann box $\square$ the function $\psi = q\phi' g\tilde B'$ extended by zero jumps at the faces wherever the $\tilde B'$-localisation g does not vanish there, so [3, (2.8)] has a boundary term; the reason that removes it in S4 (lemma 1.26: the supports of the localisation functions lie at depth $\geq 2\eta$, $\operatorname{supp} g \subset \Omega_1$) is absent here. Lemma 4.10 makes the point moot. (b) [3, (3.2)] carries no exponential factor (it comes from the external-field estimates [3, (2.5), (3.1)]); the factor $e^{-\delta_0 d(\{\square(v)\})}$ in lemma 4.10 is harmless, Step 5 showing that it costs only $e^{2\sqrt{3}\delta_0 m}$. (c) Under the cluster cut of the main paper, a chain component lies in one cluster of label boxes, so its exponential factor is bounded below by $e^{-2\delta_0 V_{\mathrm{lab}}^\sharp}$, a constant depending on $\bar n$ only.

S16, consumed form. [3, p. 433]: "the estimate (3.2) for them is a consequence of Proposition 2.1". Delivered: [3, (3.2)] for every chain class, with sup-norms (stronger), constants independent of η, k, $\square$, B_0 and the torus, by lemmas 4.8 to 4.10. The chains are bounded at $G_0 = G_k(\square, B_0)$ directly; they do not pass through S17–S19, whose $O(1)$ bound on $\delta G_k(\square,\eta\mathbb{Z}^d,0)$ holds only at depth (lemma 4.16).

4.4. S17–S19: gauge covariance on $\square$, the gauge-out of B_0, and the field $\delta G_k(\square,\eta\mathbb{Z}^d,0)$.

S17: gauge covariance, [3, p. 433].

Lemma 4.12 (gauge transformation of the operators)**.** *Let $\square$ be a union of unit blocks with Neumann conditions, A a vector field on the bonds of $\square$, and $\lambda\colon \square \to \mathbb{R}$. Put $(V_\lambda\phi)(x) := e^{-e_kq\lambda(x)}\phi(x)$ (orthogonal, commuting with q) and $A_b^\lambda := A_b + \eta^{-1}(\lambda(b_+) - \lambda(b_-))$. Then:* (a)

$D_{A^\lambda}(V_\lambda\phi)(b) = e^{-e_k q\lambda(b_-)}(D_A\phi)(b)$, *hence* $-\Delta^{\eta,N}_{A^\lambda,\square} = V_\lambda(-\Delta^{\eta,N}_{A,\square})V_\lambda^*$; (b) $(Q_k(A^\lambda)V_\lambda\phi)(y) = e^{-e_k q\lambda(y)}(Q_k(A)\phi)(y)$, *hence* $P_k(A^\lambda) = V_\lambda P_k(A)V_\lambda^*$ *and* $G_k(\square, A^\lambda) = V_\lambda G_k(\square, A)V_\lambda^*$.

Proof. (a) $U(A^\lambda_b) = U(A_b)e^{e_k q(\lambda(b_+)-\lambda(b_-))}$, all exponentials of multiples of q commuting; insert into the definition of D. Both Laplacians sum over the same bonds $b \subset \square$, so the forms agree. (b) For the contour $\Gamma_{y,x} \subset B^k(y) \subset \square$ of [1, (2.1)], $A^\lambda(\Gamma_{y,x}) = A(\Gamma_{y,x}) + \lambda(x) - \lambda(y)$ by telescoping, hence $U(A^\lambda(\Gamma))e^{-e_k q\lambda(x)}\phi(x) = e^{-e_k q\lambda(y)}U(A(\Gamma))\phi(x)$. Then $P_k = Q_k^*Q_k$, the orthogonal factor at y cancels, and the mass term commutes. □

Lemma 4.13 (gauge invariance of the expressions)**.** *For every expression E of a class after the $\tilde{B}'$-expansion, with scalar lines $G_k(\square, B_0)$, vertices* [3, (1.6)–(1.11)] *and* [3, (1.13)–(1.15)] *at B_0, and any vector fields (A'-lines, $\tilde{A}$, $\tilde{B}'$, photon lines),*

$$E(B_0^\lambda; V_\lambda\Phi_{\text{ext}}) = E(B_0; \Phi_{\text{ext}}),$$

where V_λ acts on every scalar slot of every external field (kernels in several variables included: δG-fields, [3, (1.16)]*-kernels, the fields $\phi(y)$ of* [3, (1.12)]*) and the vector fields are not transformed. This is* [3, p. 433]*: "the expression corresponding to any such renormalized class is gauge-invariant ... if the external scalar fields are simultaneously transformed".*

Proof. [3, (1.6), (1.7)] and $(\phi\cdot\phi)^2$ are invariant under pointwise orthogonal maps. [3, (1.8)]: $[D_{B_0^\lambda}(V_\lambda\phi')(b)\cdot q^m(V_\lambda\phi'')(b_-)] = [e^{-e_k q\lambda(b_-)}D_{B_0}\phi'(b)]\cdot q^m e^{-e_k q\lambda(b_-)}\phi''(b_-) = D_{B_0}\phi'(b)\cdot q^m\phi''(b_-)$ by lemma 4.12(a) and the orthogonality of $e^{-e_k q\lambda(b_-)}$, which commutes with q^m. [3, (1.9)–(1.11)]: the same; $R_{\bar{n}+1}(-\eta q e_k\tilde{A}_b)$ is a function of q and commutes. [3, (1.13)–(1.15)]: by lemma 4.12(b) each factor at y acquires $e^{-e_k q\lambda(y)}$, and the vector contour factors are scalars times powers of q and commute; a pair at the same y is invariant, and a pair with [3, (1.12)] is invariant when $\phi(y) \mapsto e^{-e_k q\lambda(y)}\phi(y)$. Lines: $G_k(\square, B_0^\lambda) = V_\lambda G_k(\square, B_0)V_\lambda^*$. The expression is a full contraction of vertex tensors with line kernels and external fields; every V_λ either cancels against the V_λ^* of the line at the same slot or is absorbed into the transformed external field. □

S18: the gauge-out of B_0, [3, p. 433].

Lemma 4.14 (removing the constant field)**.** *Let $\square = \prod_\mu\{c_\mu, c_\mu+\eta, \dots, c_\mu+K_\mu-\eta\}$ be non-wrapping, i.e. a box of the lift of the torus (the no-wrap property of the cube step), let $B_0 \in \mathbb{R}^d$ be constant and $\lambda_0(x) := -B_0\cdot(x-c)$ in the box coordinates. Then $\eta^{-1}(\lambda_0(x+\eta e_\mu)-\lambda_0(x)) = -B_{0,\mu}$ on every bond of $\square$, so $B_0^{\lambda_0} = 0$, $G_k(\square, B_0) = V_{\lambda_0}^* G_k(\square, 0)V_{\lambda_0}$ and $E(B_0; \Phi_{\text{ext}}) = E(0; V_{\lambda_0}\Phi_{\text{ext}})$.*

Proof. λ_0 is single-valued and linear on $\square$ because the sites of a non-wrapping box have unique box coordinates. Apply lemmas 4.12 and 4.13 with $\lambda = \lambda_0$. □

This is the exact content of the gauge transformation "described more precisely in the proof of Lemma II.2.4" ([2, (2.69)–(2.74)]); nothing beyond lemmas 4.12 and 4.13 is needed, and no smallness is used.

S19: the field $\delta G_k(\square, \eta\mathbb{Z}^d, 0)$, [3, p. 433].

Lemma 4.15 (images)**.** *Let $\square$ be as in lemma 4.14 with $c \in \mathbb{Z}^d$ and $K \in \mathbb{Z}^d_{\geq 1}$ (a union of unit blocks), Neumann, and let $\mathcal{G}$ be the group generated by the reflections in the planes $x_\mu = c_\mu - \eta/2$ and $x_\mu = c_\mu + K_\mu - \eta/2$. Then for $x, x' \in \square$*

$$G_k(\square, 0)(x, x') = \sum_{\gamma\in\mathcal{G}} G_k(\eta\mathbb{Z}^d, 0)(x, \gamma x'),$$

absolutely convergent. Hence $\delta G_k(\square, \eta\mathbb{Z}^d, 0) := G_k(\square, 0) - G_k(\eta\mathbb{Z}^d, 0)|_{\square\times\square} = \sum_{\gamma\neq\text{id}} G_k(\eta\mathbb{Z}^d, 0)(\cdot, \gamma\cdot)$.

Proof. This is the image identity of (A5) at $j = k$, $\xi = \eta$, $m^2 = m_k^2$, $a = a_k$: the reflections preserve the block structure (the box is a union of unit blocks), the reflected sum $\sum_\gamma G_k(\eta\mathbb{Z}^d, 0)(x, \gamma x')$ is $\mathcal{G}$-symmetric, hence satisfies the Neumann conditions, and it inverts $-\Delta^{\eta,N}_\square + m_k^2 + a_k P_k$ on functions on $\square$, so it is $G_k(\square, 0)$. The absolute convergence follows from (6): the images $\gamma x'$ lie one in each tile $\gamma\square$. □

Lemma 4.16 (bound)**.** *For $u \in \square$ let $\varrho(u) := \operatorname{dist}_\infty(u, \square^c) \geq \eta$. For $\gamma \neq \mathrm{id}$ and $u, w \in \square$: $|u - \gamma w| \geq \varrho(u) + \varrho(w) - \eta \geq (\varrho(u) + \varrho(w))/2$. Consequently, for $x_1, x_2 \in \square$ with $\varrho(x_i) \geq 2$ and Hölder partners in $\square$ with $\varrho \geq \varrho(x_i) - 1$,*

$$|\Delta_T D_S\, \delta G_k(\square, \eta\mathbb{Z}^d, 0)(x_1, x_2)| \leq C_\delta e^{-(\delta_Z/8)(\varrho(x_1)+\varrho(x_2))} \prod_{i \in T} |x_i - \tilde{x}_i|^\alpha,$$

with C_δ depending only on (d, L, a, α). At points of the 1-neighbourhood of $\square_1$ and their partners, where $\varrho \geq r^\flat - 2 - \eta \geq 2$ (so $r^\flat \geq 5$ suffices), and with the localisation functions h, h' of [3, p. 432] *($|dh|, |d^2 h| \leq c_h$),*

$$\|h\, \delta G_k(\square, \eta\mathbb{Z}^d, 0)\, h'\|_{1,\alpha} \leq C(1 + 3\sqrt{3}c_h)^2 e^{-(\delta_Z/4)(r^\flat - 1)}.$$

Proof. Step 1 (distance to images). Let $\gamma \neq \mathrm{id}$ and let μ be a coordinate in which γ acts nontrivially. In that coordinate the group acts freely and each orbit meets the interval $I_\mu = \{c_\mu, \dots, c_\mu + K_\mu - \eta\}$ once, so $z := (\gamma w)_\mu \notin I_\mu$. The segment from u_μ to z crosses an end plane b of I_μ, so $|u_\mu - z| = |u_\mu - b| + |b - z|$ with $|u_\mu - b| \geq \varrho(u) - \eta/2$. If z lies in the copy adjacent across b, then $z = 2b - w_\mu$ and $|b - z| = |b - w_\mu| \geq \varrho(w) - \eta/2$; points of farther copies are at distance $\geq K_\mu - \eta/2 \geq |b - w_\mu|$ from b. Hence $|u - \gamma w| \geq |u_\mu - z| \geq \varrho(u) + \varrho(w) - \eta$.

Step 2 (termwise bounds). γ is a lattice isometry mapping bonds of $\square$ to bonds (possibly reversed) and partners to partners at the same distance. So each term $\Delta_T D_S[G_k(\eta\mathbb{Z}^d, 0)(x_1, \gamma x_2)]$ is bounded by (6) at the image points, where $\eta + \rho_\gamma \geq 1$: $\leq C_1 e^{-\delta_Z \rho_\gamma/2} \prod |\cdot|^\alpha$, ρ_γ the minimal distance between the points involved.

Step 3 (sum). $e^{-\delta_Z \rho_\gamma/2} \leq e^{-(\delta_Z/8)(\varrho(x_1)+\varrho(x_2))} e^{-(\delta_Z/4)\rho_\gamma} e^{\delta_Z}$, the shifts by η and by the partners being absorbed in the last factor. The images γx_2 lie one in each tile $\gamma\square$, and the tiles have sides ≥ 1, so $\sum_\gamma e^{-(\delta_Z/4)\operatorname{dist}(x_1, \gamma\square)} \leq C_d(\delta_Z)$.

Step 4 (localised norm). Lemma 1.18 converts the pointwise bound at the points within $1 + \eta$ of $\operatorname{supp} h \cup \operatorname{supp} h'$ into the $(1, \alpha)$-norm; the Hölder supremum of [3, (1.32)] over all pairs is $\leq \max\{\text{local supremum}, 2 \sup |D \cdot|\}$ (conventions of section 4), so the localised bound covers it. □

S19, consumed form. [3, p. 433]: "we substitute $G_k(\square, 0) = G_k(0) + \delta G_k(\square, \eta\mathbb{Z}^d, 0)$ and we treat δG_k as an external scalar field". Delivered: the substitution is the image identity (lemma 4.15), and the external field has $(1, \alpha)$-norm $O(e^{-(\delta_Z/4)(r^\flat - 1)}) \leq O(1)$ at the localisations of the classes with an old vertex, all of which lie in $\square_1$ (S15). The field is *not* $O(1)$ at face-adjacent points: for $\varrho(x_1) = \varrho(x_2) = \eta$ the face image is at distance η, and (6) gives only $\sim (2\eta)^{-1}$; so the $O(1)$ norm needs depth of order ≥ 1, which the old-vertex classes have ($\varrho \geq r^\flat - 1$), while the chain classes, whose localisations are not confined to $\square_1$, are bounded by lemma 4.10 directly at $G_k(\square, B_0)$, without S19. No auxiliary torus and no depth input beyond the containment in $\square_1$ is used.

4.5. **S20: the norms** [3, (1.32)] **at B_0 and at $\tilde{B}$,** [3, p. 434]**.** [3, p. 434]: after S17–S19 the expressions are estimated by [3, (3.3)] "with norms equal to (1.32) with $\tilde{B} = 0$" of the transformed external scalar fields, and "Thus we get the estimate (3.2) exactly". The transformed fields are $V_{\lambda_0}\Phi$; their norms at 0 are the norms of Φ at B_0 (lemma 4.17), and the norms at B_0 are compared with those at $\tilde{B}$ (lemma 4.18). Here $\|f\|_{1,\alpha;A}$ is the sum form of [3, (1.32)] with the Hölder supremum over all pairs.

Lemma 4.17 (covariance of the norm)**.** *For λ on $\square$ and any function f of one scalar slot, $\|V_\lambda f\|_{1,\alpha;A^\lambda} = \|f\|_{1,\alpha;A}$. In particular $\|V_{\lambda_0} f\|_{1,\alpha;0} = \|f\|_{1,\alpha;B_0}$, slot by slot for functions of several variables.*

Proof. The sup term is unchanged, V_λ being orthogonal. Derivative term: by lemma 4.12(a), $|D_{A^\lambda}(V_\lambda f)(b)| = |D_A f(b)|$. Hölder term: $U(A^\lambda(\Gamma_{x,x'})) = e^{-e_k q\lambda(x)} U(A(\Gamma_{x,x'})) e^{e_k q\lambda(x')}$ by telescoping along Γ; with the factor $e^{-e_k q\lambda(x')}$ from lemma 4.12(a) at the bond starting at x', the transported difference equals $e^{-e_k q\lambda(x)}$ times the old one. □

Lemma 4.18 (B_0 versus $\tilde B$)**.** *Let $B = B_0 + B'$ on $\square$ with B_0 constant, $\sigma_0 := \sup|B'|$, $\sigma_1 := \sup|\partial^\eta B'|$ over the bonds of $\square$, and $e_k(\sigma_0+\sigma_1) \le 1$. Then for every f supported in $\square$ (one slot),*

$$\|f\|_{1,\alpha;B_0} \le \big(1+C_{20}(f)\,e_k(\sigma_0+\sigma_1)\big)\|f\|_{1,\alpha;B}, \qquad C_{20}(f) := C(d,\|q\|)\,(1+D_f)^{1-\alpha}, \quad D_f := \operatorname{diam}\operatorname{supp} f,$$

and the same with B and B_0 exchanged. For several slots the factor is raised to the number of slots. The diameter factor cannot be dropped from this form of the bound: for B' constant and $D_B f$ a transported constant the B-Hölder term vanishes while the B_0-quotient at distance D is of order $e_k\sigma_0 D^{1-\alpha}|Df|$.

Proof. Write $D := D_B$, $D_0 := D_{B_0}$. For the bond b at x in direction μ,

$$D_0 f(x) = Df(x)+E(x), \qquad E(x) := U(B_b)\Phi(B'_b)f(x+\eta e_\mu), \qquad \Phi(\beta) := \eta^{-1}(U(-\beta)-1) = \eta^{-1}(e^{-e_k\eta q\beta}-1).$$

For antisymmetric X, $|e^X-1| \le |X|$, and $d\Phi/d\beta = -e_k q e^{-e_k\eta q\beta}$; so $|\Phi(\beta)| \le e_k\|q\||\beta|$ and $|\Phi(\beta)-\Phi(\beta')| \le e_k\|q\||\beta-\beta'|$.

Step 1. $\sup|D_0 f| \le \sup|Df| + e_k\|q\|\sigma_0 \sup|f|$.

Step 2 (the Hölder term, pairs with $|x-\tilde x| \le 1+D_f$). Let $\Gamma = \Gamma_{x,\tilde x}$ be shortest, $U_0 := U(B_0(\Gamma))$ and $U := U(B(\Gamma)) = U_0 U(B'(\Gamma))$. Then

$$U_0 D_0 f(\tilde x) - D_0 f(x) = [UDf(\tilde x) - Df(x)] + (U_0 - U)Df(\tilde x) + [U_0 E(\tilde x) - E(x)].$$

The first bracket is $\le |x-\tilde x|^\alpha$ times the Hölder term of $\|f\|_{1,\alpha;B}$. Next, $|U_0-U| = |U(B'(\Gamma))-1| \le e_k\|q\|\sigma_0|x-\tilde x|_1$. For the last bracket, with $\tilde b \parallel b$ at $\tilde x$, $|B'_{\tilde b} - B'_b| \le \sigma_1|x-\tilde x|_1$; write $U_0 U(B_{\tilde b})f(\tilde x+\eta e_\mu) = U(-B'(\Gamma))[U(B(\Gamma))U(B_{\tilde b})f(\tilde x+\eta e_\mu)]$. The bracket differs from $U(B_b)f(x+\eta e_\mu)$ by at most $|x-\tilde x|_1 \sup|Df| + |\mathsf{H}-1|\sup|f|$, where H is the holonomy of the loop $b \cup \Gamma' \cup \tilde b^{-1} \cup \Gamma^{-1}$, Γ' the translate of Γ by ηe_μ, and $|\mathsf{H}-1| \le 2e_k\|q\|\sigma_1\eta|x-\tilde x|_1$, since the plaquette flux is $\eta^2(\partial B'_\nu - \partial B'_\mu)$, B_0 being constant. Altogether $|U_0E(\tilde x) - E(x)| \le Ce_k\|q\|(\sigma_0+\sigma_1)|x-\tilde x|_1(\sup|f| + \sup|Df|)$. For $|x-\tilde x| \le 1+D_f$ one has $|x-\tilde x|_1 \le \sqrt{d}\,|x-\tilde x|^{1-\alpha}|x-\tilde x|^\alpha \le \sqrt{d}(1+D_f)^{1-\alpha}|x-\tilde x|^\alpha$, so the second and third terms contribute at most $C(1+D_f)^{1-\alpha}e_k(\sigma_0+\sigma_1)(\sup|f|+\sup|Df|)$ to the B_0-quotient.

Step 3 (pairs with $|x-\tilde x| > 1+D_f$). Df, E and $D_0 f$ vanish outside $S_\mu := \{y : y \in \operatorname{supp} f$ or $y+\eta e_\mu \in \operatorname{supp} f\}$, and $\operatorname{diam} S_\mu \le D_f+\eta \le 1+D_f$, so one of the two points, say x, lies outside S_μ. Then $U_0 D_0 f(\tilde x) - D_0 f(x) = U_0 Df(\tilde x) + U_0 E(\tilde x)$, so the B_0-quotient at this pair is at most the B-quotient at the same pair ($= |Df(\tilde x)|/|x-\tilde x|^\alpha$) plus $|E(\tilde x)|/|x-\tilde x|^\alpha \le e_k\|q\|\sigma_0\sup|f|$ (Step 1 and $|x-\tilde x| \ge 1$); if $\tilde x$ lies outside S_μ the roles are exchanged.

Step 4. In both cases the Hölder term of $\|f\|_{1,\alpha;B_0}$ is at most that of $\|f\|_{1,\alpha;B}$ plus $C(1+D_f)^{1-\alpha}e_k(\sigma_0+\sigma_1)(\sup|f|+\sup|Df|)$. Adding Step 1 gives the claim; the argument is symmetric in B, B_0. □

Remark 4.19 (a diameter-free form). Bounding instead the third bracket of Step 2 by $2\sup|E| \le 2e_k\|q\|\sigma_0\sup|f|$ for $|x-\tilde x| \ge 1$, and the second by $\sup|Df|\min\{2, e_k\|q\|\sigma_0|x-\tilde x|_1\}/|x-\tilde x|^\alpha \le 2^{1-\alpha}(\sqrt{d}e_k\|q\|\sigma_0)^\alpha\sup|Df|$, gives $\|f\|_{1,\alpha;B_0} \le (1+C(d,\|q\|)[(e_k\sigma_0)^\alpha + e_k(\sigma_0+\sigma_1)])\|f\|_{1,\alpha;B}$, symmetric in B_0, B, with no diameter; the order $(e_k\sigma_0)^\alpha$ is sharp by the example in lemma 4.18. The weakest form sufficing for corollary 4.20 is that the all-pairs Hölder supremum is $\le \max\{$local supremum, $2\sup|f|\}$ so that the factor is $\le 3 + Ce_k(\sigma_0+\sigma_1) \le 3+C$. Lemma 4.18 is the form used.

Corollary 4.20 ("Thus we get the estimate (3.2) exactly")**.** *If* [3, (3.3)] *holds for the transformed expressions, with the norms* [3, (1.32)] *at $\tilde B = 0$ of the external scalar fields $V_{\lambda_0}\Phi$, then* [3, (3.2)] *holds with the norms* [3, (1.32)] *at $\tilde B$, and the constant is multiplied by $(1+C_{20}(f)e_k(\sigma_0+\sigma_1))^s$, s the number of external scalar slots. For the h-localised external fields of* [3, (3.2)]*, $D_f = O(1)$:* [3, (1.33)] *multiplies the external legs by h, with $h = 1$ on $\square(v)$ and $h = 0$ outside a neighbourhood. In general $D_f \le \operatorname{diam}\square \le \sqrt{3}(3r^\flat + 2M)$, and the factor is $\le (1+C(d,\|q\|))^s = O(1)$ under* (cap)*, which reads $e_k(\sigma_0+\sigma_1)(1+\sqrt{3}(3r^\flat+2M)) \le 1$; with $\sigma_0 \le c(r^\flat+M)p(t)$ and $\sigma_1 \le c_1 p(t)$ this is $Ce\,t^{1/2}(1+\ln t^{-1})^{\bar p} \le 1$ for $t \le t_0$, a cap of the kind of* [3, p. 412].

Proof. Lemma 4.14, then lemma 4.17 (the 0-norm of $V_{\lambda_0}\Phi$ is the B_0-norm of Φ), then lemma 4.18 (the B_0-norm is at most the factor times the $\tilde B$-norm). External vector fields carry the norm with $\tilde B = 0$ by definition [3, p. 420] and are untouched. □

The cost of this step is thus a size factor $(1+O(e_k r^{\flat} p(t)))^{\#\text{slots}} \to 1$ of the printed kind; with the factor e_k inside U it is not $(1+O(r^{\flat}p))^{\#\text{slots}}$.

4.6. **S21, I: the identities of the renormalisation step, [3, pp. 434–445].** [3, p. 434] states [3, (3.5)]: for each class G_{ren} (after S13–S20, so with scalar lines $G_k(0)$ on $\eta\mathbb{Z}^d$, photon lines on the torus, vertices localised in $\square_1$ and all other objects external),

$$\sum_{G\in G_{\text{ren}}} \sum_{\tilde{l}} \sum_{j\in J(\tilde{l})} E(G(j)) = \sum_{G'\in G'_{\text{fin}}} \sum_{\tilde{l}'} \sum_{j\in J(\tilde{l}')} \gamma(G',\tilde{l}')E'(G'(j)),$$

with generalised graphs G' and sets of orderings (indicator γ) along which every subgraph of the sequence has positive degree; [3, Prop. 2.2] (theorem 2.4) then gives [3, (3.3)]. Bałaban writes out the lowest-order members of each divergent type and says "the same procedure" [3, p. 438], "the same applies" [3, p. 437], "in the way described several times" [3, p. 444] for the rest. This subsection and the next prove the identities and the one-loop classes which the renormalisation theorem of the main paper (Step 3(D_4)) uses; the two-loop remainder terms, the transport of legs across the subgraphs [3, (2.4)] and the induction of [3, pp. 444–445] are proved there. Throughout $d=3$, and $\alpha\in(0,\frac{1}{2})$ is Bałaban's Taylor exponent [3, p. 435], so that a loss of α never makes a half-integer degree non-positive. Every Ward–Takahashi identity below is applied to full lines (cumulative propagators summed over all internal scales below the external scale, then replaced by the free C^{ξ}), never to a fixed ordering or a fixed scale assignment, and every remainder of the replacement is bounded explicitly.

Lemma 4.21 (degree formula and the divergent types)**.** *Let G be connected and built from [3, (1.6), (1.8), (1.10)] only. Let m_v be the number of vector legs at v (internal and external), $\delta_v\in\{0,1\}$ with $\delta_v=1$ iff the differentiated scalar leg of a [3, (1.8)]-vertex is internal, V_6 the number of [3, (1.6)]-vertices, and $E=E_s+E_v$ the number of external scalar and vector legs. Then*

$$D(G) = V_6 + \sum_{(1.8)} \big(1+\tfrac{m_v}{2}-\delta_v\big) + \sum_{(1.10)} \tfrac{m_v}{2} + \tfrac{E}{2} - 3.$$

The vertex weights are $\geq\frac{1}{2}$ for [3, (1.8)] (equality iff $m_v=1$, $\delta_v=1$) and ≥ 1 for [3, (1.6), (1.10)]; E_s is even. Mass-counterterm vertices count as the graph they replace [3, (2.3)], and graphs containing a block vertex [3, (1.13)–(1.15)], an R-vertex or a finite counterterm have $D>0$ (corollary 2.11). Hence $D(G)\leq 0$ only in the following cases: $E\geq 5$ never; $E=4$ exactly the graph [3, (2.4)]; $E=3$ with $(E_s,E_v)=(0,3)$, which vanishes by lemma 4.22, or $(2,1)$, the $\phi\phi A$ class [3, (2.18)–(2.21)]; $E=2$ with $(2,0)$, the scalar self-energies, $D=-1$ only for [3, (3.6)] (the two tadpoles and [3, (3.8)]), and $(0,2)$, the vector self-energies, $D=-1$ only for the four graphs [3, (3.25)]; $E=1$ with $(0,1)$ vanishes by lemma 4.22, and $(1,0)$ is impossible; $E=0$ the vacuum graphs. Divergent graphs with $E\geq 2$, in particular every proper divergent subgraph, have at most four vertices; vacuum graphs can have up to six (six [3, (1.8)]-vertices with $m=1$ on one scalar loop with three photon lines have $D=0$ at order e^6), which matches the vacuum counterterms up to $\alpha+2\beta\leq 6$ of [3, p. 418].

Proof. $D_G(v)=(\eta\text{-power})-(\text{internal legs})/2-\delta_v$ in $d=3$, with η-powers 3, $3+m-1$, $3+m-2$ for [3, (1.6), (1.8), (1.10)] (lemma 2.6). Use $\sum_v(\text{internal legs})=\sum_v(\text{legs})-E$ with 4, $2+m$, $2+m$ legs; then $D=\sum_v D_G(v)-3$ gives the formula. $D\leq 0$ forces $\sum_v w_v\leq 3-E/2$, and $w\geq\frac{1}{2}$ bounds the number of vertices by $6-E$: by 4 for $E\geq 2$ and by 6 for $E=0$. $E=4$, $\sum w\leq 1$: a single vertex with a self-line has $E\leq 2$; two vertices need $w=\frac{1}{2}$ each, i.e. [3, (1.8)] with $m=1$, $\delta=1$, joined by the scalar line that carries both derivatives, which is [3, (2.4)]. $E=5$, $\sum w\leq\frac{1}{2}$: one [3, (1.8)]-vertex with $\delta=1$ needs an internal scalar line, hence a second vertex or a self-line, both impossible. $E=2$, $D=-1$, $\sum w=1$: a single [3, (1.6)] with a self-line; a single [3, (1.10)] with $m=2$ (the photon tadpole); or two [3, (1.8)] with $m=1$, $\delta=1$, i.e. [3, (3.8)] for $E_s=2$ and the bubbles of [3, (3.25)] for $E_v=2$; a single [3, (1.8)] with $m=2$ has $\delta=0$ when both scalar legs are external ($w=2$) and $\delta=1$ when they form a self-line ($E_v=2$, the fourth graph of [3, (3.25)]); [3, (1.10)] with $m=2$, $E_v=2$ is the third graph of [3, (3.25)]. □

Lemma 4.22 (C-parity; [3, p. 434]: "$\operatorname{tr} q^{2n+1} = 0$")**.** *After S18–S19 every scalar line is a scalar kernel times $\mathbf{1}_N$. If a graph without R-vertices has no external scalar leg and an odd number of external vector legs, its expression vanishes.*

Proof. At [3, (1.8)]/[3, (1.10)] the two scalar legs are joined by q^{m_v}; at [3, (1.6)] ($(\phi\cdot\phi)^2$) and [3, (1.7)] the legs are paired by δ_{ab}. With no external scalar legs the $\mathbb{R}^N$-index contractions close into loops, and each loop gives $\operatorname{Tr}\prod_v q^{m_v} = \operatorname{Tr} q^{M_\ell}$, M_ℓ the number of vector legs on the loop (all transporters are 1 at $B_0 = 0$). $\sum_\ell M_\ell = \sum_v m_v = 2I_v + E_v$ is odd, so some M_ℓ is odd, and $\operatorname{Tr} q^{\mathrm{odd}} = \operatorname{Tr}((q^{\mathrm{odd}})^T) = -\operatorname{Tr} q^{\mathrm{odd}} = 0$. □

Lemma 4.23 (Taylor's formula [3, (3.10)])**.** *For f on $\eta\mathbb{Z}^d$ and a shortest (monotone) contour $\Gamma_{x,y}$ from x to y,*

$$f(y) = f(x) + \sum_{\mu}(y_\mu - x_\mu)(\partial^\eta_\mu f)(x) + \sum_{b\subset\Gamma} s_b \eta\big[(\partial^\eta_{\mu_b} f)(b^{\mathrm{lo}}) - (\partial^\eta_{\mu_b} f)(x)\big],$$

b^{lo} the lower endpoint of b, μ_b its direction, $s_b = \operatorname{sgn}(y-x)_{\mu_b}$. The remainder is bounded by $|y-x|_1^{1+\alpha}$ times the α-Hölder constant of ∂f on the box spanned by x, y.

Proof. Telescoping along Γ: all μ-bonds of a monotone Γ have the same orientation, and there are $|y_\mu - x_\mu|/\eta$ of them. (Bałaban's $(b)_x$ is the forward bond at x parallel to b.) □

Lemma 4.24 (max-scale regrouping)**.** *Let $\mathcal{L}$ be the line set of a graph, $I \subset \mathcal{L}$ the internal lines of a subgraph G_0, and $\mathcal{E} \subset \mathcal{L}\setminus I$, $\mathcal{E}\neq\emptyset$, its external lines (the lines of the graph with an end at a vertex of G_0). Fix a strict total order $\prec$ of $\mathcal{L}$. For an index assignment $j\colon \mathcal{L}\to\{0,\dots,k-1\}$ let $\tilde l(j)$ order $\mathcal{L}$ by increasing (j_l, l) lexicographically, and put $J(\tilde l) := \{j : \tilde l(j) = \tilde l\}$; this is an admissible choice of the sets of [3, (2.7)] (they partition all assignments, and $J(\tilde l) \subset \{j_{l(1)} \le \cdots \le j_{l(m)}\}$). Let $\mathcal{M}$ be any set of orderings of $\mathcal{L}\setminus I$ and $\mathcal{L} := \{\tilde l : \tilde l|_{\mathcal{L}\setminus I} \in \mathcal{M}$, and every $l \in I$ precedes every $e \in \mathcal{E}\}$. Then for every function F of the assignment*

$$\sum_{\tilde l\in\mathcal{L}}\sum_{j\in J(\tilde l)} F(j) = \sum_{j_{\mathcal{L}\setminus I}:\ \tilde l(j_{\mathcal{L}\setminus I})\in\mathcal{M}}\ \sum_{j_I\in\prod_{l\in I}\{0,\dots,J_l-1\}} F(j),$$

where $e^* := \operatorname{argmin}_{e\in\mathcal{E}}(j_e, e)$ is the lexicographic minimum in $(j_e, \prec)$, which depends on $j_{\mathcal{E}}$, and $J_l := j_{e^*} + [l \prec e^*] \in \{j_{e^*}, j_{e^*}+1\}$. If F is multilinear in the slice propagators of the lines of I, the inner sum replaces every $l \in I$ by the cumulative propagator $\sum_{i<J_l} G_{(i)} = \mathsf{T}_{J_l}^{-1}$ (corollary 4.3(i); for the photon lines the same with the vector covariance). This is the form of "the same expression appears for all orderings ... with the only condition that they are earlier than the external lines" [3, p. 437] and "we sum over admissible orderings and indices" [3, p. 444].

Proof. The $J(\tilde l)$ partition the assignments, so the left side is the sum over $\{j : \tilde l(j) \in \mathcal{L}\}$. The order induced on $\mathcal{L}\setminus I$ depends on $j_{\mathcal{L}\setminus I}$ only. "Every $l \in I$ precedes every $e \in \mathcal{E}$" is equivalent to $(j_l, l) <_{\mathrm{lex}} (j_{e^*}, e^*)$ for all $l \in I$, and given $j_{\mathcal{E}}$ this is the product set $j_l \le J_l - 1$. □

The lemma is a statement about sums over complete index ranges of the internal lines below the external scale; it asserts no identity for a fixed ordering or scale assignment.

Lemma 4.25 (rescaling of the regrouped lines and coefficients)**.** *Let $j_0 := j_{e^*}$, $s := L^{j_0}\eta$ and $\xi := \eta/s = L^{-j_0}$. Rescaled by s (corollary 4.3, Step 1), a regrouped scalar line $\mathsf{T}_{J_l}^{-1}$ becomes $s^{2-d}\breve G_l(\cdot/s, \cdot/s)$, where $\breve G_l$ is of the type of lemma 4.7: mass $m^2s^2 \in [0,1]$, block term $a_{j_0}P$ on unit blocks if $J_l = j_0$, and $a_{j_0+1}L^{-2}P$ on L-blocks if $J_l = j_0+1$ (the case $J = k'+1$ of corollary 4.3). The same holds for the photon lines (componentwise; mass $\mu_0^2 t^2 s^2$). A local coefficient of degree D built from such lines (a sum over all but one vertex of a product of line kernels, derivatives and moments $(x'-x)^\beta$) equals s^D times the same expression on the ξ-lattice; in $d = 3$ with $D = 0$ it is "the same expression but on the scale ξ" [3, p. 444].*

Proof. Each line contributes s^{2-d} times its rescaled kernel, each derivative s^{-1}, each moment factor s, each vertex sum $\eta^d = s^d\xi^d$ and each vertex η-power s; the total exponent is D by [3, (2.1)–(2.3)]. □

Lemma 4.26 (the Ward–Takahashi identities, generated at full lines)**.** *Let Y be $\xi\mathbb{Z}^d$ or a periodic box, $C := C^\xi \otimes \mathbf{1}_N$ (lemma 4.5), $C_{\rm ph} := C^\xi \otimes \mathbf{1}_d$, A a finitely supported external bond field, and F any polynomial in the $|\phi(x)|^2$ (for instance $\lambda\sum :|\phi|^4:$ and $\frac12\sum \delta m^2(x) :|\phi(x)|^2:$, Wick-ordered with respect to any constants; mass counterterm vertices included). Put $Z(A) := \int d\mu_{C_{\rm ph}}(A') \int d\phi \exp[-\frac12\langle\phi,(-\Delta^\xi_{A+A'}+1)\phi\rangle - F(\phi)]$ and $W := \log Z$ as formal power series in $(e,\lambda,\delta m^2,A)$. The Taylor coefficients of W are the sums of all connected graphs of the given order and external structure with full lines C, $C_{\rm ph}$ (the vertices are those of* [3, (1.8)–(1.11)] *with unrestricted n, n', which agree with Bałaban's below order $\bar n+1$). Then:* (a) *on a periodic box, $Z(A+\partial\lambda)=Z(A)$ for every periodic λ;* (b) *every Taylor coefficient W_n (symmetric n-linear in A at fixed order in the couplings) obeys $W_n(\partial\lambda,A_2,\dots,A_n)=0$;* (c) *(b) holds on $\xi\mathbb{Z}^d$ for finitely supported λ;* (d) *the two-vector-leg kernel $\Pi_{\mu\nu}(y,y')$ of every order obeys $\sum_{y'}\xi^d\Pi_{\mu\nu}(y,y')=0$ for all μ,ν,y.*

Proof. (a)–(c) are lemmas 2.13 to 2.15 with $M^2=1$, the block term absent and $B=0$ (remark 2.18(c)): substituting $\phi=V_\lambda\psi$ has Jacobian 1, $-\Delta_{A+\partial\lambda+A'}=V_\lambda(-\Delta_{A+A'})V_\lambda^*$ (lemma 4.12(a); a periodic box has no boundary), and F is invariant, which gives (a); comparing powers of s in $W((sA)+s\partial\lambda)=W(sA)$ and polarising gives (b); at fixed order a coefficient is a finite sum of connected graphs whose lines are periodisations of the $\xi\mathbb{Z}^d$ lines (exponential decay, lemma 4.5), and with A, λ finitely supported each graph sum converges absolutely to its $\xi\mathbb{Z}^d$ value as the box grows, which gives (c). (d) is lemma 2.16(b) for Π: by (c), $\sum_{y',\nu}\xi^d\Pi_{\mu\nu}(y,y')(\lambda(y'+\xi e_\nu)-\lambda(y'))/\xi=0$ for all finitely supported λ, i.e. $\sum_\nu[\Pi_{\mu\nu}(y,z-\xi e_\nu)-\Pi_{\mu\nu}(y,z)]=0$ for every z; multiply by z_κ and sum over z (absolutely convergent by the exponential decay); the shift gives $\xi\sum_w\xi^d\Pi_{\mu\kappa}(y,w)=0$. □

Lemma 4.27 (lattice reflections; the symmetry behind [3, (3.27)–(3.28), (3.37), (3.38)])**.** *For the theory of lemma 4.26 on $\xi\mathbb{Z}^d$ with constant coefficients, the reflection R_κ ($x_\kappa\mapsto -x_\kappa$) acts by $\phi\mapsto\phi\circ R_\kappa$, $A_\mu(b)\mapsto A_\mu(R_\kappa b)$ ($\mu\neq\kappa$) and $A_\kappa(b)\mapsto -A_\kappa(b^r)$, where for $b=(x,x+\xi e_\kappa)$ we put $b^r := (R_\kappa x-\xi e_\kappa, R_\kappa x)$. This leaves Z invariant. Consequently, for the full fixed-order classes with lines C, $C_{\rm ph}$:* (a) *the zero-momentum $\phi\phi A$ coefficient $\sum_{x_1,x_2}\xi^{2d}\Gamma_\kappa(x_1,x_2;b)$ vanishes;* (b) *if $\sum_{y'}\xi^d\Pi_{\mu\nu}(y,y')=0$, then the first moment $\sum_{y'}\xi^d\Pi_{\mu\nu}(y,y')(y'_\lambda-y_\lambda)$ vanishes for all μ,ν,λ ($d\geq 2$).*

Proof. Invariance. $|U(A_b)\phi(b_+)-\phi(b_-)|=|U(-A_b)\phi(b_-)-\phi(b_+)|$ by orthogonality, so reversing a bond with $A\mapsto -A$ preserves the action; C is even, and $C_{\rm ph}$ is invariant under $A_\kappa\mapsto -A_\kappa$. *(a)* $\Gamma_\kappa(R_\kappa x_1,R_\kappa x_2;b^r)=-\Gamma_\kappa(x_1,x_2;b)$, and by translation invariance $\sum_{x_1,x_2}\Gamma_\kappa(x_1,x_2;b)$ does not depend on b; so the sum equals minus itself. *(b)* Write $m(b)$ for the bond midpoint. Translation invariance gives $\Pi_{\mu\nu}(b,b')=\pi_{\mu\nu}(m(b')-m(b))$ and the invariance gives $\pi_{\mu\nu}(R_\kappa u)=s^\kappa_\mu s^\kappa_\nu\pi_{\mu\nu}(u)$ with $s^\kappa_\mu=-1$ iff $\mu=\kappa$. Since $y'_\lambda-y_\lambda=(m(b')-m(b))_\lambda+\frac\xi2(\delta_{\mu\lambda}-\delta_{\nu\lambda})$ and $\sum_u\pi_{\mu\nu}(u)=0$ by hypothesis, the first moment is $M:=\sum_u\xi^d\pi_{\mu\nu}(u)u_\lambda$. Reflection in $\kappa=\lambda$ gives $M=-s^\lambda_\mu s^\lambda_\nu M$, so $M=0$ unless exactly one of μ,ν equals λ; reflection in $\kappa\neq\lambda$ gives $M=s^\kappa_\mu s^\kappa_\nu M$, so $M=0$ unless, for every $\kappa\neq\lambda$, $\mu=\kappa\Leftrightarrow\nu=\kappa$. Both conditions together are impossible: if $\mu=\lambda\neq\nu$, then $\kappa=\nu$ violates the second. □

[3, (3.27)–(3.28)] (a computation for the one-loop bubble) and [3, (3.37), (3.38)] ("the functions which we are summing are odd") are instances of (b) and (a).

4.7. **S21, II: the one-loop classes and the subtraction identities.** Notation: j_0 is the lowest index of the external legs of the class, $\xi=L^{-j_0}$, and after lemmas 4.24 and 4.25 every internal line is $\breve G_l=C^\xi+R_l$ with R_l as in lemma 4.7; $\hat G_s$, $\hat G_{\rm ph}$ denote the rescaled regrouped scalar and photon lines. Sums over y' carry the weight ξ^3. The bounds $|C^\xi|,|\hat G|\leq C(\xi+\rho)^{-1}e^{-\delta\rho}$ (lemma 4.5, corollary 4.3), $|\partial R|\leq C(1+|\log(\xi+\rho)|)e^{-\delta\rho}$ and $|\partial R\partial^*|\leq C(\xi+\rho)^{-1}e^{-\delta\rho}$ (lemma 4.7) are used throughout.

Lemma 4.28 (scalar self-energies with $D=-1$, [3, (3.6)–(3.17)])**.** (a) *The two tadpoles of* [3, (3.6)] *and the last two counterterms of* [3, (3.7)] *cancel exactly ("Wick ordering"): the class*

graph has the loop line $G_k(0)(x,x)$ or the torus photon line at (x,x) and the localisation of [3, (1.6)] *to unit cubes of Ω_1, resp. the factors $g_k(x)^2$; the counterterms* [3, (1.29)] *carry the same localisation and the same lines once the torus scalar line is identified with $G_k(0)$, the difference being an image term, an external field of norm $O(e^{-\delta S_k})$ (lemma 4.35).* (b) *For* [3, (3.8)] *minus its counterterm, i.e. the expression* [3, (3.9)]*: Taylor's formula (lemma 4.23) applied to the leg with the smaller index* [3, p. 435] *gives* [3, (3.11)]*. Its second term is the generalised graph* [3, (3.12)]*, of degree $-1+1+\alpha=\alpha$; the factor $|x'-x|^{1+\alpha}$ multiplies every subgraph containing a line of G_0, since both lines join x and x', so its* [3, (2.4)]*-subgraph has degree $1+\alpha>0$. Its first term, regrouped (lemma 4.24) and rescaled (lemma 4.25, $D=0$), is the vertex* [3, (3.15)] *with coefficient*

$$c_\mu(y)=\sum_{y'}\xi^3\sum_{\nu} q(\partial_\nu\hat G_s\partial_\nu^*)(y,y')\,q\,g(y)\hat G_{\rm ph}(y,y')g'(y')\,(y'_\mu-y_\mu),$$

and $|c_\mu|\le C$ uniformly in ξ, j_0, k. This is [3, (3.17)].

Proof. (a) is the identity of the lines and localisations stated. (b): write $\hat G_s=C^\xi+R_s$. Since C^ξ is translation invariant, $\sum_\nu\partial_\nu C^\xi\partial_\nu^*=-\Delta^\xi C^\xi=\delta^\xi-C^\xi$ and $\delta^\xi(y'-y)(y'_\mu-y_\mu)=0$, so

$$c_\mu=-q^2\sum_{y'}\xi^3C^\xi(y-y')\,g\hat G_{\rm ph}g'\,(y'-y)_\mu+q^2\sum_{y'}\xi^3\Big(\sum_\nu\partial_\nu R_s\partial_\nu^*\Big)(y,y')\,g\hat G_{\rm ph}g'\,(y'-y)_\mu,$$

which is [3, (3.16)] (whose first line prints $g(y')$ for $g'(y')$). The first sum is $\le C\sum\xi^3(\xi+\rho)^{-2}\rho e^{-\delta\rho}\le C$, and by lemma 4.7 the second is bounded in the same way. □

Lemma 4.29 (scalar self-energies with $D=0$, [3, (3.18)–(3.20)])**.** *If $\Sigma(x,x')$ is the expression of a graph G_0 of the class with external legs at x, x', and its counterterm graph is G_0 with both legs at x ([3, p. 417]: "represented by the same graph G but with both external legs localized in x"; so it has the same lines, slices and orderings), then* [3, (3.19)] *holds exactly:*

$$\sum\eta^{2d}\phi(x)\cdot\Sigma(x,x')\phi'(x')-\sum\eta^d\phi(x)\cdot\Big(\sum_{x'}\eta^d\Sigma(x,x')\Big)\phi'(x)=\sum\eta^{2d}\phi(x)\cdot\big(\Sigma(x,x')|x'-x|^\alpha\big)\frac{\phi'(x')-\phi'(x)}{|x'-x|^\alpha}.$$

Along a chain of vertices $x=x_0,\dots,x_l=x'$ of G_0, $|x'-x|^\alpha\le\sum_i|x_{i+1}-x_i|^\alpha$ (concavity), which distributes the gain $+\alpha$ to the lines of a path from x to x'. A subgraph containing no line of the path is a proper subgraph of G_0, hence positive when G_0 is primitively divergent within the ordering; the Hölder quotient of the external leg costs α on that leg, and $D\ge\frac12>\alpha$ for every positive subgraph containing it.

Proof. The identity is the definition of the counterterm graph; the degree bookkeeping is as stated. □

Lemma 4.30 (the differentiated-leg graphs [3, (3.21)–(3.24)])**.** (a) *The first graph of* [3, (3.21)] *(a* [3, (1.8)]*-vertex with $n=2$ whose two A'-legs are contracted) has a coefficient of the type $\eta e_k^2G_{\rm ph}(b,b)$, bounded uniformly: by (6) at $\rho=0$, $|G(x,x)|\le C_1\eta^{-1}$ in $d=3$; this is the printed "$\eta G_k(x,x)$ is convergent".* (b) *The second graph, after Taylor at order 0, is* [3, (3.22)]*: a remainder of degree α plus the vertex* [3, (3.23)]*, whose coefficient (of the vertex $\phi\cdot\partial_\mu\phi'$) is bounded uniformly in ξ, j_0, k.*

Proof. (b) Write $g'(x')=g'(x)+(g'(x')-g'(x))$; the difference carries $|\partial g'||x'-x|$ (degree $+1$) and is convergent. The main term, rescaled and with $\hat G=C^\xi+R$, is $q^2g(x)g'(x)[\sum_y\xi^3(\partial_\mu C^\xi)(y)C^\xi(y)+$ (terms with an R)]. Exactly, $\sum_y\xi^3(\partial_\mu C^\xi)(y)C^\xi(y)=-\frac\xi2\sum_y\xi^3|(\partial_\mu C^\xi)(y)|^2$, by symmetrising with $\sum_y C(y+\xi e)^2=\sum_y C(y)^2$; by lemma 4.5 this is $\le\frac\xi2C\sum\xi^3(\xi+|y|)^{-4}\le C$. The R-terms are $\le C\sum\xi^3[(1+|\log(\xi+\rho)|)(\xi+\rho)^{-1}+(\xi+\rho)^{-2}]e^{-\delta\rho}\le C$ by lemmas 4.5 and 4.7. ([3, (3.24)] shows a finite limit; only boundedness is used.) □

Lemma 4.31 (vector self-energies with $D=-1$, [3, (3.25)–(3.30)])**.** *Let the class be all order-e_k^2 graphs with two external vector legs: the two bubbles and the two tadpoles of* [3, (3.25)]*. Apply Taylor to first order on the leg $g'A'$ at x' around x ([3, (3.26)]). The Hölder remainder is a generalised graph of degree $-1+1+\alpha=\alpha$. After regrouping and rescaling the zero-order*

coefficient is $\Pi^{(\eta)}_{\mu\mu'}(x) = (L^{j_0}\eta)^{-1}\Pi^{(\xi)}_{\mu\mu'}(y)$ *and the first-moment coefficient is* $\Pi^{(\eta)}_{\mu\mu'\nu}(x) = \Pi^{(\xi)}_{\mu\mu'\nu}(y)$, *with lines* $\breve{G}_l = C^\xi + R_l$. *Then:* (i) *the all-C part of* $\Pi^{(\xi)}_{\mu\mu'}$ *vanishes (lemma 4.26(d); Bałaban's* [3, (3.29)] *via* [3, (2.26)] *at* $M^2 = 1$*), and the all-C part of* $\Pi^{(\xi)}_{\mu\mu'\nu}$ *vanishes (lemma 4.27(b);* [3, (3.27)–(3.28)]*);* (ii) *the terms with at least one R-line are bounded uniformly in* ξ, j_0, k, *both for* $\Pi^{(\xi)}_{\mu\mu'}$ *and for* $\Pi^{(\xi)}_{\mu\mu'\nu}$. *So* [3, (3.30)] *holds: the class equals two generalised graphs of degree* α, *plus the vertex* $A \cdot \Pi_{\mu\mu'\nu}\partial A$ *with a bounded coefficient, plus the vertex* $A \cdot \Pi_{\mu\mu'}A$ *with coefficient* $(L^{j_0}\eta)^{-1}O(1)$. *The last is a zero-momentum remainder vertex of power* -1 *at the scale* $L^{j_0}\eta$ *of the external legs; it is kept and bounded, not claimed to vanish.*

Proof. (i) is as stated. (ii) The R-terms of the bubbles are of four kinds. $(C\partial^*)(R\partial^*)$ and $(R\partial^*)(R\partial^*)$: bounded by $C\sum \xi^3(\xi+\rho)^{-2}(1+|\log(\xi+\rho)|)e^{-\delta\rho} \le C$. $C\cdot(\partial R\partial^*)$: bounded by $C\sum \xi^3(\xi+\rho)^{-2}e^{-\delta\rho} \le C$. $R\cdot(\partial R\partial^*)$: bounded likewise. $R(y,y')(\partial_{\mu'}C\partial^*_\mu)(y'-y)$: here absolute values give only a logarithm. Write $R(y,y') = R(y,y) + [R(y,y') - R(y,y)]$. Then $\sum_{y'}\xi^3(\partial_{\mu'}C\partial^*_\mu)(y'-y) = 0$ exactly (the sum over y' of a forward difference in y' of a decaying function), and $|R(y,y')-R(y,y)| \le C'\rho(1+|\log(\xi+\rho)|)$ for $y' \ne y$, by integrating the pointwise bound $|\partial R(y,z)| \le C(1+|\log(\xi+|z-y|)|)$ of lemma 4.7 along a monotone path from y to y' (the integral is $\le C(\xi+\rho)(2+|\log(\xi+\rho)|) \le C'\rho(1+|\log(\xi+\rho)|)$ since $\rho \ge \xi$; the cruder bound $|y'-y|_1 \sup|\partial R|$ would give a $\log(1/\xi)$). This leaves $\sum \xi^3\rho(1+|\log(\xi+\rho)|)(\xi+\rho)^{-3}e^{-\delta\rho} \le C$. The tadpoles with R are bounded by $|R(y,y)| \le C$ and $\xi|(R\partial^*)(y,y)| \le C\xi(1+|\log\xi|)$. For the first moment there is an extra factor ρ, and all four kinds are bounded by absolute values; the fourth gives $\sum \xi^3\rho(\xi+\rho)^{-3}e^{-\delta\rho} \le C$. □

Remark 4.32. Bałaban's "If at least one propagator $G_{j_0}(0)$ is replaced by $G_{j_0}(0)(1-m^2-aP)C^\xi$, then we get a convergent expression" [3, p. 441] is true, but for the term $R\cdot\partial C\partial^*$ of the bubble it needs the cancellation $\sum_{y'}\partial_{\mu'}(\cdots) = 0$ above, not power counting with absolute values: in position space R is bounded but does not vanish at coincident points. The cancellation uses only the bounds of lemma 4.7 on the R-line and the forward-difference structure of the other line.

Lemma 4.33 (the subtraction identities of the $D = 0$ vector and $\phi\phi A$ classes, [3, (3.31)–(3.38)])**.** (a) *For the remaining divergent two-vector-leg graphs (orders* e_k^4, $e_k^2\lambda$, *with counterterm insertions; at most four vertices by lemma 4.21),* [3, (3.31)] *is an exact identity: the Hölder term has degree* $+\alpha$, *and the local term has coefficient* $\sum_{x'}\eta^d\Pi_{\mu\mu'}(x,x')$. *After regrouping and rescaling, the all-*C^ξ *part of this coefficient vanishes by lemma 4.26(d) for the full class of that order (the classes "containing all possible connected graphs ... of a fixed order"* [3, p. 431]*); the localisation functions of internal photon legs are first moved to* x *by Taylor in* g, *which costs a degree* $+1$ *(*[3, p. 444], [3, (3.36)]*). The terms with at least one R-line gain at least* 1 *in the superficial degree:* $|R| \le C$ *against* $(\xi+\rho)^{-1}$, $|\partial R| \le C(1+|\log|)$ *against* $(\xi+\rho)^{-2}$, $|\partial R\partial^*| \le C(\xi+\rho)^{-1}$ *against* $(\xi+\rho)^{-3}$. (b) *For the* $\phi\phi A$ *class:* [3, (3.34)] *is an exact rewriting (transport the legs with the lowest indices to the vertex with the highest-index leg, through Hölder differences); the localisation functions are moved to the vertex* x' *by Taylor in* g, *costing degree* $+1$ *(*[3, (3.36)]*); after regrouping, rescaling and C-replacement the all-C zero-momentum coefficient* $\sum_{y,y''}\xi^{2d}\Gamma''_\mu$ *vanishes for the full class of each order by lemma 4.27(a), of which Bałaban's class-by-class vanishings are instances or consequences:* [3, (2.21a)] *by lemma 4.22;* [3, (2.18)] *"the same expressions with opposite signs";* [3, (2.19)] *by* [3, (3.37)]*;* [3, (2.20)] *by* [3, (3.38)]*;* [3, (2.21c)] *by translation invariance;* [3, (2.21b), (2.21g)] *"with the proper counterterm". The R-terms gain* ≥ 1 *as in (a).*

Proof. The identities are Taylor's formula (lemma 4.23) at order 0 and the definitions; the vanishings are lemma 4.26(d) and lemma 4.27; the gains of the R-terms are lemma 4.7 against lemma 4.5. □

What remains of S21, and where it is proved. The identities above reduce [3, (3.5)] to three statements about the regrouped totals: that the R-terms of the two-loop $D = 0$ classes (the superficial gain of lemma 4.33 made into a bound with the inner subtractions) are bounded

uniformly in ξ; that the legs can be transported across the subgraphs [3, (2.4)] so that each is either made convergent by a Hölder factor or admits the summation by parts [3, (2.8)] without a derivative on an external field ([3, p. 443]); and the induction of [3, pp. 444–445] over the orderings, with the vacuum classes cancelled exactly by the vacuum counterterms [3, (1.24)] up to the image terms of lemma 4.35 (vacuum graphs of at most six vertices, lemma 4.21). These are the renormalisation theorem of the main paper (Step 3(D_4)), which takes lemmas 4.21 to 4.31 and 4.33 and corollary 4.3 and lemmas 4.5 to 4.7 as its inputs. In it, every inner processing is an exact rewriting (lemmas 4.23 and 4.24) and the counterterm graphs carry the same lines ([3, p. 417]), so the total over the regrouped index range is the original total and lemma 4.26(d), lemma 4.27 and the exact counterterm cancellation apply to it.

4.8. **The seven further sketched places F1–F7.**

Lemma 4.34 (F1: the Ω_2 branch of the cube split, [3, p. 433]: "[or $G_k(\Omega_2,\tilde B)$]")**.** *If every old vertex has depth $> D^\flat$ in Ω_2 (the depth parameter of the cube step), then $\square \subset \Omega_2$ and $G_k(\Omega_2,\tilde B) = G_k(\square,\tilde B) + \delta G_k(\Omega_2,\square,\tilde B)$. At points of $\square_1$, $\delta G_k(\Omega_2,\square,\tilde B)$ obeys the two-domain estimate* [1, (2.26)] *in its restricted form for the pair $(\square,\Omega_2)$ (proposition 1.23 and corollary 1.24), exactly as the cube step does for $(\square,\Omega)$. Every later step (S13–S21, F2–F7) treats the split line $G_k(\square,\tilde B)$ and never refers to which of Ω, Ω_2 was split.*

Proof. The cube step of the main paper is stated and proved for both branches ("depth $> D^\flat$ in Ω (resp. Ω_2)"): the containment $\square \subset \Omega_2$ follows from the depth hypothesis and the side of $\square$, and the two-domain estimate for $(\square,\Omega_2)$ is proposition 1.23 with Ω_2 in the role of the larger domain. No new step is involved. □

Lemma 4.35 (F2: torus versus $\eta\mathbb{Z}^d$ propagators in the counterterms, [3, pp. 418–419] against [3, pp. 433–445])**.** *Let S_k be the torus side in scale-k units, a multiple of L, with $S_k \geq \max\{6\sqrt{3}\,r(t_k)+2,\ 8/\delta_Z\}$, δ_Z the constant of lemma 4.1. (The second entry can be dropped at the cost of a constant depending on δ_Z, since $\sum_{m\neq 0} e^{-\delta_Z S_k|m|/4} \leq C(d)(1-e^{-\delta_Z/4})^{-d}e^{-\delta_Z S_k/4}$ for all $S_k \geq 1$; it is kept so that the constants below are those of corollary 4.3.) Then:*

(a) Periodisation. *$G_k(T_\eta,0)(x,x') = \sum_{n\in S_k\mathbb{Z}^d} G_k(\eta\mathbb{Z}^d,0)(\tilde x,\tilde x'+n)$ for lifts $\tilde x,\tilde x'$, and the same for each slice and for the photon covariance; the sum converges absolutely.*

(b) The image term on a cube. *On a non-wrapping cube $\square$ of side $3r(t_k)$, $\delta K := G_k(T_\eta,0) - G_k(\eta\mathbb{Z}^d,0)$ obeys $\|\delta K\|_{1,\alpha} \leq Ce^{-\delta_Z S_k/4}$ on $\square\times\square$.*

(c) The counterterms in class form. *Expand every scalar line of a counterterm graph* [3, (1.26)–(1.31)]*, transformed as on* [3, p. 433]*, as $G_k(\eta\mathbb{Z}^d,0)+\delta K$. The term without δK is the counterterm in the class form used in* [3, (3.7)–(3.9), (3.19)] *and on* [3, pp. 444–445]*: same lines, vertices and localisation, the photon lines being torus lines on both sides. Every other term carries at least one external field δK of norm $\leq Ce^{-\delta_Z S_k/4} \leq Ce^{-\delta_Z r(t_k)}$, and with that line removed it is bounded by* [3, (3.3)] *for the classes into which it decomposes (the regrouping after a cut, proved in the main paper), by induction on the number of lines, lower-order counterterms being treated first* [3, p. 417].

Proof. (a) is the periodisation identity of (A5) for the periodic box, with the absolute convergence from (6) (the images lie one in each period cell). (b) Every image point is at distance $\geq S_k/2 \geq 1$ from $\square$, so (6) applies termwise with $\rho \geq S_k/2$, and the sum over $n\neq 0$ is bounded as in lemma 4.16, Step 3; the pointwise bounds give the $(1,\alpha)$-norm since the all-pairs Hölder supremum is $\leq \max\{\text{local supremum}, 2\sup|D\cdot|\}$. (c) is the stated expansion and induction. □

Lemma 4.35 makes the cancellations "exact" in lemma 4.28(a), lemma 4.29 and in the vacuum clause of the induction, up to these image terms. The condition $S_k \geq 8/\delta_Z$ is a lower bound on the torus side of the same kind as $S_k \geq 6\sqrt{3}\,r(t_k)+2$, fixed after L and before ε_0; the main paper lists both.

Lemma 4.36 (F3: "We repeat some of the operations done before", [3, p. 433])**.** (i) *(*[3, (3.1)] *on $\square$ at the constant field.) For localisation functions h, h' with $\operatorname{dist}(\operatorname{supp}h,\operatorname{supp}h') \geq 1$ and $|dh|,|d^2h| \leq c_h$: $\|hG_k(\square,B_0)h'\|_{1,\alpha} \leq C(1+3\sqrt{3}c_h)^2e^{-(\delta_Z/4)\operatorname{dist}(\operatorname{supp}h,\operatorname{supp}h')}$.* (ii) *(*[3, (1.16)]

on $\square$.) For the remainders $K = [G_0 V]^n \tilde{G} [V G_0]^{n'}$ of the $\tilde{B}'$-expansion, with $G_0 := G_k(\square, B_0)$, $\tilde{G} := G_k(\square, \tilde{B})$, $V := G_0^{-1} - \tilde{G}^{-1}$, $n, n' \geq 2$ ([3, p. 414]: "for n, n' sufficiently large"), and either $m := n + n' \geq 4$ and $\alpha < \frac{1}{2}$, or $m \geq 6$ and $\alpha < 1$: $\|hKh'\|_{1,\alpha} \leq C(C\sigma)^{n+n'} e^{-\beta \operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')}$ with $\sigma := e_k(\sigma_0 + \sigma_1) \leq 1$.

Proof. (i) $G_k(\square, B_0) = V_{\lambda_0}^* G_k(\square, 0) V_{\lambda_0}$ (lemma 4.14), and the $(1, \alpha)$-norms at B_0 equal those of the gauge-transformed kernel at 0 (lemma 4.17 in each slot). (6) on $X = \square$ (type (N)) gives the mixed bounds at every pair of points with $\rho \geq 1$, and lemma 1.18 localises them. No depth inside $\square$ is needed, because $\square$ is a parallelepiped at zero field. (ii) The ℓ^2 sandwich (proposition 1.17) applies on $X = \square$ with $(\tilde{A}, \tilde{B}) \to (\tilde{B}', B_0)$, all its inputs being available at every point of $\square$: (H0) from the coercivity of section 1.1; the weighted bounds (W) of lemma 1.11 for any regular field and any boundary condition; the pointwise bounds (K), needed only for the end factors, which are G_0 (since $n, n' \geq 2$), from lemma 4.14 and (6) on $\square$ at every point; and the symmetry. The vertex lemma lemma 1.14 holds with $(\tilde{A}, \tilde{B}) \to (\tilde{B}', B_0)$, both operators living on the same bonds and blocks of $\square$, with $\sigma = e_k(\sigma_0 + \sigma_1) \leq 1$. If an end factor were $\tilde{G}$ (n or $n' \leq 1$), pointwise bounds for $\tilde{G}$ at a non-constant field near $\partial\square$ would be needed; this case does not occur, the remainders being taken in the two-sided form with $n, n' \geq 2$ (part III, S14). $\square$

Lemma 4.37 (F4: "all necessary Ward–Takahashi identities", [3, pp. 430–431])**.** *The identities are those of lemma 4.26 (section 2.3): they cover every gauge-invariant polynomial F, every order in A, the limit of free boundary conditions (part (c)) and the zero-momentum form used on* [3, pp. 441–442] *(part (d)). The classes to which they are applied are the full fixed-order classes: all connected graphs of a given order and external-leg type with agreeing localisations (per label, as in section 4.9).* [3, p. 431] *allows this explicitly for the vector self-energies, and every smallest class of* [3, pp. 429–431] *is contained in one; the consumers of Proposition 1 (*[1, 2] *and* [13]*) use only sums over all graphs, so Proposition 1 for the full classes is what is consumed.*

Proof. Lemma 4.26; the class choice is a definition. $\square$

Lemma 4.38 (F5: "the remaining powers of $L^j\eta$ connected with the graph G_1 give us $(L^j\eta)^{D(G_1)}$", [3, p. 427])**.** *In* [3, (2.14)] *let $G_1 = \{l(1); v, v'\}$, $v \neq v'$, $j := j_{l(1)}$ the lowest index, so $j(v) = j(v') = j$ and $|\Delta(v)| = |\Delta(v')| = (L^j\eta)^d$. Perform $\sum_{\Delta(v')}$ against the exponential of $l(1)$ (an $O(1)$ sum, the cubes having side $L^j\eta$), then $\sum_{\Delta(v) \subset \Delta(v_1)}$ using the factor $(L^j\eta)^d$ of v, which produces the factor $(L^{j(v_1)}\eta)^d$ of the new vertex v_1 (the cubes are L-adically nested). The factors left over from G_1 are $(L^j\eta)^{d+p_{v'}}$ from v' and $(L^j\eta)^{p_v}$ from v, where p_v is the η-power of v excluding the volume factor η^d ("a proper power of $L^{j(v)}\eta$"); $(L^j\eta)^{-(d-2)/2}$ for each of the two legs of $l(1)$; and $(L^j\eta)^{-1}$ for each derivative on $l(1)$. The legs of the other lines at v, v' keep their own factors and now belong to v_1. The product is $(L^j\eta)^e$ with $e = d + p_v + p_{v'} - (d-2) - \delta_v - \delta_{v'} = D_{G_1}(v) + D_{G_1}(v') - d = D(G_1)$ by* [3, (2.1)–(2.2)]*. The general step* [3, (2.16)] *is the same count with v, v' replaced by shrunk vertices whose factors are those of* [3, (2.14)] *for the shrunk graph (lemma 2.3, Step 1(iv)).*

Lemma 4.39 (F6: "In the case $v = v'$ we make the same operations beginning with the summation over $\Delta(v)$", [3, p. 428])**.** *If $l(1)$ is a self-line at v, its exponential factor is 1 and there is no $\Delta(v')$-sum. $\sum_{\Delta(v) \subset \Delta(v_1)}$ uses $(L^j\eta)^d$ and produces $(L^{j(v_1)}\eta)^d$. The leftover is $(L^j\eta)^{p_v - (d-2) - \delta_v} = (L^j\eta)^{D_{G_1}(v) - d} = (L^j\eta)^{D(G_1)}$. Then $\sum_{j \leq j_{l(2)}} (L^j\eta)^{D(G_1)} \leq (1 - L^{-D})^{-1} (L^{j_{l(2)}}\eta)^{D(G_1)}$, since $D(G_1) \geq \frac{1}{2}$. This is* [3, (2.15)] *in that case (lemma 2.3, Step 2).*

Proof of lemmas 4.38 and 4.39. Bookkeeping on [3, (2.14)], as stated; the two lemmas are the first step of lemma 2.3 in Bałaban's notation. $\square$

Lemma 4.40 (F7: components separately; "We can extract also all coupling constants", [3, p. 433])**.** (a) *If the graphs of a class are unions of components sharing no line, the expression is the product of the component expressions: the vertex sums, the lines and the external fields factorise, and the class of a union is the product of the component classes by construction* [3,

pp. 431–432]. (b) *Every vertex carries the monomial* $e_k^{d_v(v)}\lambda(t)^{d_s(v)}$ *times a bounded combinatorial factor depending on* $\bar{n}$ *only; all graphs of a class have the same orders* [3, p. 420]*, so* $e_k^{d_v(G_{\mathrm{ren}})}\lambda^{d_s(G_{\mathrm{ren}})}$ *factors out, the counterterm coefficients being defined graph by graph with the order of their graph* [3, p. 417]. (c) *For a connected class localised in a cube of side* $O(\bar{n})$ *(the cluster side* l_N *of the main paper),* $d(\{\square(v)\}) \le O(\bar{n})$*, so* $e^{-\delta_0 d} \ge e^{-O(\bar{n})\delta_0}$*. Hence* [3, (3.3)] *for the components, times their coupling monomials, gives* [3, (3.2)]*, and then* [3, (1.33)] *with the exponential factor from* [3, (3.1)] *(proposition 2.22;* [3, p. 432]*).*

Proof. Bookkeeping, as stated. □

4.9. **S14.4: the classes after the $\tilde{B}'$-expansion,** [3, p. 433]**.** [3, p. 433]: the $\tilde{B}'$-expansion of a renormalised class gives an expression "which can be decomposed into a sum of renormalized classes". Part III proves (S14) that the untouched divergent subgraphs of an expanded graph are renormalised by terms of the expansion of the class, and that a touched divergent subgraph changes type: a 2ϕ subgraph touched once becomes a $2\phi 1A$ subgraph of degree 0, which needs the partner with its two external scalar legs exchanged; a $1A$ subgraph touched once, or a vacuum component touched twice, becomes a $2A$ subgraph of degree ≤ 0, which needs its Ward–Takahashi class. The closure property this requires of the class family is proved here.
Classes. Localisation labels are attached to each leg of a vector field (the support box of its partition function, a grid box of side ≤ 2), to each [3, (1.6)]/[3, (1.7)] vertex (its unit cube) and to each block factor [3, (1.13)–(1.15)] (the block $B^k(y)$); the counterterm graph $T_\gamma G$ of a 2ϕ subgraph γ carries the labels of G ([3, p. 417]: the legs are attached at one point, the internal vertices keep their localisations). Two graphs are "in agreement" [3, p. 432] if they carry the same label multiset, and the *per-label full class* of a given external type and order is the set of all connected graphs of that type and order with a given label multiset; equivalently, with the partition functions written as multipliers $t_y H_y$ on the vector legs and $t_Q \mathbf{1}_Q$ on the [3, (1.6)]/[3, (1.7)] vertices, it is the coefficient of one monomial $\prod t$ in the expansion of the generating functional [3, (1.5)]. This is the class choice of lemma 4.37, permitted by [3, p. 431], and it is the one the main paper uses for the regrouping after the cut and for the renormalisation. After the $\tilde{B}'$-expansion the new vertices are labelled in the same way (one label per $\tilde{B}'$-leg).

Lemma 4.41 (exchange of the external scalar legs)**.** *A per-label full class is closed under the exchange of the two external scalar legs of any* 2ϕ *or* $\phi\phi A$ *subgraph, the whole graph included.*

Proof. The class is a t-monomial coefficient of [3, (1.5)], and the Wick expansion sums over all pairings of legs. Exchanging which of two legs is paired with which outside slot is a bijection of pairings; it preserves orders and labels (labels sit on legs and vertices, not on pairings), and it maps graphs of the class to graphs of the class. For the whole graph the two external slots are the arguments of the external field $\Phi_{\mathrm{ext}}(x, x')$, and the sum over pairings contains both assignments. □

Lemma 4.42 (the $\tilde{B}'$-Taylor coefficients of a class)**.** *Let* $\mathcal{C}$ *be the per-label full class of type* $1A$ *(resp. vacuum) and order* o*, at background* $\tilde{B} = B_0 + \tilde{B}'$ *on* $\square$*. Its first (resp. second)* $\tilde{B}'$*-Taylor coefficient is the per-label full class of type* $2A$ *and the same order, with one (resp. two) external legs equal to* $\tilde{B}'$ *carrying their partition labels; the combinatorial weights are those of the Taylor coefficients.*

Proof. On $\square$ the action $S(\phi; B_0 + \tilde{B}' + X)$ depends on $\tilde{B}'$ and on the other gauge-field legs X (A' with its g and partitions, $\tilde{A}$) only through the total field $B_0 + \tilde{B}' + X$ ([3, (1.4)]: both the kinetic term and the averaging term depend on the total field only), and $\tilde{B}'$-legs carry partition labels like every vector leg. Let $W(\tilde{B}', X) :=$ log of the Gaussian integral at background B_0; W is a function of $\tilde{B}' + X$, so $\partial/\partial\tilde{B}'(y) = \partial/\partial X(y)$ on every Taylor coefficient, label by label. By the linked-cluster theorem (lemma 2.14) the Taylor coefficients of W are the sums of all connected graphs, whose vertices are the joint Taylor coefficients of S in the total field and whose lines are $G_k(\square, B_0)$. The $\tilde{B}'$-expansion of part III produces exactly these joint Taylor coefficients: insertions on lines are the Taylor coefficients of $G_k(\square, B_0 + \tilde{B}')$, and expanded old vertices are

the $\tilde{B}'$-Taylor coefficients of the old vertices; the factorised form of part III and the resolvent expansion [1, (3.44)] are two groupings of the same polynomials. So the first $\tilde{B}'$-coefficient of the $1A$ class is the coefficient of one X-leg and one $\tilde{B}'$-leg of $\partial^2 W/\partial X^2$, which is the $2A$ class, and the second coefficient of the vacuum class is the same with two $\tilde{B}'$-legs. (The parity statement that odd total orders in $(A, \tilde{B}')$ vanish plays no role here: the bijection holds for the full polynomials.) □

Lemma 4.43 (compatibility with the cut and with the renormalisation)**.** (a) *The expanded class is again a per-label full class. Its cut on* □ *(the lines between different clusters of label boxes become external fields) is the cluster cut of the main paper on its label boxes, and the regrouping theorem there applies verbatim. A touched* 2ϕ *subgraph and its mirror (lemma 4.41) have the same labels, as do a touched* $1A$ *subgraph and its* $2A$ *completion (lemma 4.42); so partners stay in one block of the cut or are split together.* (b) *The renormalisation step (sections 4.6 and 4.7 and the main paper) is applied to per-label full classes, the class choice of lemma 4.37. The* $\tilde{B}'$*-legs are external fields A in the Ward–Takahashi identities (lemma 4.26), which hold for every external field and every additional field X; after the gauge-out S18,* $B_0 = 0$ *and the lines are* $G_k(0)$.

Proof. (a) The clusters are determined by the label multiset, which the exchange and the completion preserve (the moved ends stay at vertices whose labels lie in the same cluster), so the cut set is unchanged. (b) As stated. □

S14.4, consumed form. With lemmas 4.41 to 4.43, the partners required by the touched subgraphs of part III are terms of the expansion of the class: the exchange partner of a touched 2ϕ subgraph S is $(S^\pi)_\tau = (S_\tau)^\pi$ with $S^\pi \subset G^\pi$ in the class (lemma 4.41), and the $2A$ Ward–Takahashi class at the order of a touched $1A$ subgraph or vacuum component is the set of all single (resp. double) $\tilde{B}'$-attachments to all graphs of its type (lemma 4.42). The expansion of a per-label full class is therefore a sum of per-label full classes, which is the decomposition "into a sum of renormalized classes" of [3, p. 433]; S15 ("again a sum of renormalized classes") inherits it. Bałaban's one sentence hides a closure property of the class family under differentiation in the background, which the per-label full classes have.

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